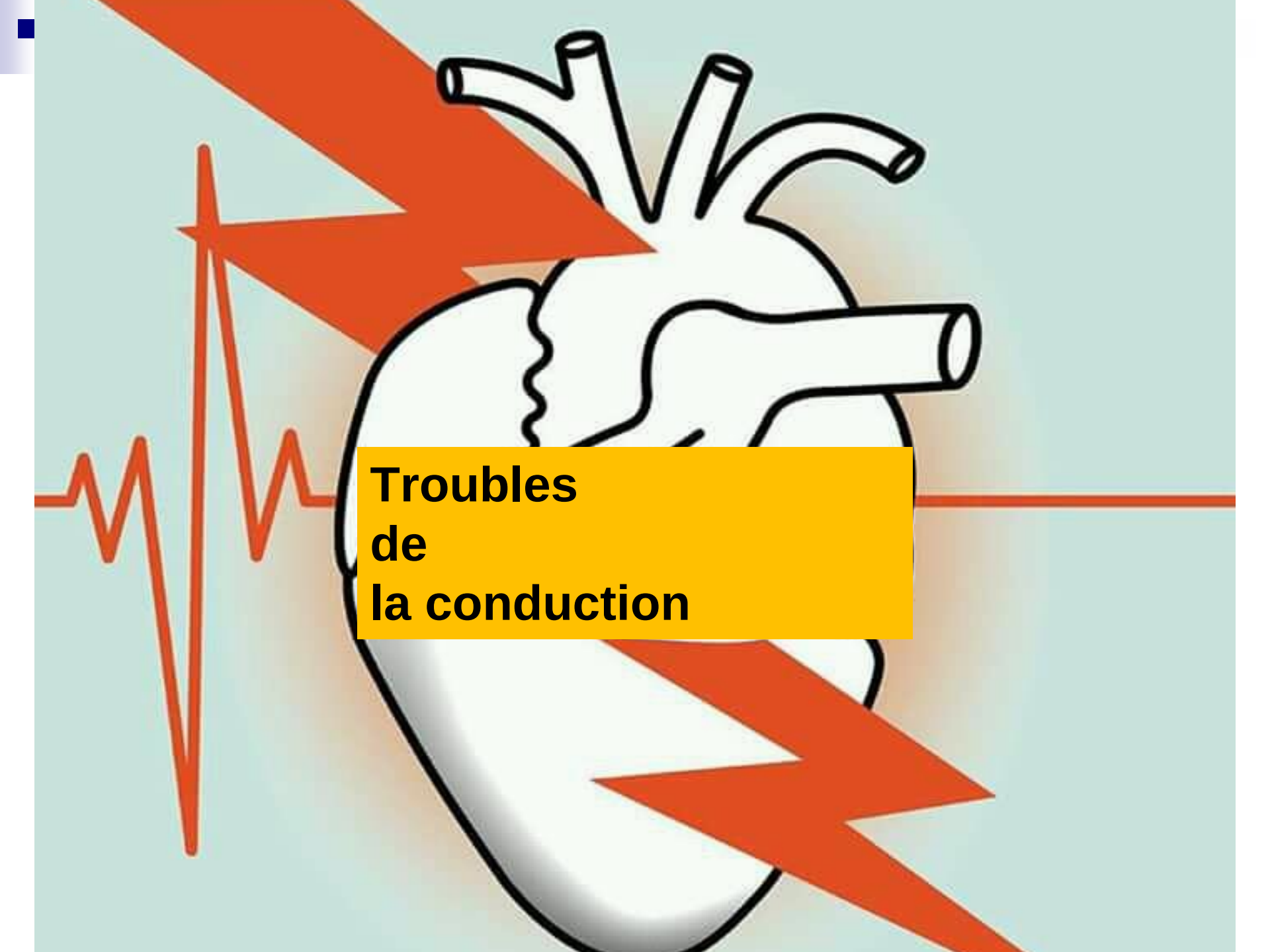




Trouble de la conduction

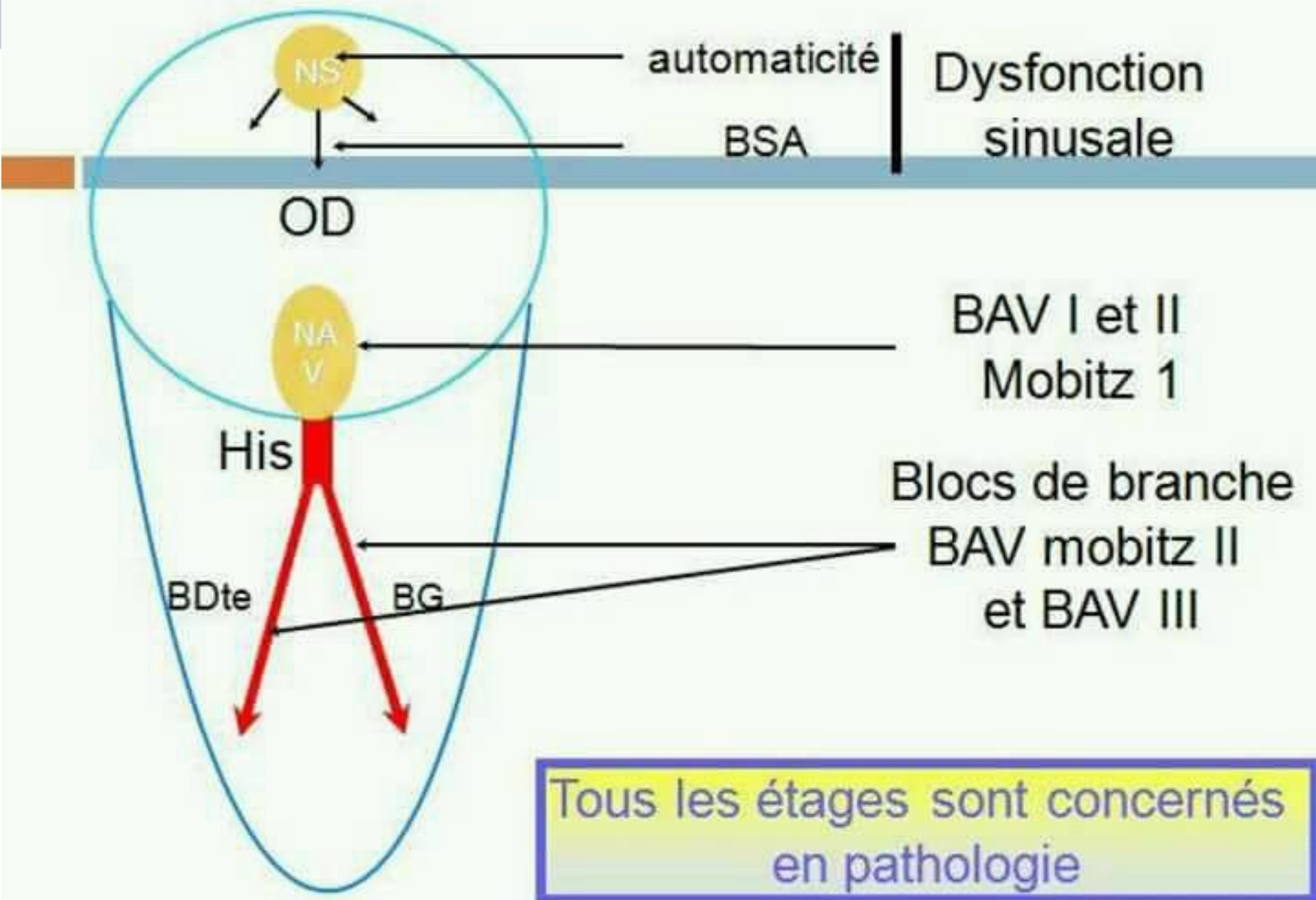
Dr Chelouache Tarek
Service de cardiologie
HMRUC.

A stylized illustration of a human heart in white with black outlines. A large, thick red lightning bolt strikes the heart from the top left. To the left of the heart, a red ECG line is shown. A yellow rectangular box is superimposed over the middle of the heart.

Troubles de la conduction

Les troubles de conduction

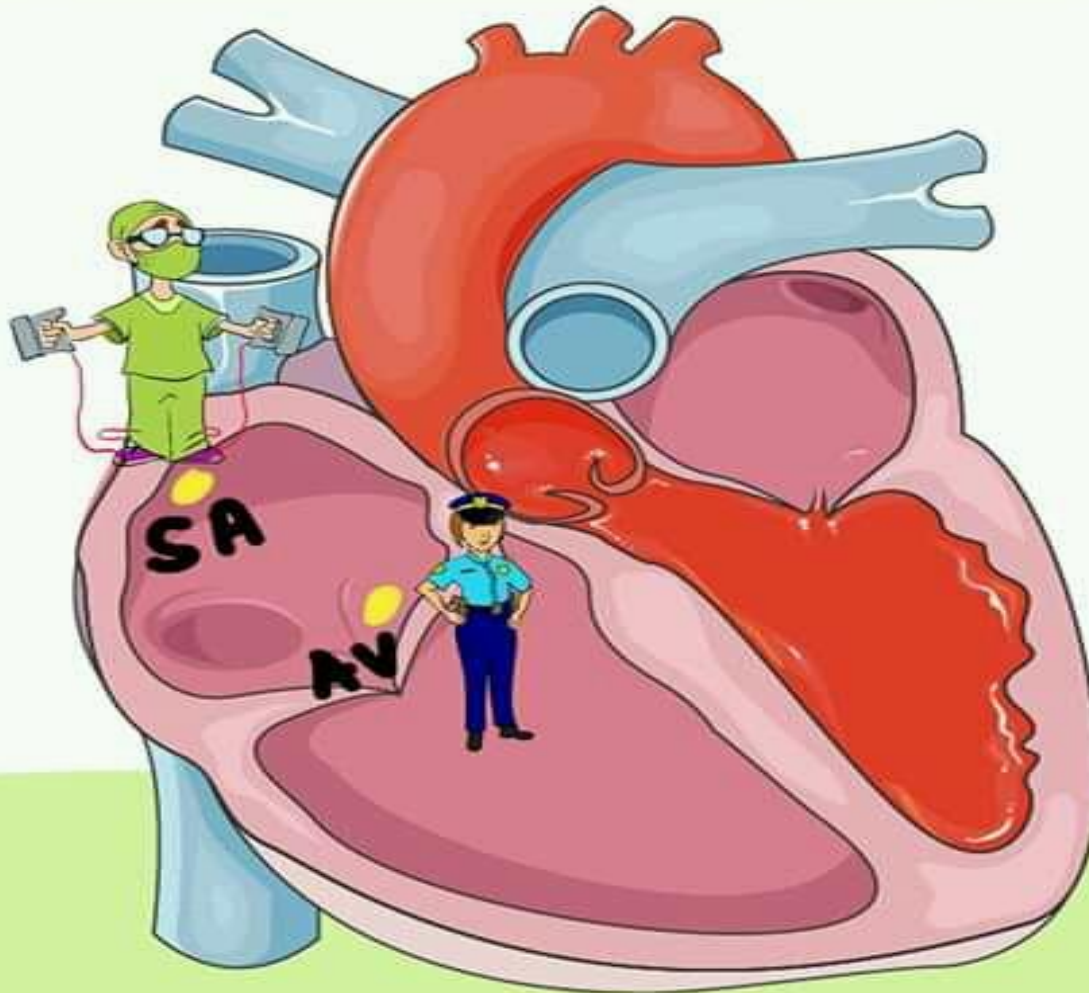
- Nœud sinusal :
 - ☐ bradycardie sinusale
 - ☐ Paralysie sinusale
 - ☐ bloc sino-auriculaire (BSA) de type 1, 2 et 3.
- Nœud auriculo-ventriculaire :
 - ☐ Bloc auriculoventriculaire (BAV) de type 1, 2, ou 3
- Branches du faisceau de His :
 - ☐ Bloc de branche droit (BBD)
 - ☐ Bloc de branche gauche (BBG)
- Branche gauche du faisceau de His :
 - ☐ Hémibloc antérieur gauche (HBAG)
 - ☐ Hémibloc postérieur gauche (HBPG)



SA node Vs AV node

www.medinaz.com

SA node is the “**pacemaker**” of the heart
AV node is the “**gatekeeper**” of the heart



Les blocs sinoauriculaires

BSA I



P-P constants
Bloc non détectable

BSA II
Type I croissant



P-P croissant
jusqu' à pause

BSA II
Type I décroissant



P-P décroissant
jusqu' à pause

BSA II
Type II



P-P constants
une seule pause

BSA II
haut degré



P-P constants
 ≥ 2 pauses

BSA III



P absente
Rythme d' échappement

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P. Taboulet

Plan

- Les Bloc auriculo-ventriculaire (B.A.V):
 - Quelques rappels:
 - _ Rappel anatomique.
 - _ Caractéristiques électrophysiologiques.
 - Situations cliniques.
 - Diagnostique des B.A.V.
 - Prise en charge thérapeutique.



Objectif :

- Connaître l'anatomie du tissu nodal.
- Comprendre la vascularisation du tissu nodal.
- Connaître les caractéristiques électrophysiologiques du TC.

Objectif :

- Enumérer les différentes étiologies des B.A.V.
- Apprendre les manifestations cliniques des B.A.V
- Comprendre les aspects électriques des B.A.V

Objectif :

- Savoir interpréter un tracé ECG des différents B.A.V
- Connaître la démarche Dgc et thérapeutique médicale et interventionnelle des B.A.V.

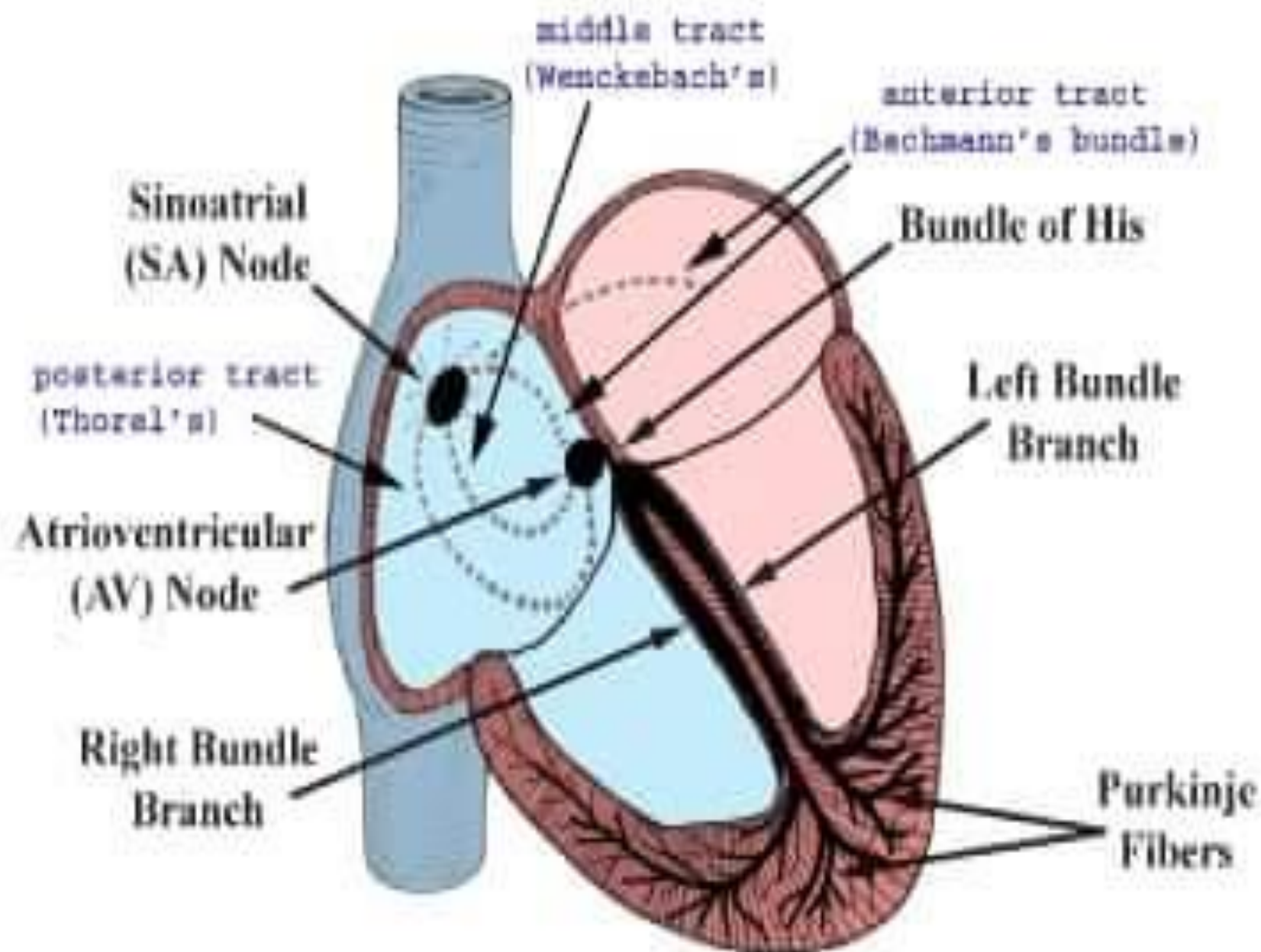
Bloc auriculo ventriculaire

- troubles du cheminement de l'activation sur la voie de conduction auriculo-ventriculaire
 1. soit dans le noeud d'Aschoff- Tawara,
 2. soit dans le tronc du faisceau de His,
 3. soit dans ses deux branches.
- prolongement de la période réfractaire



Anatomie du Tissu de conduction.

**Schématisation des éléments
composants le Tissu Nodal.**



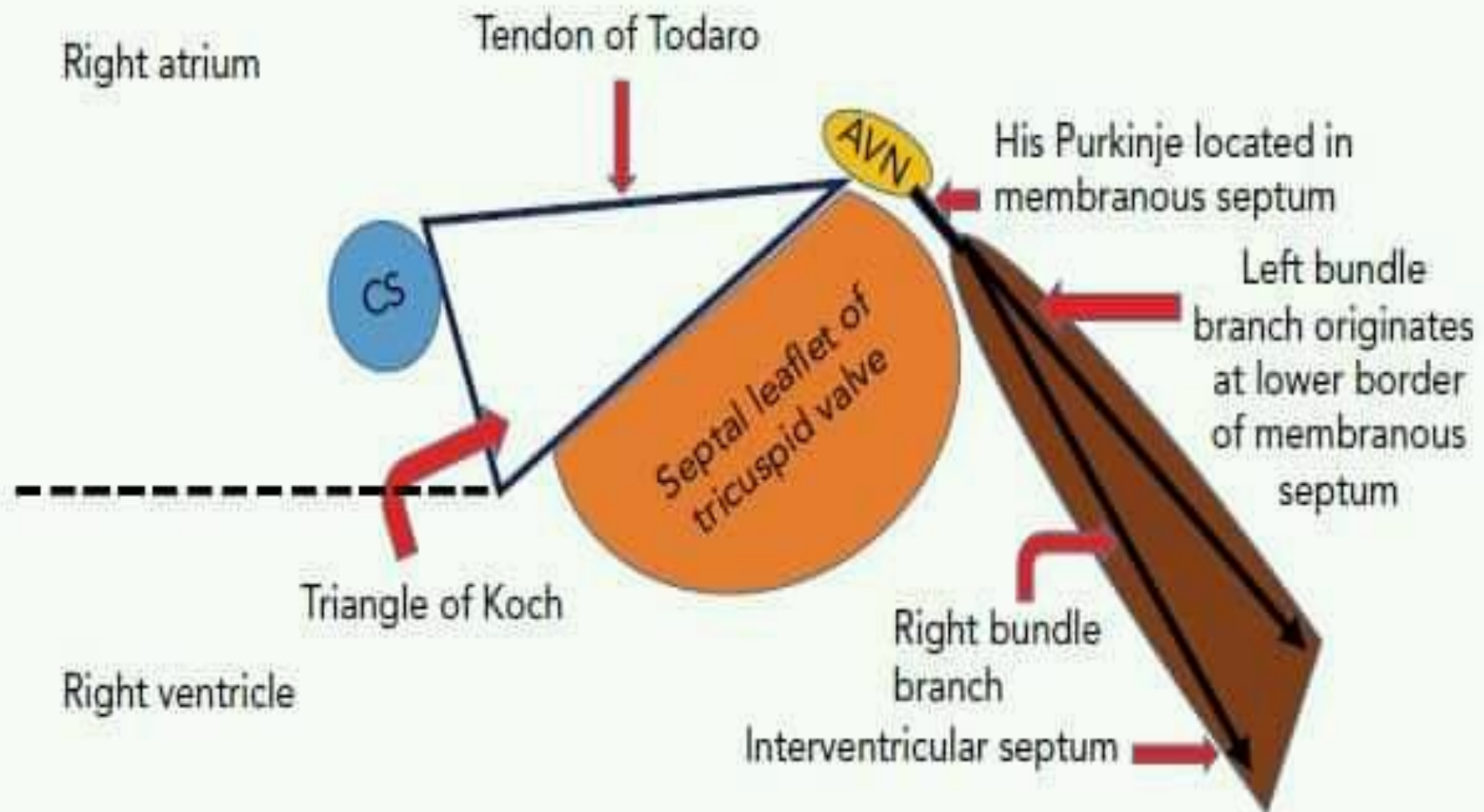
Anatomie du Tissu Nodal.

- **Noeud Sinusal : sino-atrial de Keith et Flack**, structure sous épicardique située sur la jonction entre l'auricule droit et la VCS le long de la crête terminale.
- Mesurant 10 à 20mm de large avec des spécificités individuelles.
- Dotée d'inervation automatique .
- Véritable Pacemaker physiologique.

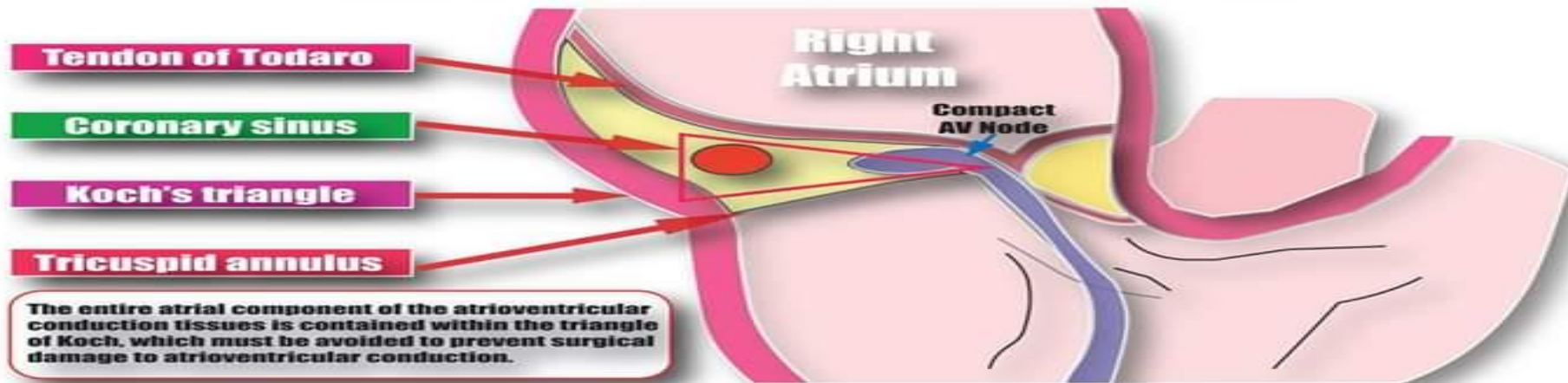
Anatomie du Tissu Nodal.

- **Noeud auriculo-ventriculaire : d'Aschof et Tawara** : structure sous endocardique située dans le triangle de **Koch**.
- Mesurant 5 a 7mm de long et 2 a 5mm de large.
- Véritable filtre rythmique.

Figure 1: Schematic Illustration of Intracardiac Anatomy and Relationship to Atrioventricular Node



Koch's Triangle



Koch's triangle is an anatomical area located in the superficial paraseptal endocardium of the right atrium, which its angles are **coronary sinus** orifice, **tendon of Todaro** and atrioventricular node. Also the elements anatomically near to it, are the insertion of the **tricuspid valve**, membranous septum, and the Eustachian ridge. This triangle ends at the site of the **coronary sinus** orifice inferiorly and, continuous with the sub-Eustachian pouch. **The tendon of Todaro** forms the hypotenuse of the triangle and the base is formed by the CSO and the vestibule of the right atrium. Variations in the size of **Koch's triangle** are common among people with different age and gender. The Atrioventricular node or the AV node is located at the apex of this triangle which depicts its anatomical importance .

Anatomie du Tissu Nodal.

- **Système His-Purkinje:** formé de :
- Faisceau de His: fins tubes de fibres minces qui cheminent tout au long du SIV qui se subdivise en deux branches.
- Branche droite du faisceau de His franchissant le VD
- Branche gauche du faisceau de His franchissant le VG subdivisée en deux hemibranches.

Anatomie du Tissu Nodal.

- Hemibranche antérieure gauche (HBAG)
- Hemibranche postérieure gauche (HBPG)
- **Réseau de Purkinje** : terminisant des deux branches infiltrant le myocarde des deux ventricules.

Barorécepteurs
aortiques

AA

VCS

VP

1

Nœud sinusal ou
nœud de Keith et
Flack

2

Nœud septal ou
auriculo-ventriculaire
ou nœud d'Ashow
Tawara

OD

OG

3

Les 2 branches du
faisceau de His

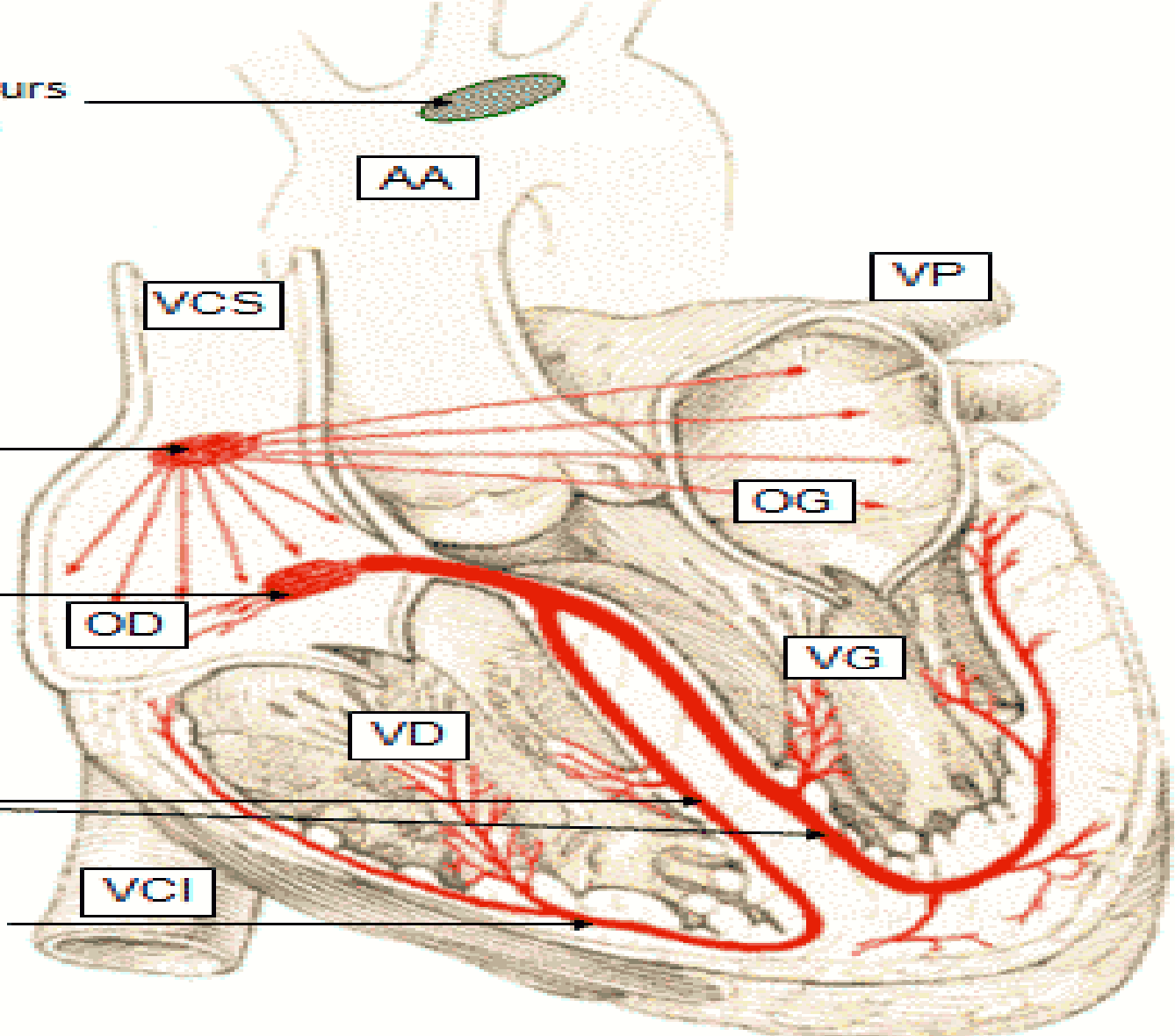
VD

VG

4

Réseau de Purkinje

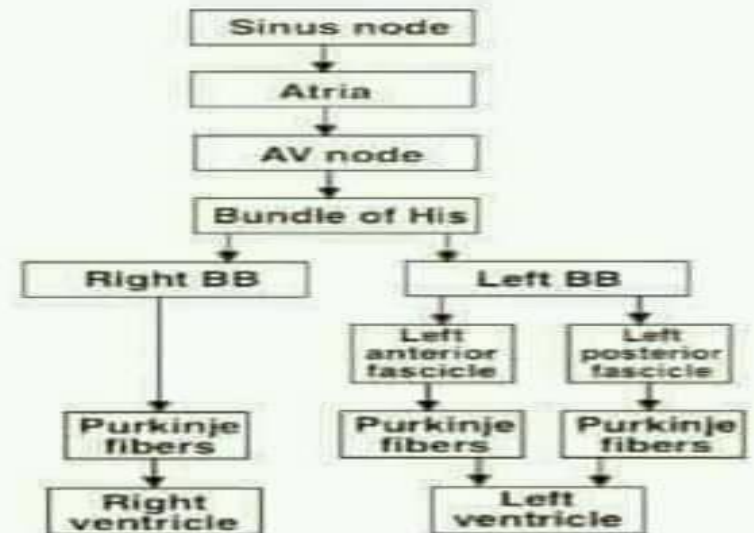
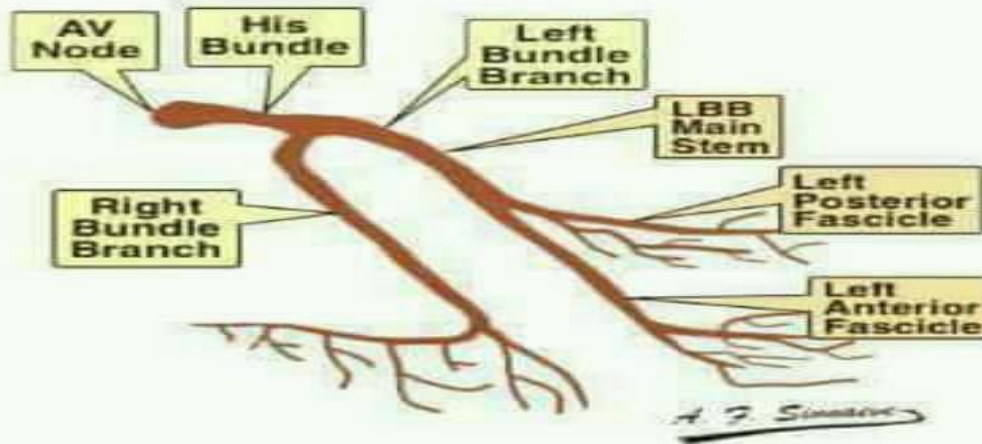
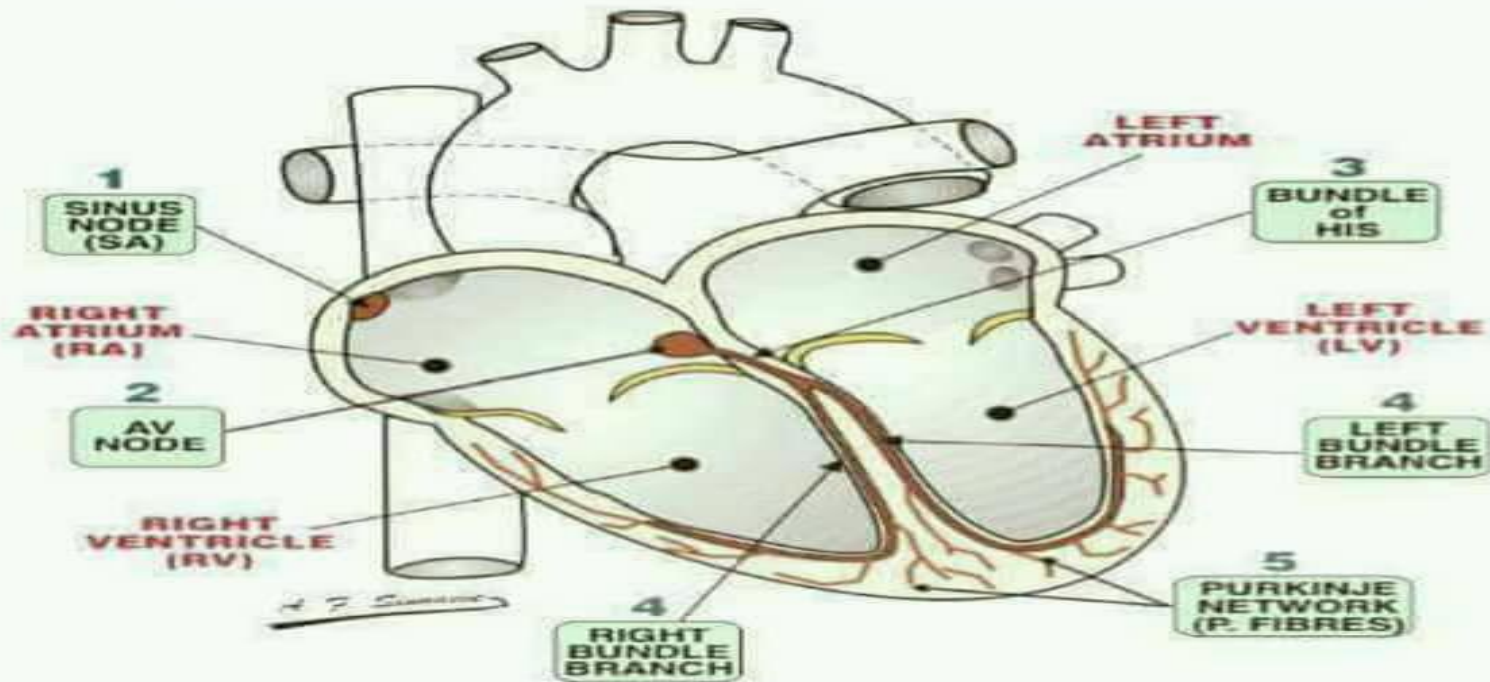
VCI



Tissu nodal et progression des courants intracardiaques

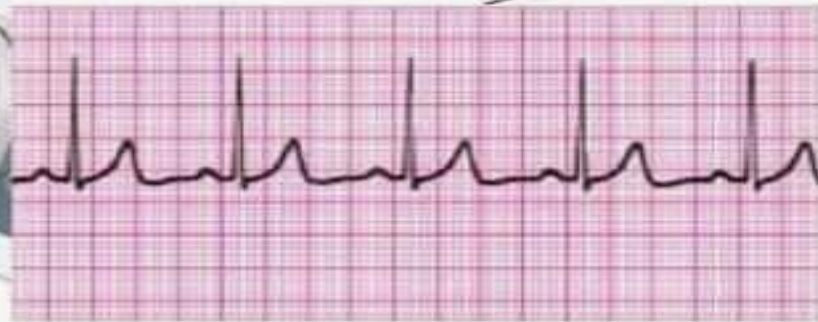
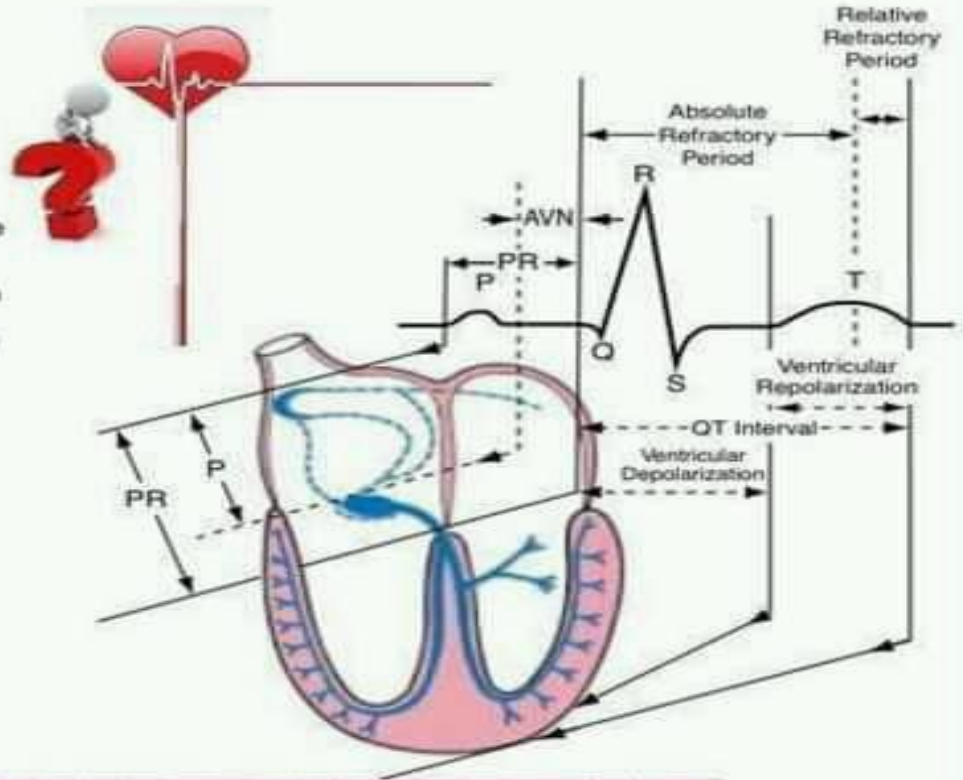
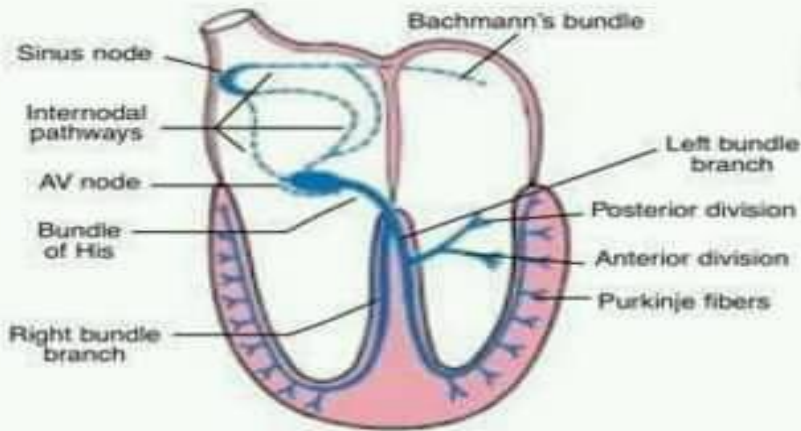
OD : oreillette droite ; VD : ventricule droit ; OG : oreillette gauche ; VG : ventricule gauche ; VCS : veine cave supérieure ; VCI : veine cave inférieure ; AA : artère aorte ; VP : veines pulmonaires (4) ; l'artère pulmonaire n'est pas représentée.

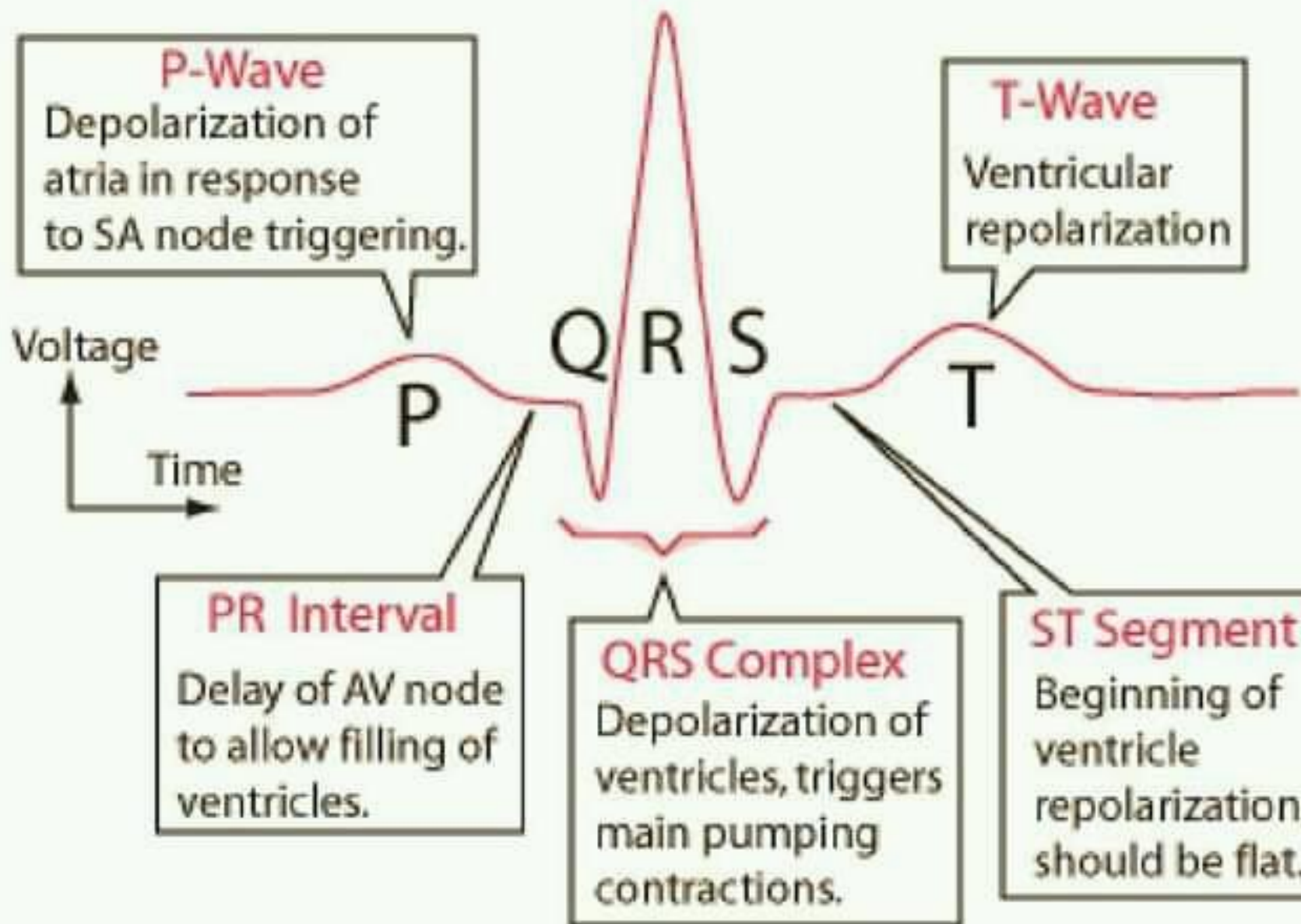
THE CONDUCTION SYSTEM OF THE HEART

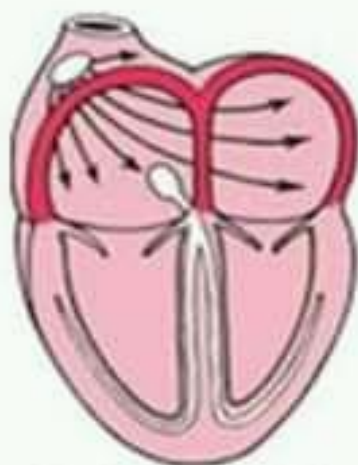
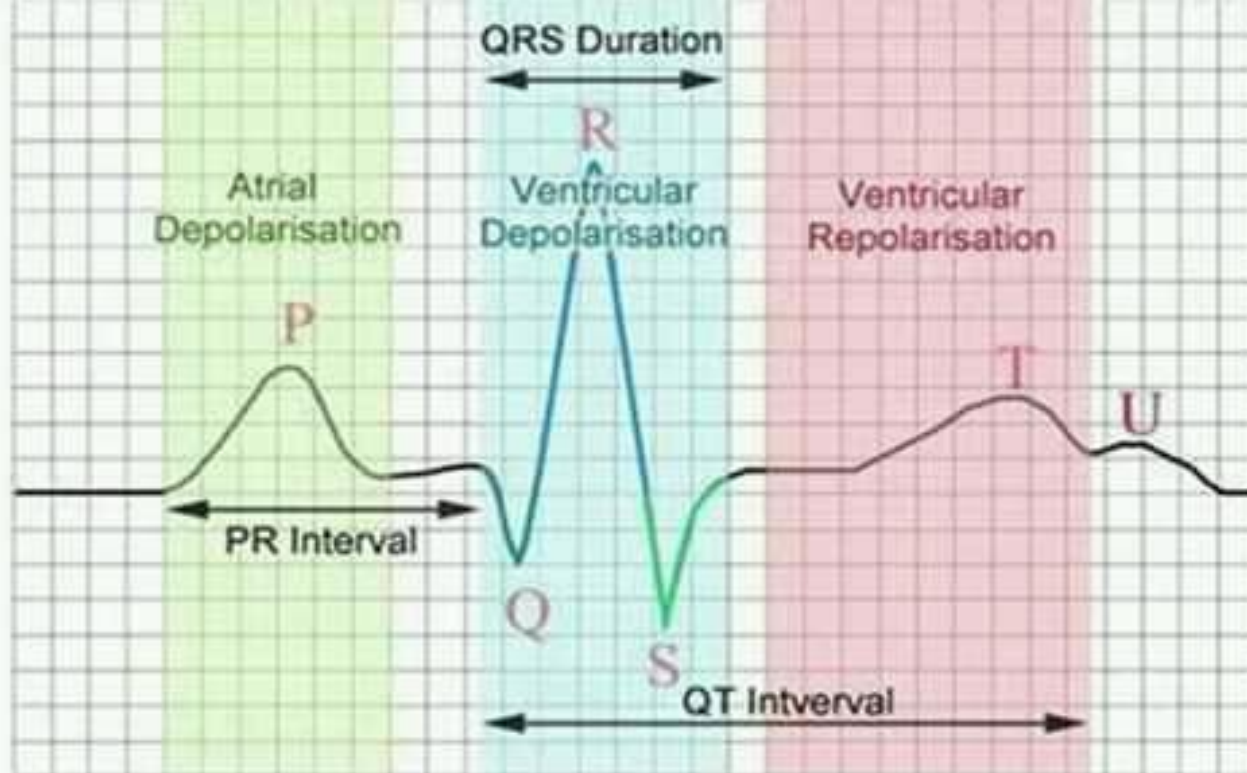


ACLS Rhythms

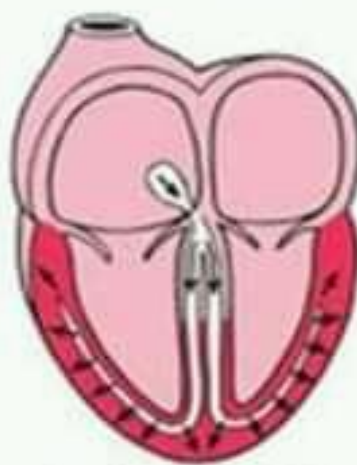
Algorithms







Activation of the
atria



Activation of the
ventricles



Recovery wave

START

P wave: atrial depolarization

P

The end

R

Q_s

P

T

P

Atria contract.

Repolarization

**ELECTRICAL
EVENTS
OF THE
CARDIAC CYCLE**

R

P

Q_s

T

P

Q wave

Q

ST segment

R

Q_s

P

Ventricles contract.

R wave

R

P

Q

R

Q_s

S wave

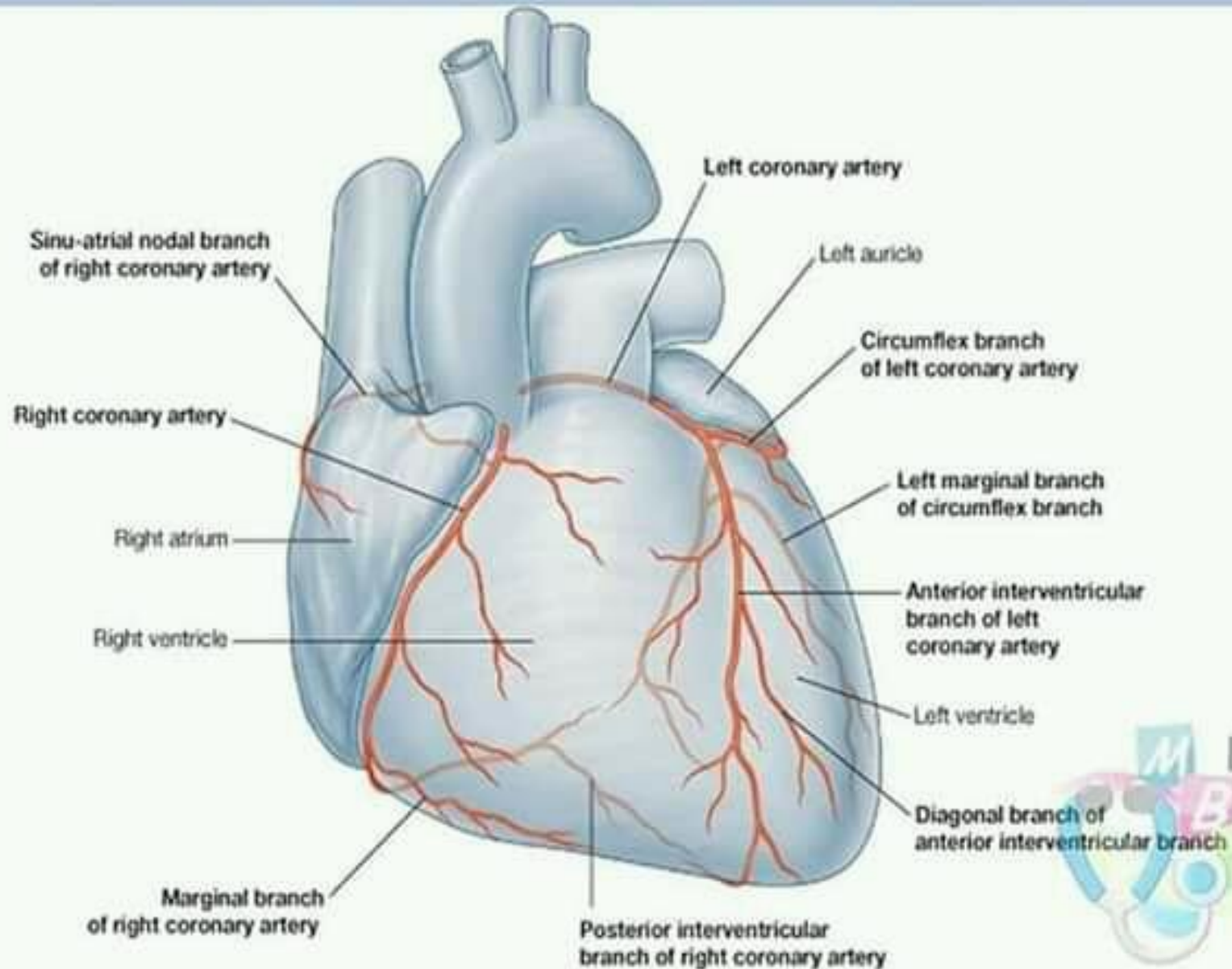
P



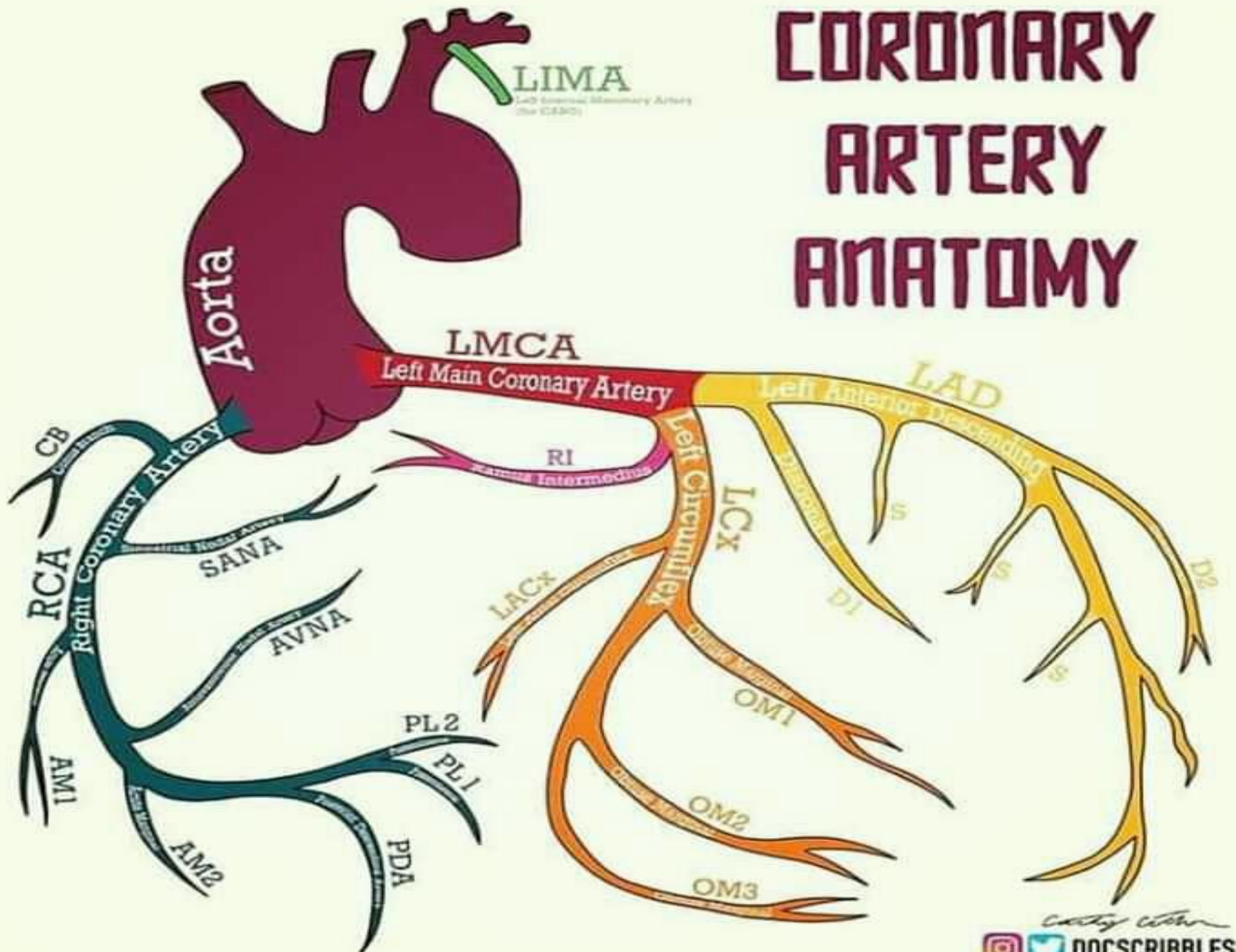
Vascularisation du Tissu Nodal.

**Coronaire responsable de l'irrigation
du Tissu Nodal.**

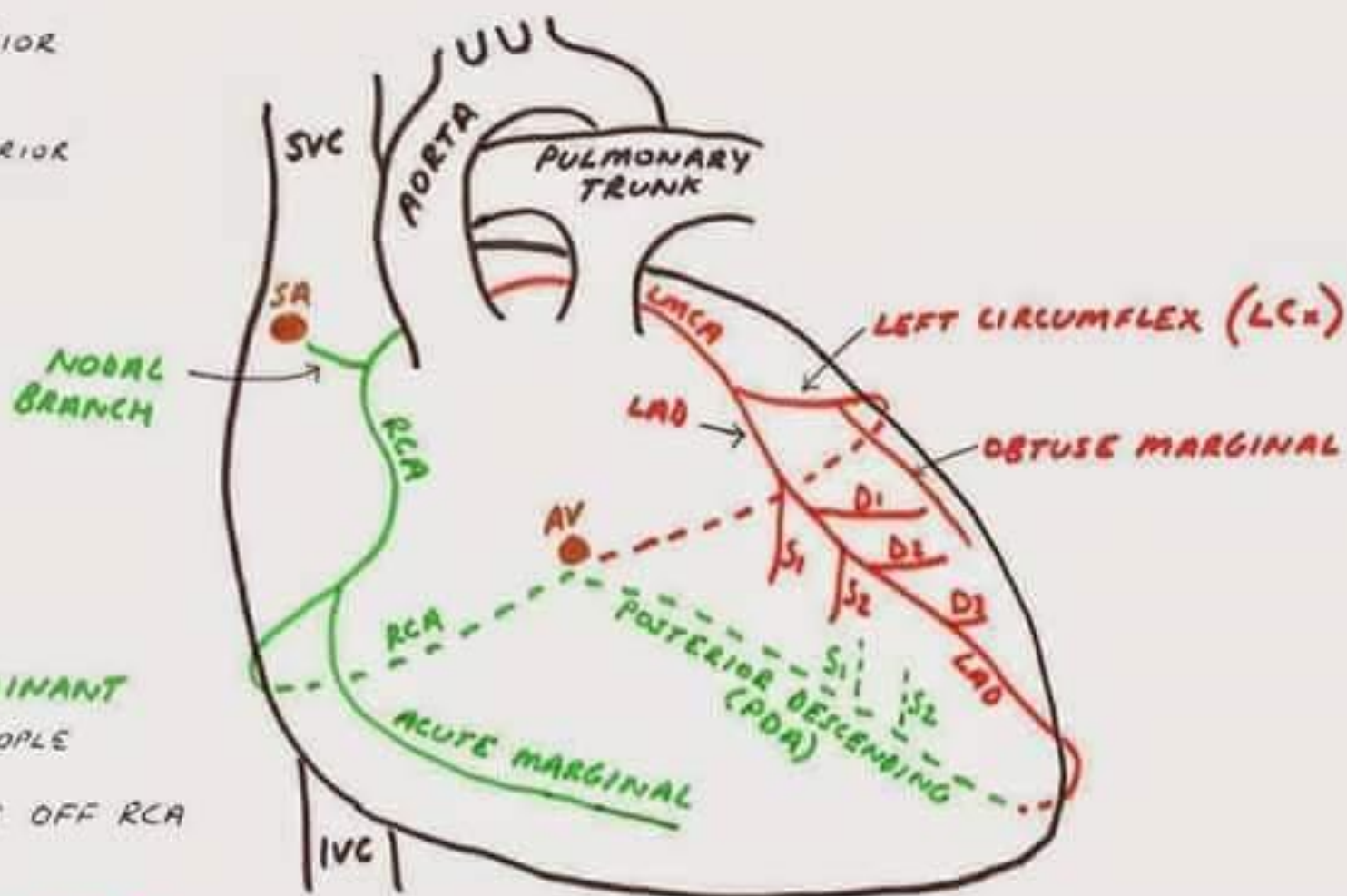
Coronary Arterial System



CORONARY ARTERY ANATOMY



 } ANTERIOR
 } POSTERIOR



RIGHT DOMINANT

- $\frac{2}{3}$ OF PEOPLE
- PDA COMES OFF RCA

LEFT DOMINANT

- PDA COMES OFF LCx

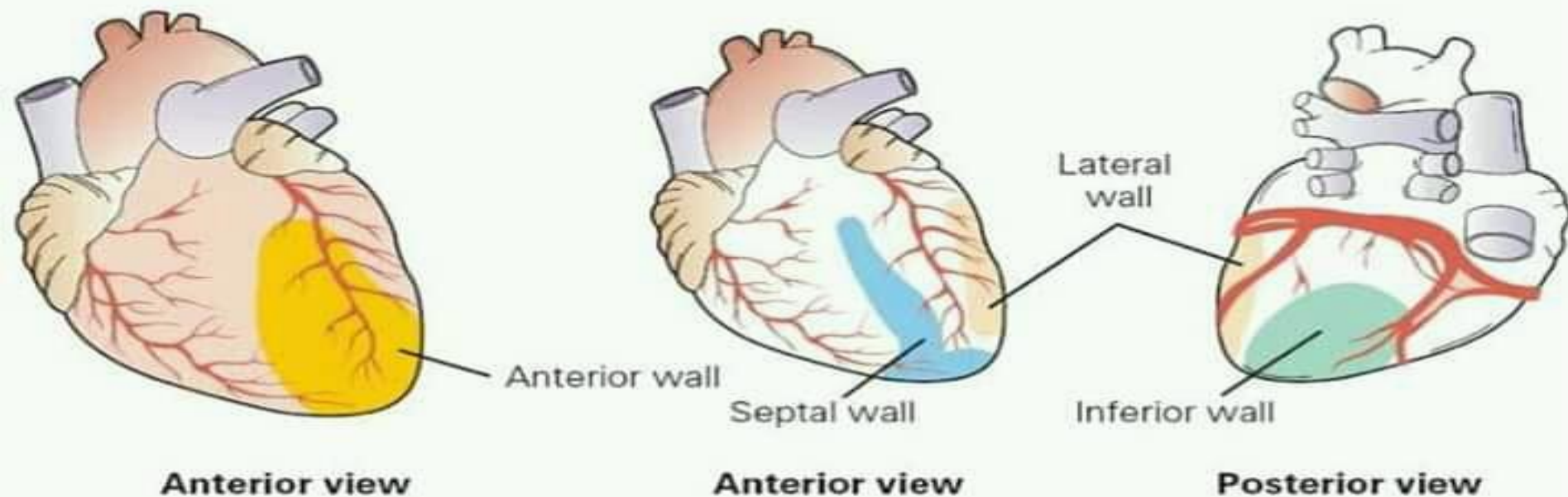
CO-DOMINANT

- PDA COMES OFF BOTH RCA + LCx

D_1, D_2, D_3 = DIAGONAL BRANCHES OF LAD

S_1, S_2 = SEPTAL BRANCHES OF LAD

S_1, S_2 = SEPTAL BRANCHES OF POSTERIOR DESCENDING ARTERY

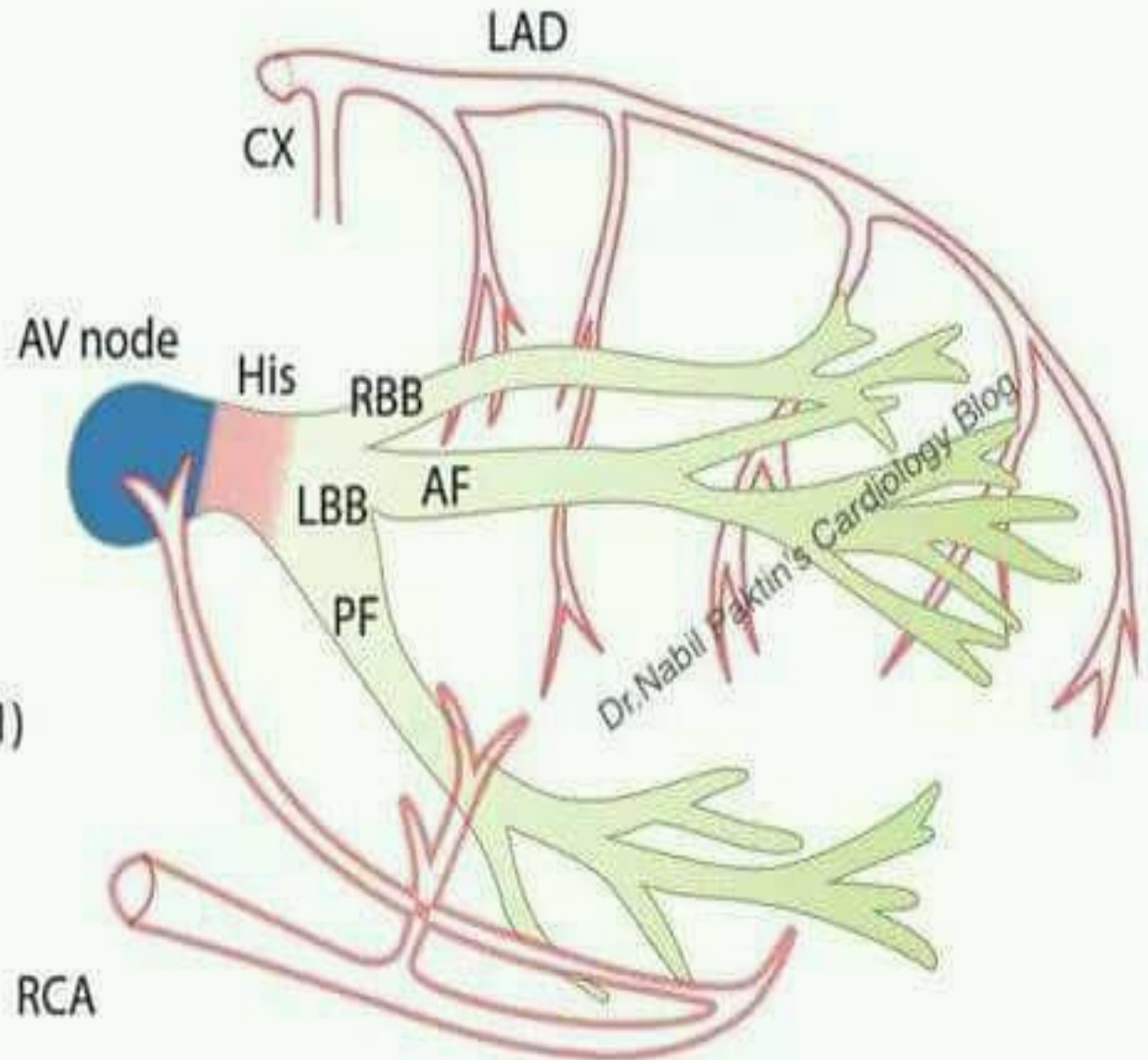


Location of MI by ECG Leads

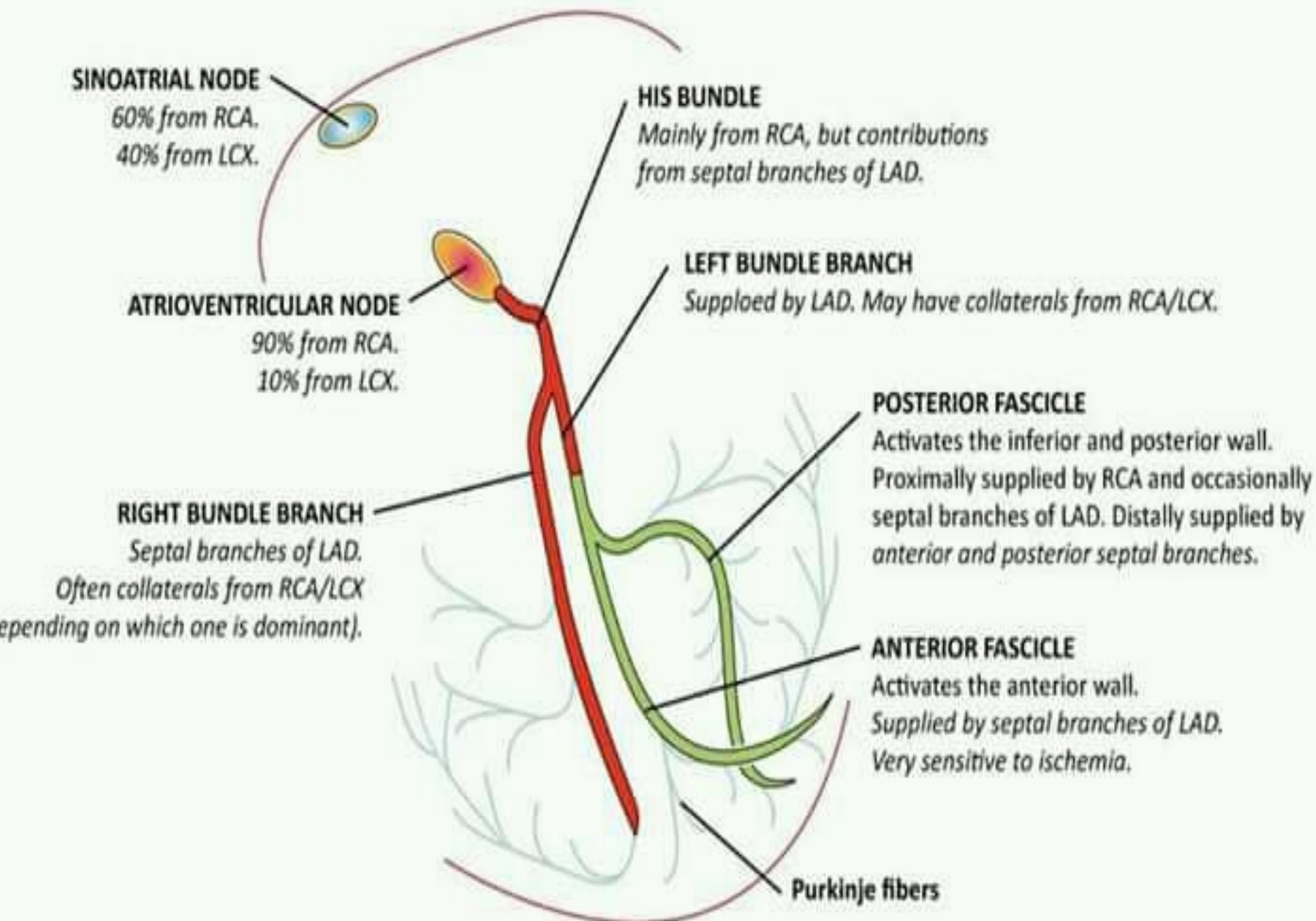
I lateral	aVR	V ₁ septal	V ₄ anterior
II inferior	aVL lateral	V ₂ septal	V ₅ lateral
III inferior	aVF inferior	V ₃ anterior	V ₆ lateral

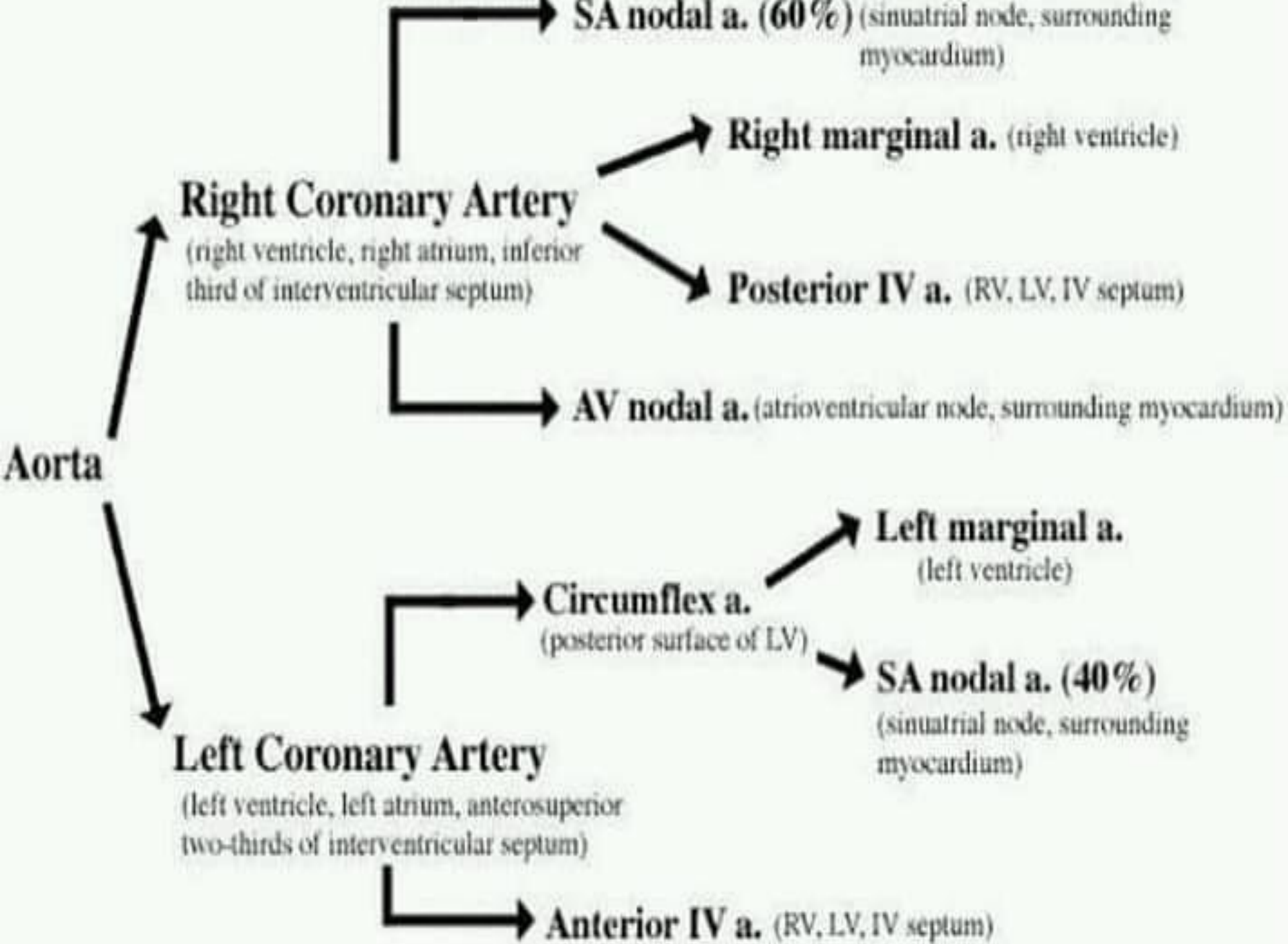
BLOOD SUPPLY CARDIAC CONDUCTION SYSTEM

- SA node: RCA in 55-60 %
AV node: RCA in 90 %
His bundle : RCA / LAD
Right bundle (RBB): LAD (S1)
Left bundle (LBB):
- Anticus (AF): LAD
 - Posticus (PF): LAD/ RCA



The components of the conduction system and their vascular supply



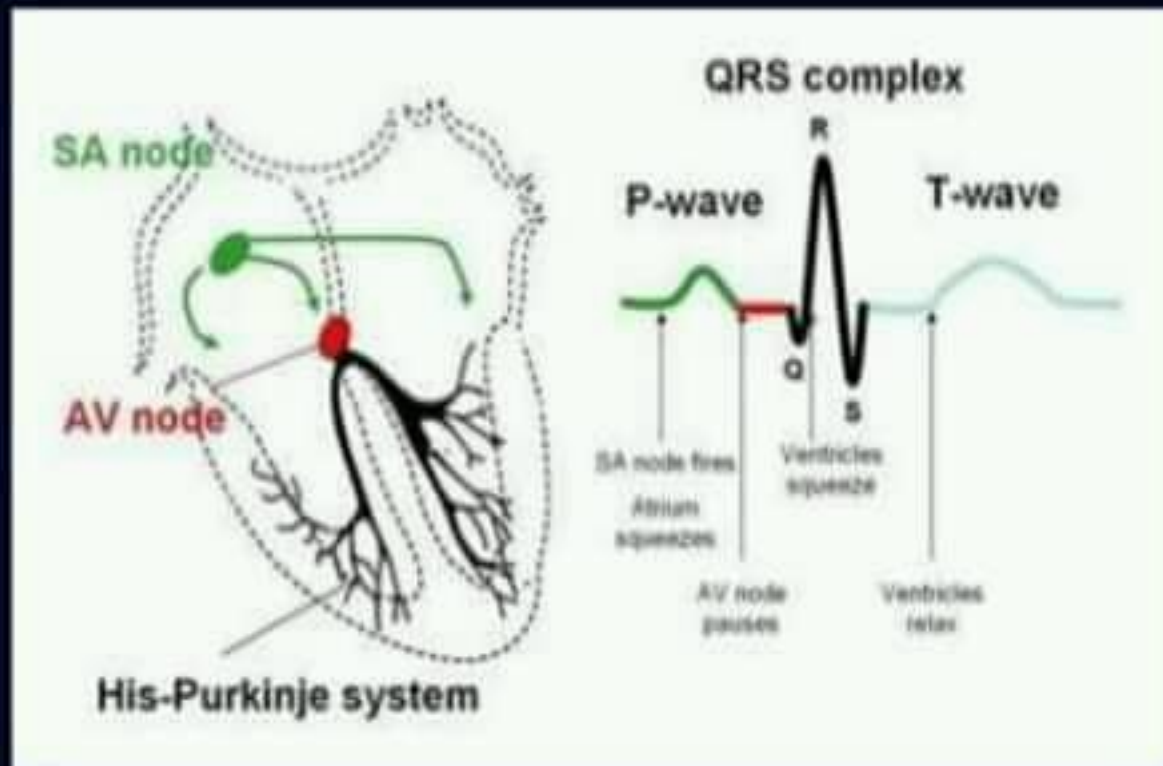




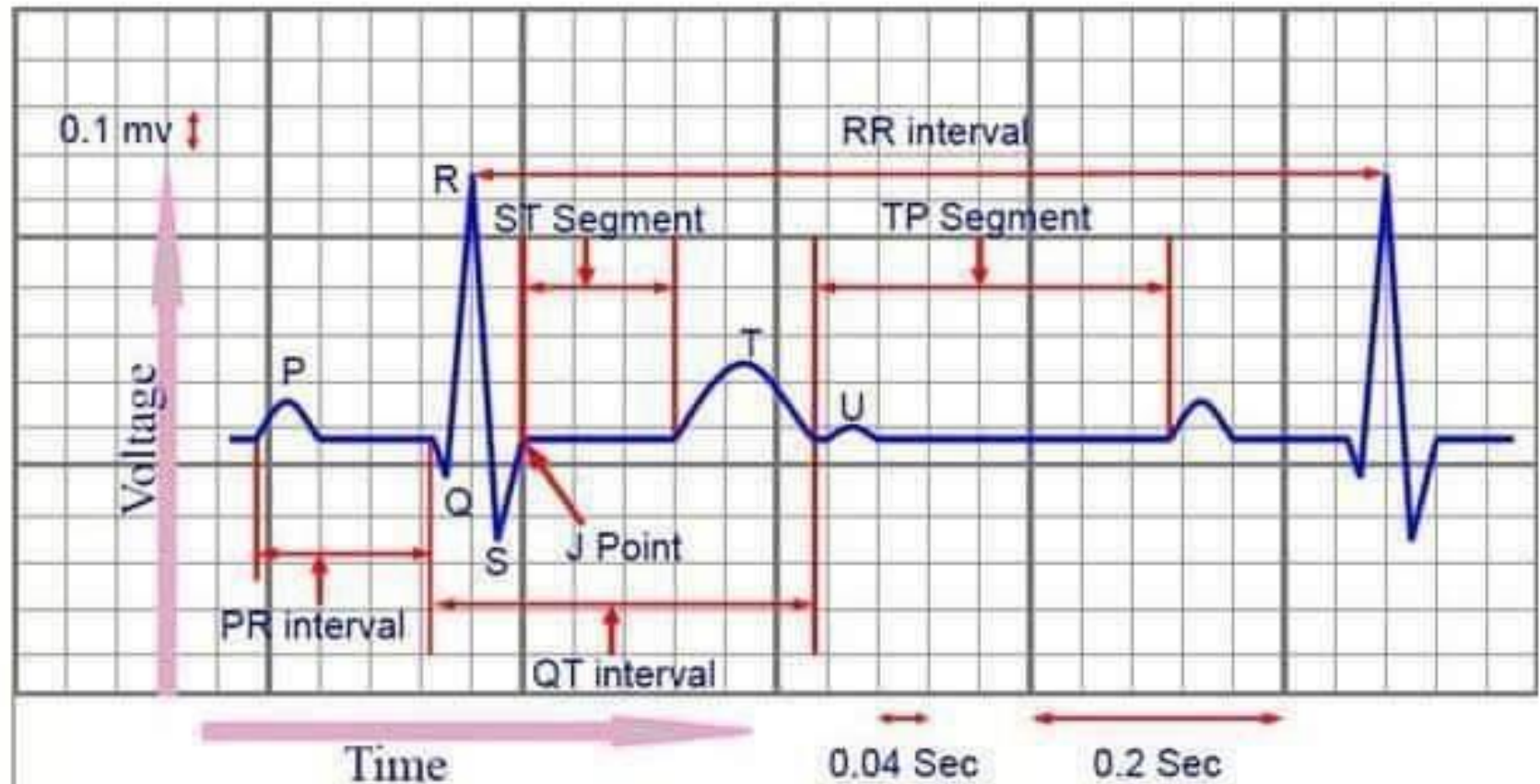
Caractéristiques électrophysiologiques du Tissue Nodal.

Vitesse de conduction de flux et phénomène d'échappement (relais).

Normal Impulse Conduction



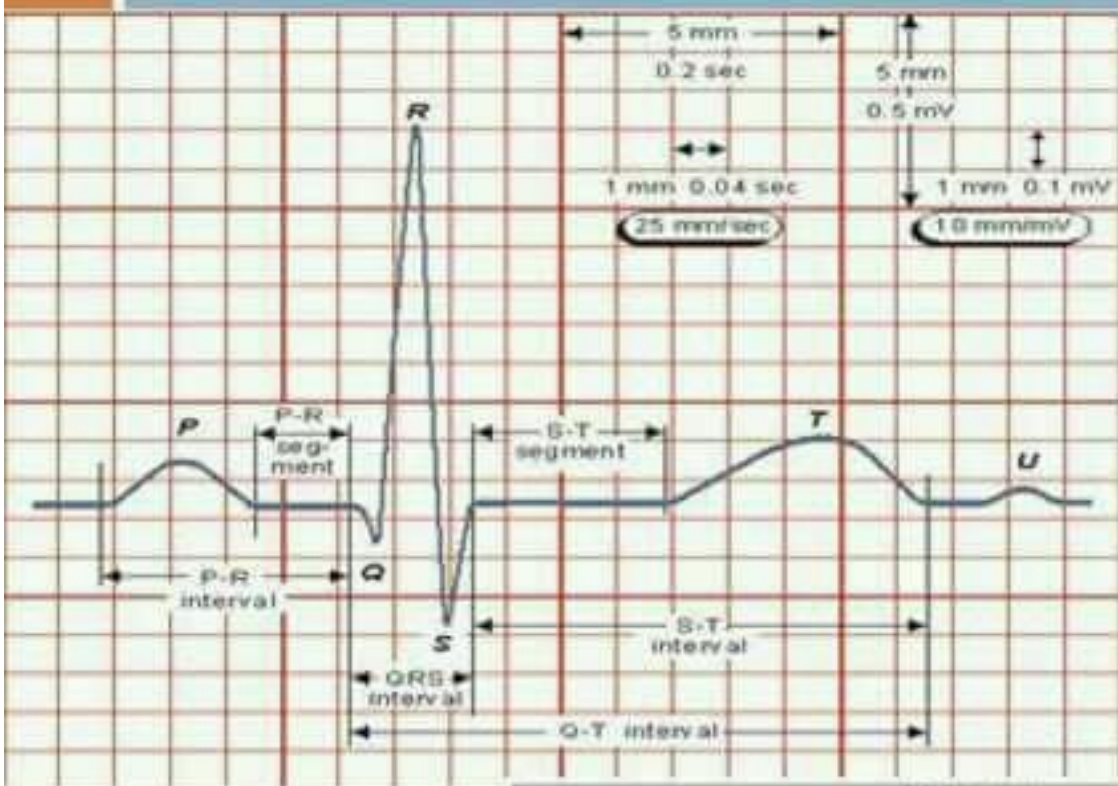
Normal ECG



- PR interval 0.12 – 0.20 sec
- QRS duration 0.08 – 0.10 sec

- QT interval 0.4 – 0.43 sec
- RR interval 0.6 – 1.0 sec

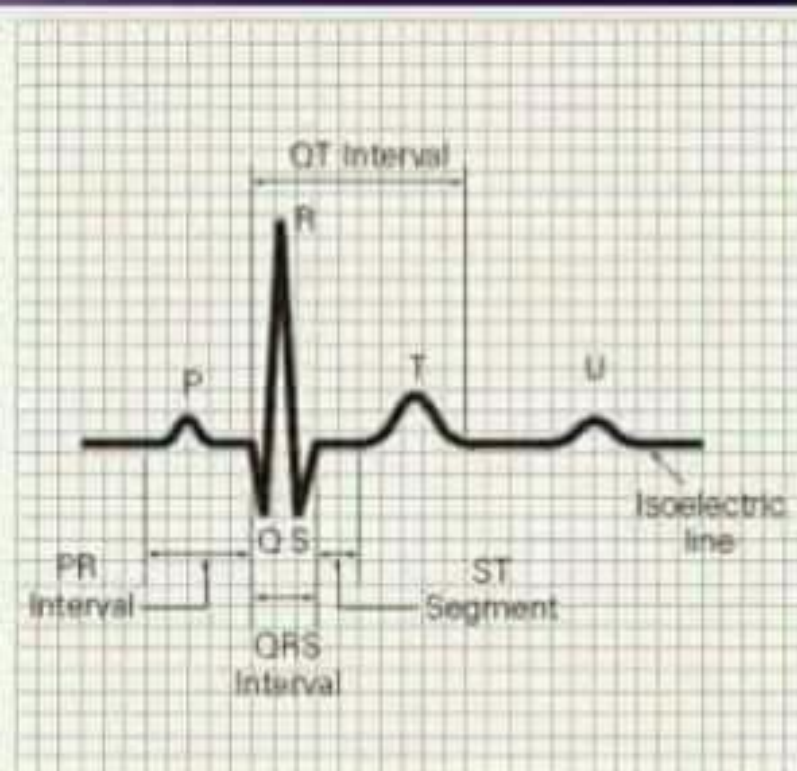
Interprétation: ECG normal



	Durée (sec)	Amplitude (mm)	Axe
P (dépolarisation des oreillettes)	< 0.12 (> 0.12 HAG ou bloc inter auriculaire)	< 2 mm (> 2.5 mm HAG) + précordiales sauf pfs V1	20° à 80° P rétrograde - D2 D3 AVF
PR	0.12 à 0.20		
Q	< 0.04 (si > 0.03 nécrose T)		
QRS (dépolarisation des ventricules, de l'endocarde vers l'épicarde)	0.08 à 0.10 0.10-0.12 sec bloc incomplet > 0.12 sec complet	Q1 + Q2 + Q3 > 15 (sinon microvoltage)	-30° à +110° < -30° axe gauche > 110° axe droit
QT (phase II du potentiel d'action)	0.4 x RR ± 0.04		

Leads (R I - R III) + (S II - S I) : HV D = -14 mm < normal < 17 mm < HV G

Sokolow-Lyon (SV1 + RV5), Blondeau - Hefer (SV2 + RV7) : normale < 36 mm, si > 35 mm : HV G

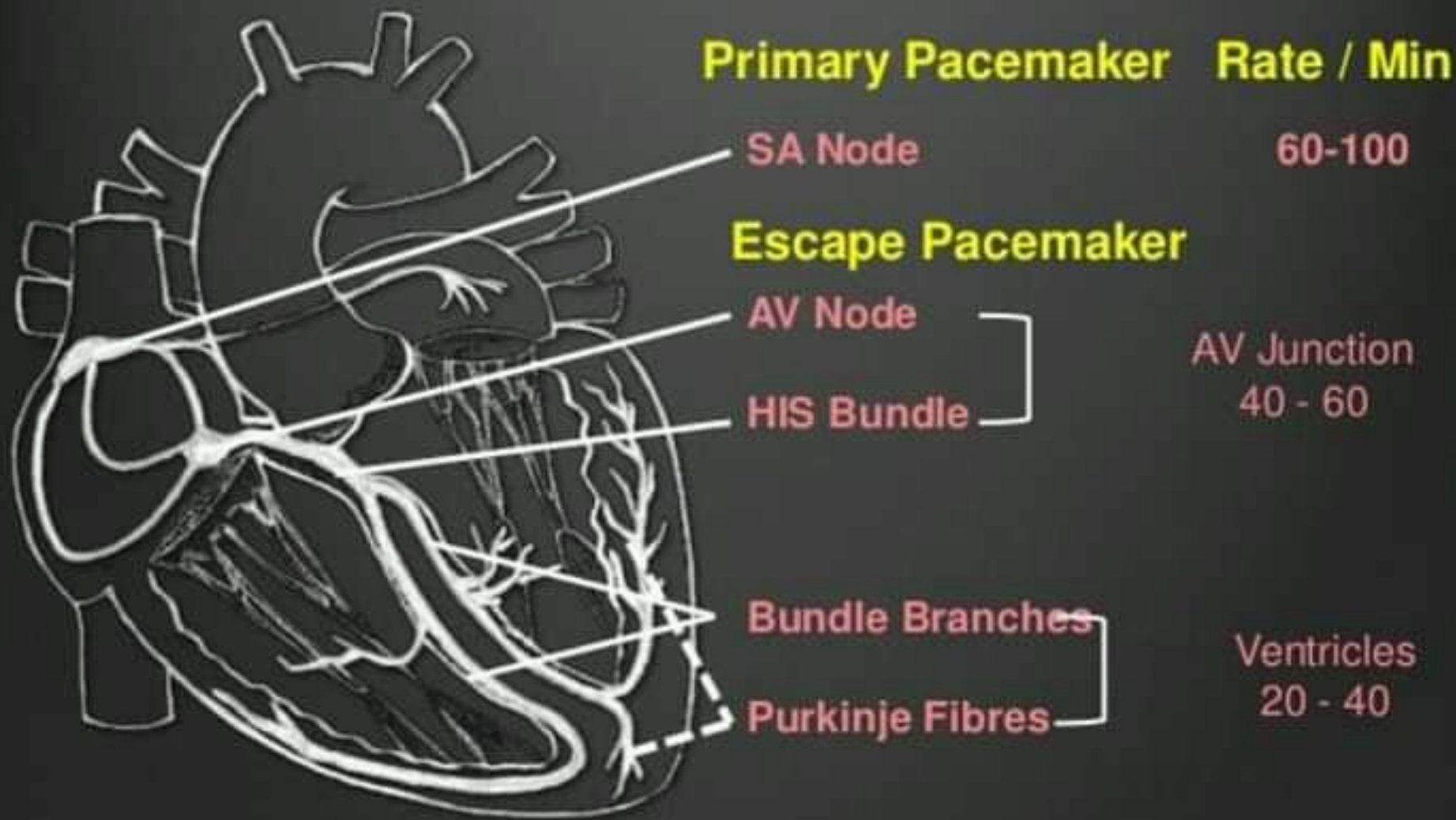


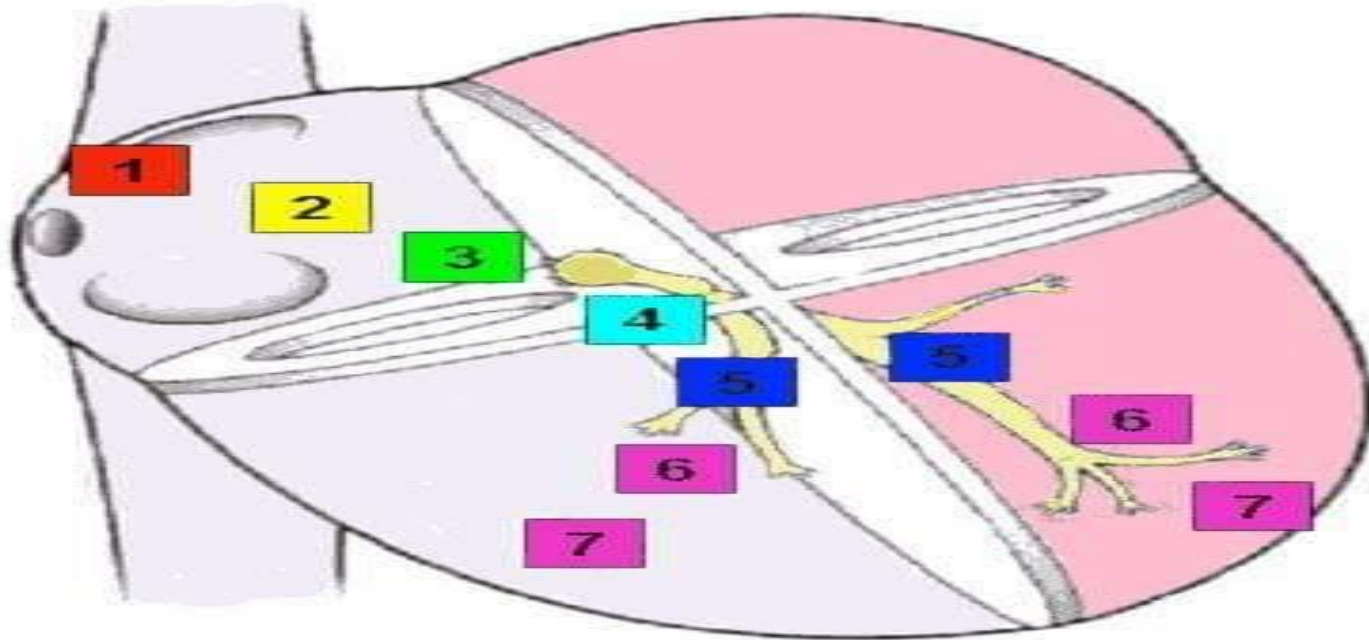
Electrical Components	
Deflection	Description
P Wave	First wave seen Small rounded, upright (positive) wave indicating atrial depolarization (and contraction)
PR Interval	Distance between beginning of P wave and beginning of QRS complex Measures time during which a depolarization wave travels from the atria to the ventricles
QRS Interval	Three deflections following P wave Indicates ventricular depolarization (and contraction) Q Wave: First negative deflection R Wave: First positive deflection S Wave: First negative deflection after R wave
ST Segment	Distance between S wave and beginning of T wave Measures time between ventricular depolarization and beginning of repolarization
T Wave	Rounded upright (positive) wave following QRS Represents ventricular repolarization
QT Interval	Measured from beginning of QRS to end of T wave. Represents total ventricular activity.
U Wave	Small rounded, upright wave following T wave Most easily seen with a slow HR. Represents repolarization of Purkinje fibers.

Pacemakers of the Heart

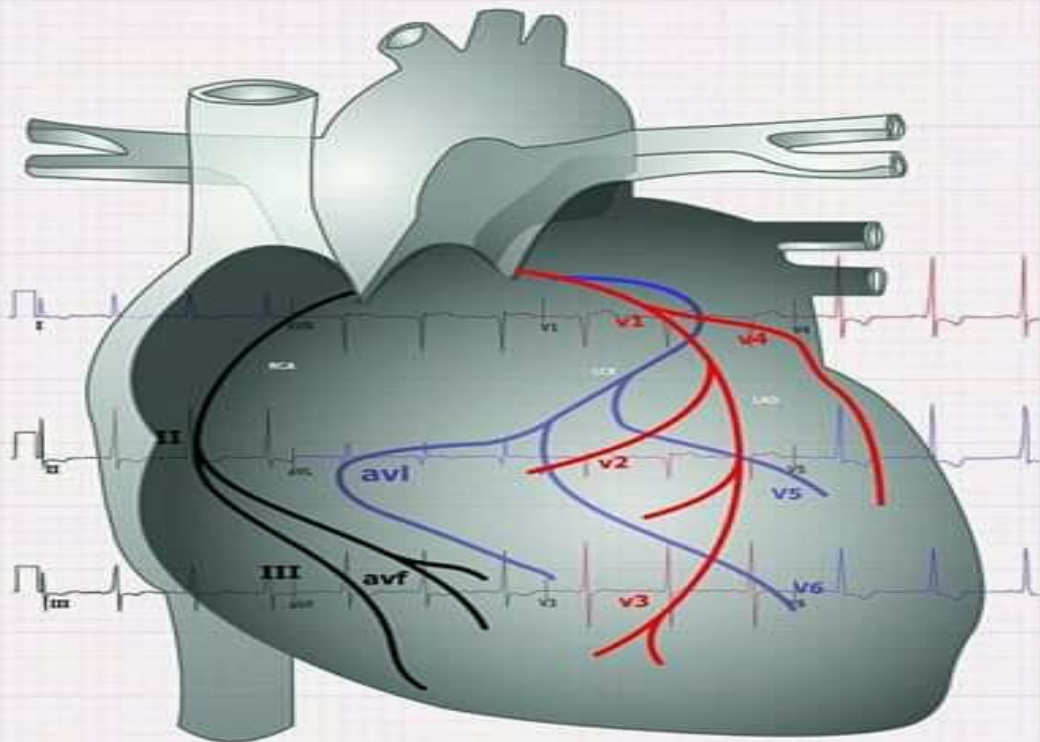
- **SA Node** - Dominant pacemaker with an intrinsic rate of **60 - 100** beats/minute.
- **AV Node** - Back-up pacemaker with an intrinsic rate of **40 - 60** beats/minute.
- **Ventricular cells** - Back-up pacemaker with an intrinsic rate of **20 - 45** bpm.

Conduction System





Normal Activation Sequence	Structure	Conduction velocity (m/sec)	Pacemaker rate (beats/min)
1	SA node	< 0.01	60 – 100
2	Atrial myocardium	1.0 – 1.2	None
3	AV node	0.02 – 0.05	40 – 55
4	Bundle of His	1.2 – 2.0	25 – 40
5	Bundle branches	2.0 – 4.0	25 – 40
6	Purkinje network	2.0 – 4.0	25 – 40
7	Ventricular myocardium	0.3 – 1.0	None



Coronary Circulation to the Conduction System

RCA	- SA node - AV node - Bundle of His - Posterior fascicle of Left bundle branch
LAD	- AV node - Bundle of His - Right bundle branch - Anterior fascicle of left bundle
LCx	Posterior fascicle of Left bundle branch

Coronary Circulation and 12 Lead EKG

Leads	Anatomical area	Coronary artery
I, aVL, VS, V6	Lateral wall	Circumflex artery
II, III, aVF	Inferior wall	Right coronary artery
V1, V2	Septum	Left anterior descending artery
V3, V4	Anterior wall	Left anterior descending artery

Classification of Antiarrhythmic Drugs

Class I	Sodium Channel blockers	
	Class I A	- Procainamide - Quinidine - Disopyramide
	Class IB	- Lidocaine - Mexiletine
	Class IC	- Flecainide - Propafenone
Class II	Beta blockers	- Carvedilol - Metoprolol
Class III	Potassium channel blockers	- Amiodarone - Sotalol - Dronedarone - Ibutilide - Dofetilide
Class IV	Calcium channel blockers	- Verapamil - Diltiazem
Class V		- Digoxin - Adenosine - Magnesium sulphate



Étiologies des Blocs auriculo-ventriculaire.

**Connaître et savoir faire la différence
entre causes réversible et irréversible.**

Quelles situations cliniques.

■ Les étiologies des B.A.V :

- ☐ Dégénératif: syndrome de Lev et Lenégre (50% des indications stimulations cardiaques suite a l'installation progressive de la fibrose sur TC chez les sujets âgés +/- l'hypothèse héréditaire mutation du gène SCN5A).
- ☐ La maladie de Lyme : ou borréliose de Lyme, est transmise lors d'une piqûre de tique infectée par une bactérie de la famille des spirochètes.
- ☐ Cardiomyopathie ischémique (IDM ant ou inf).
- ☐ Congénitaux: CIA,PCA,TGV, maladie d'Ebstein.
- ☐ Valvulopathie (RAC)
- ☐ Neuro (dysautonomie, HTIC...)
- ☐ Médicamenteux : (BB,AA,CCB,digitalique).

Situations cliniques.

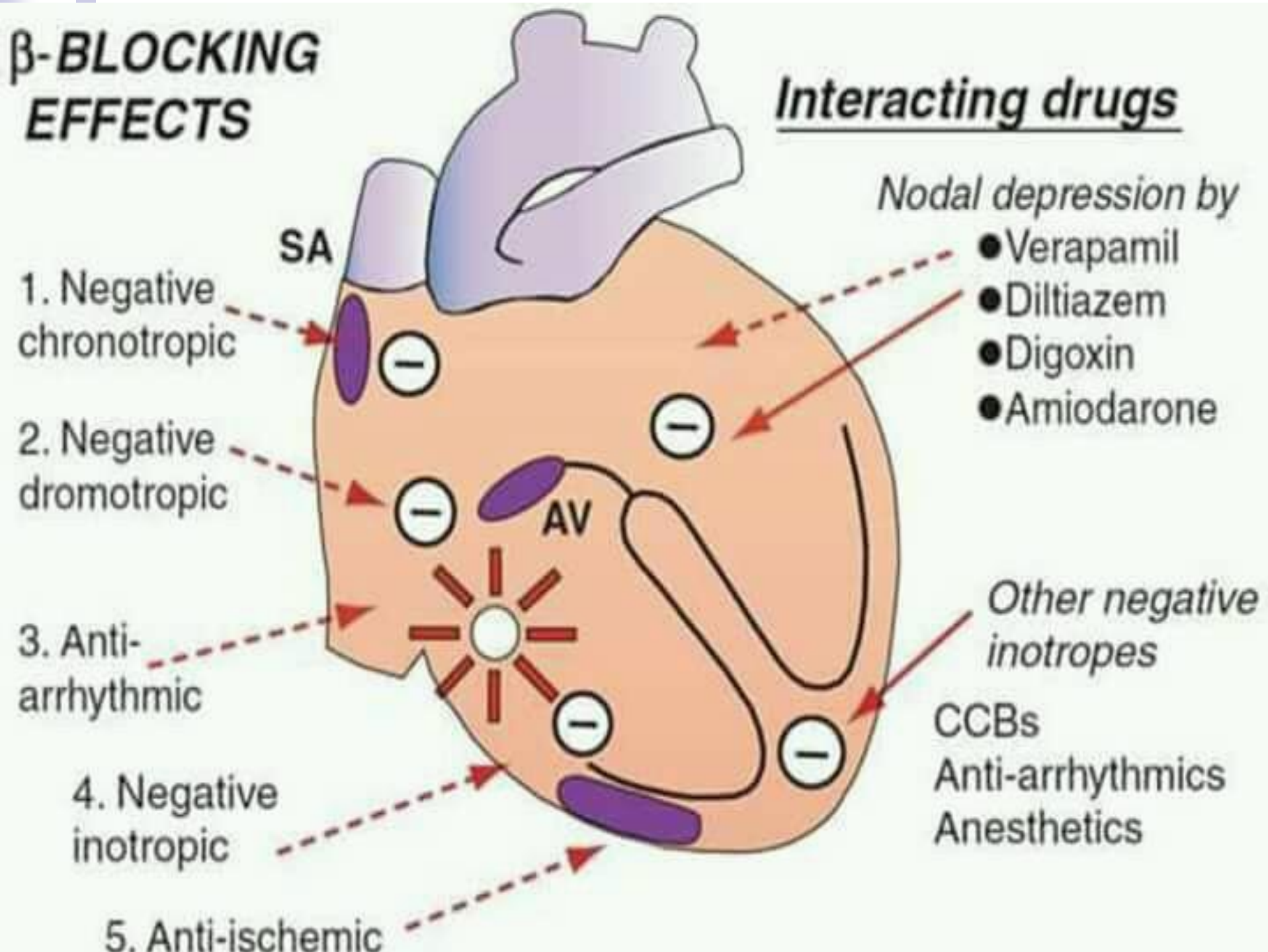
■ Les étiologies des B.A.V :

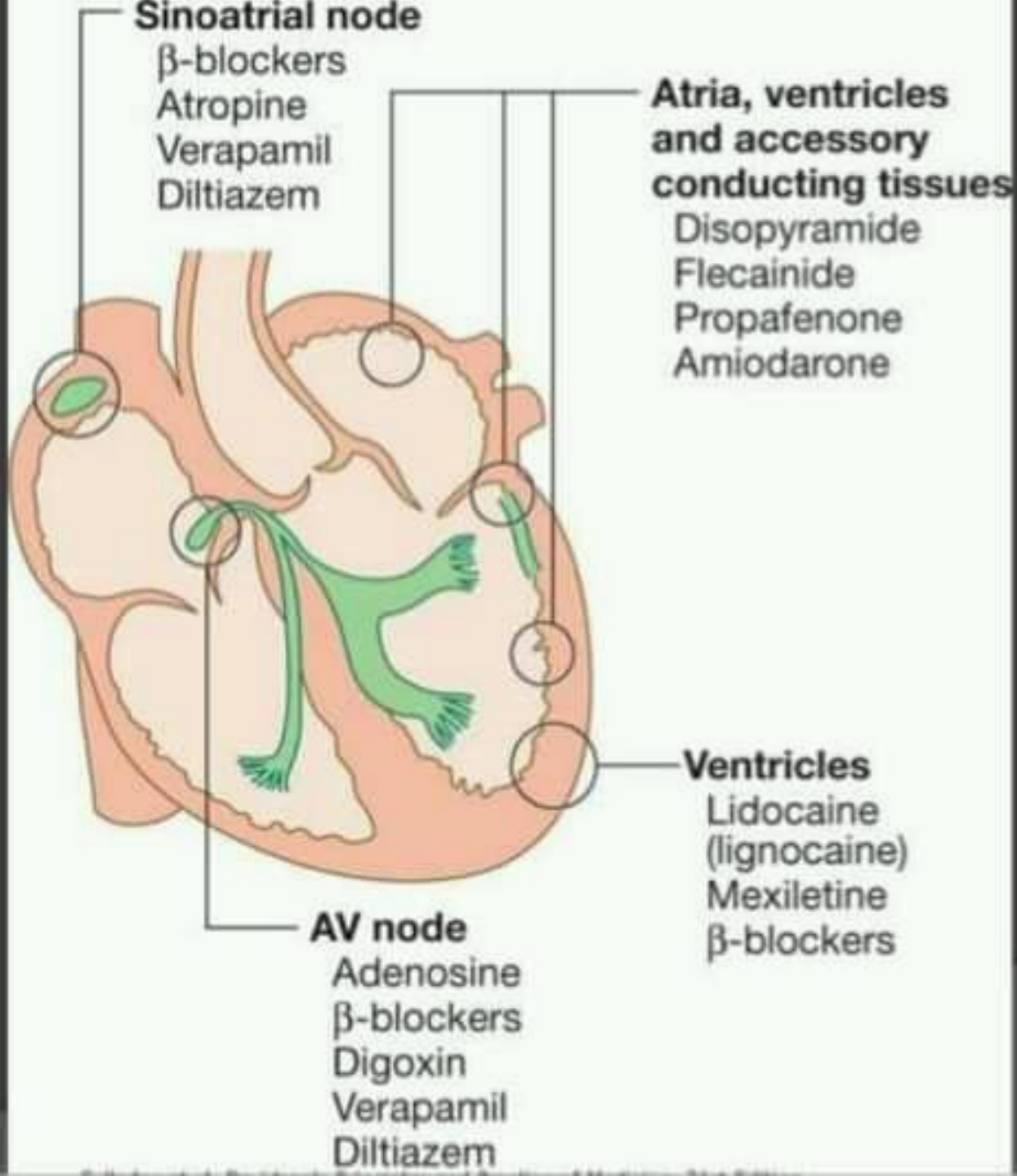
- Cardiomyopathie infiltrative : Amylose, Hémochromatose.
- Vaso-vagal.
- Troubles électroioniques(Hk+)et Métaboliques
- Cardiomyopathie valvulaire : RAC,EI.
- Myocardite: Virale, diphtérie, maladie de Chagas
- Cardiomyopathie hypertrophique (CMH).
- Post opératoire : chirurgie cardiaque d'une CIA,CIV,RV.
- Maladies de systèmes :Sclérodermie, Sarcoïdose ,SLE, arthritisme rhumatoïde.

Causes of atrioventricular block

- Acute myocardial infarction or ischemia
- Drugs (β -blockers, digitalis, Ca^{2+} -blockers)
- Conduction system fibrosis (Lev and Lenegre syndromes)
- Increased vagal tone
- Trauma or following cardiac surgery
- Hypothyroidism (rarely thyrotoxicosis)
- Hypothermia
- Hyperkalemia and other electrolyte abnormalities
- Hypoxia
- Infiltrative diseases (amyloidosis)
- Valvular disease (aortic stenosis, incompetence, endocarditis)
- Myocarditis (diphtheria, rheumatic fever, viral, Chagas' disease, tick-borne)
- Associated with neuromuscular disease, i.e., myotonic dystrophy.
- Collagen vascular disease (SLE, rheumatoid arthritis, scleroderma)
- Cardiomyopathies (hemochromatosis, amyloidosis)
- Granulomatous disease (sarcoid)
- Congenital heart block
- Congenital heart disease (ASD, Ebstein's, PDA, L-transposition)

β -BLOCKING EFFECTS







Manifestations cliniques :

**Symptômes et signes
fonctionnels**

Situations cliniques.

- Grand nombre de B.A.V mais peu de symptomatiques :
 - Les B.A.V de haut grade :
 - Bloc auriculoventriculaire type 2 mobitz II et type 3
 - Les autres « parlent » beaucoup plus rarement

Manifestations cliniques

- Maître symptôme : Lipothymie et syncope (*perte de connaissance brutale sans prodrome, sans phase post critique, sans convulsion ni morsure de langue ni perte d'urines*). **Syncope d'Adams stocks.**
- Les appelants décrivent :
 - ☐ Perte de connaissance brutale à l'emporte-pièce.
 - ☐ Malaise, confusion (baisse du débit sanguin cérébral).
 - ☐ Dyspnée (OAP sur bradycardie sévère).
 - ☐ Palpitations, bradycardie.
 - ☐ Angor rare.
- Terrain : très variable en fonction de l'étiologie

A-V conduction defects

degree

symptoms

location

1st degree

difficulty in AV conduction

asymptomatic

2nd degree

some P waves fail to be conducted to the ventricles

- Mobitz I (Wenckebach)
- Mobitz II
- 2:1 2nd degree AV block
- advanced 2nd degree AV block

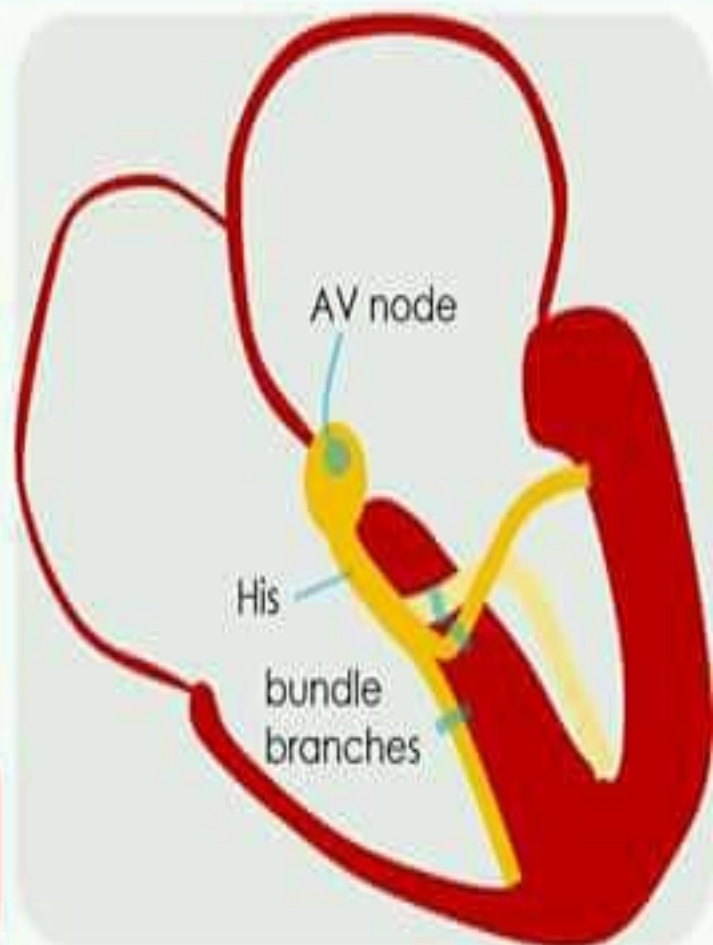
fatigue & dyspnea

dizzy spells

3rd degree

none of the atrial impulses are conducted to the ventricles

syncope or Stokes-Adams
sudden death



Anomalies de la conduction

BAV

```
graph TD; BAV[BAV] --> BAV1[BAV de 1° degré]; BAV --> BAV2[BAV de 2° degré]; BAV --> BAV3[BAV de 3° degré]; BAV2 --> Mobitz1[Type Mobitz 1]; BAV2 --> Mobitz2[Type Mobitz 2];
```

A hierarchical flowchart on a blue background. At the top is a light green box labeled 'BAV'. A horizontal line below it branches into three boxes: 'BAV de 1° degré' (yellow), 'BAV de 2° degré' (orange), and 'BAV de 3° degré' (blue). Below the orange box, another horizontal line branches into two red boxes: 'Type Mobitz 1' and 'Type Mobitz 2'.

BAV de 1° degré

BAV de 2° degré

BAV de 3° degré

Type Mobitz 1

Type Mobitz 2

degrees of A-V conduction defects

First-degree A-V conduction defect

not a real block but a conduction delay, all the P-waves are conducted to the ventricles but with a prolonged PR interval (>0.20 seconds)

Second-degree A-V block

characterized by an occasional failure in A-V conduction so that at least one P-wave is not conducted to the ventricles

Third-degree A-V block (complete A-V block)

situation in which none of the P waves is conducted to the ventricles; there is complete A-V dissociation

Les troubles de la conduction siège du bloc.

■ Noeud auriculo-ventriculaire et faisceau de HIS :

- BAV 1 : Allongement constant de PR(>200ms), Une onde p devant chaque QRS. **Nodal**



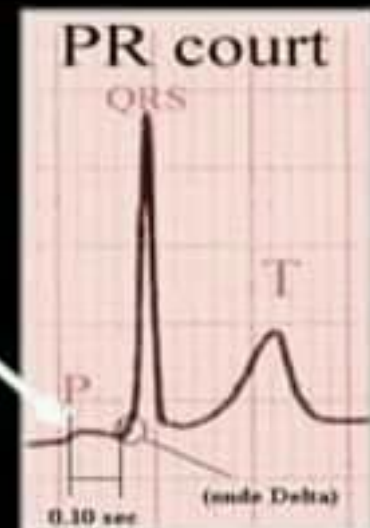
- BAV 2 :
 - Mobitz I : périodes de Luciani-Wenckebach. **Nodal**
 - Mobitz II : blocage complet intermittent. **Infra-nodal**
- BAV 3 : Blocage complet permanent. **Infra-nodal**. Niveau du bloc à confirmer par EEP.

BAV 1degré

- Ralentissement de la conduction auriculo-ventriculaire.
- Allongement de PR sup. à 0.20

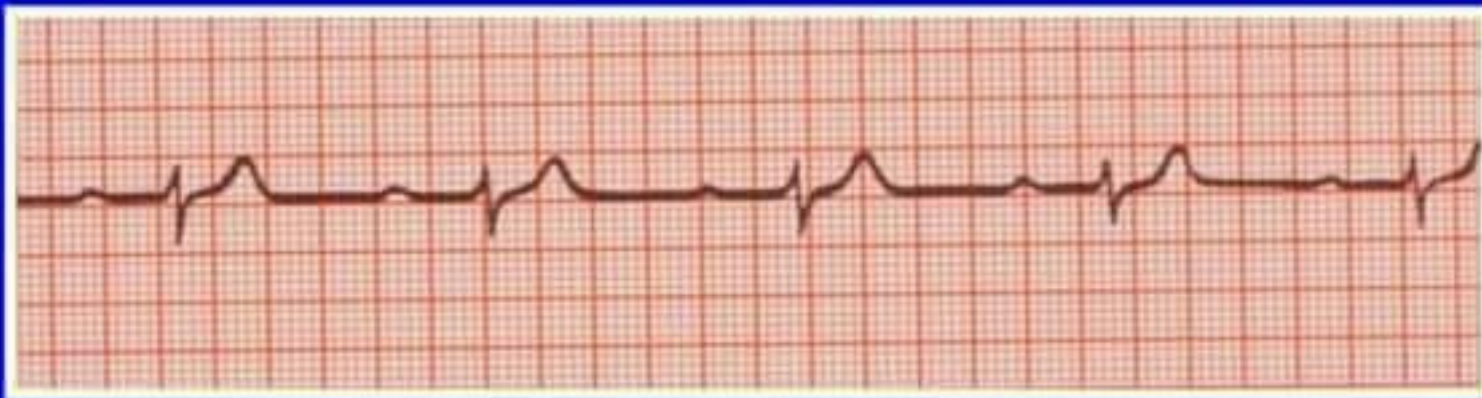
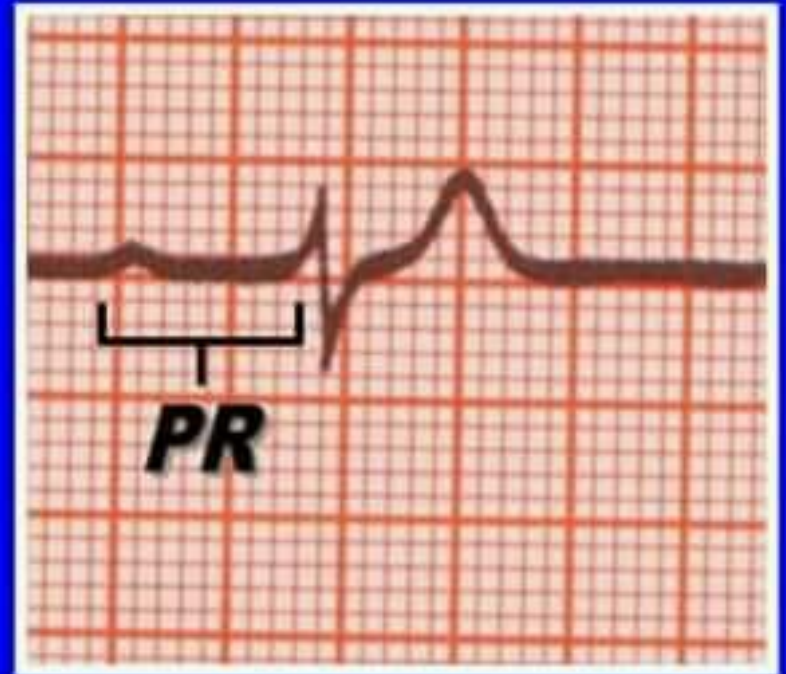


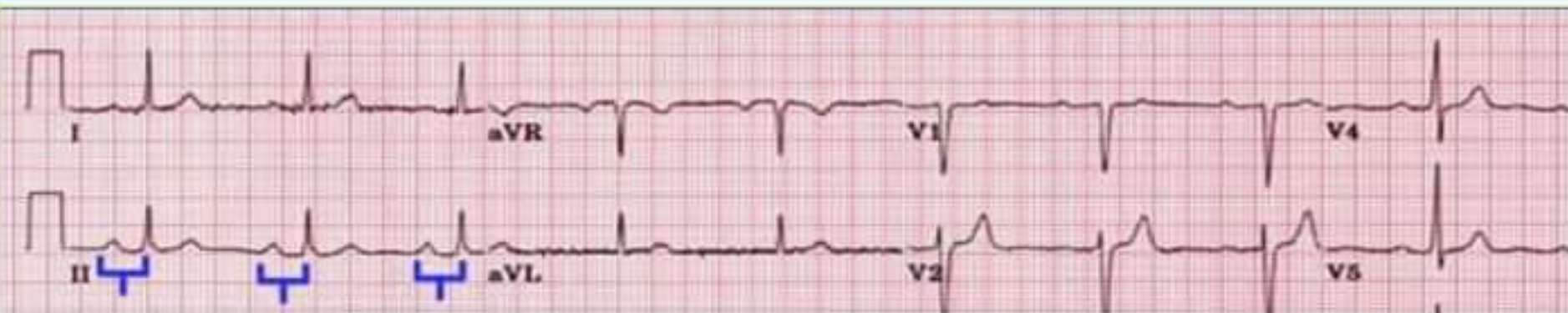
Rappel



BAV 1degré

PR sup. à 0,20 constant
Rythme sinusale



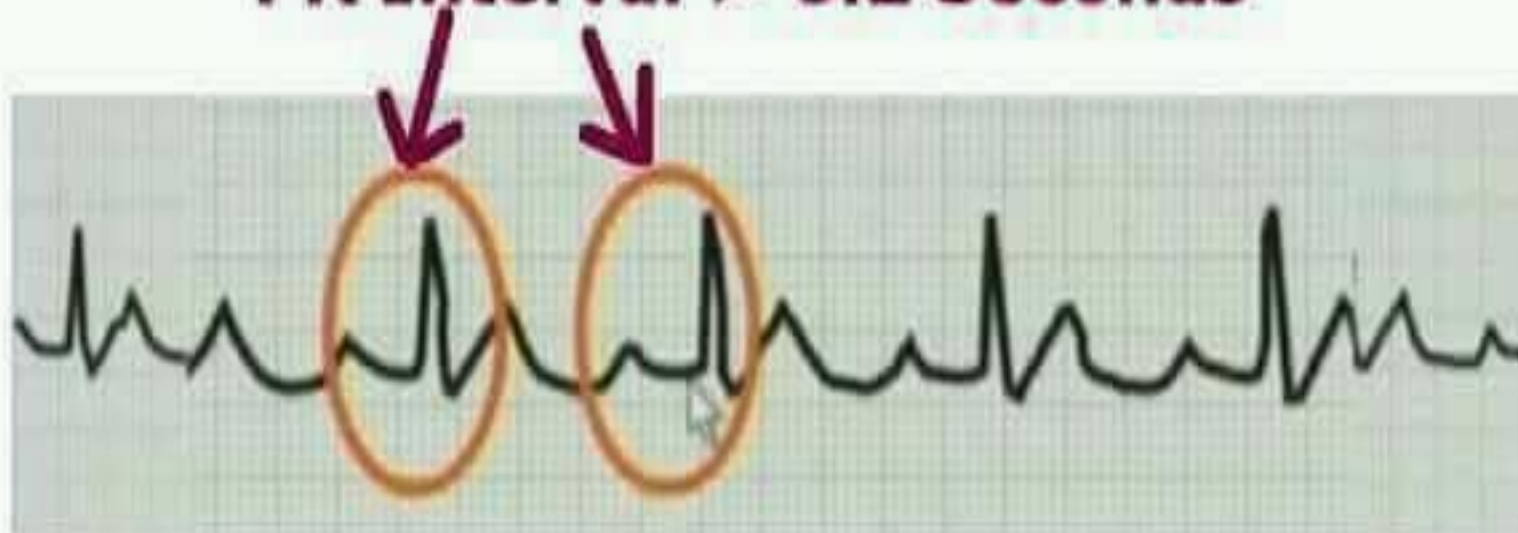


⌞ *Allongement constant de PR sup. à 0.20s*

BAV 1 degré

1st Degree AV Block

PR Interval > 0.2 Seconds

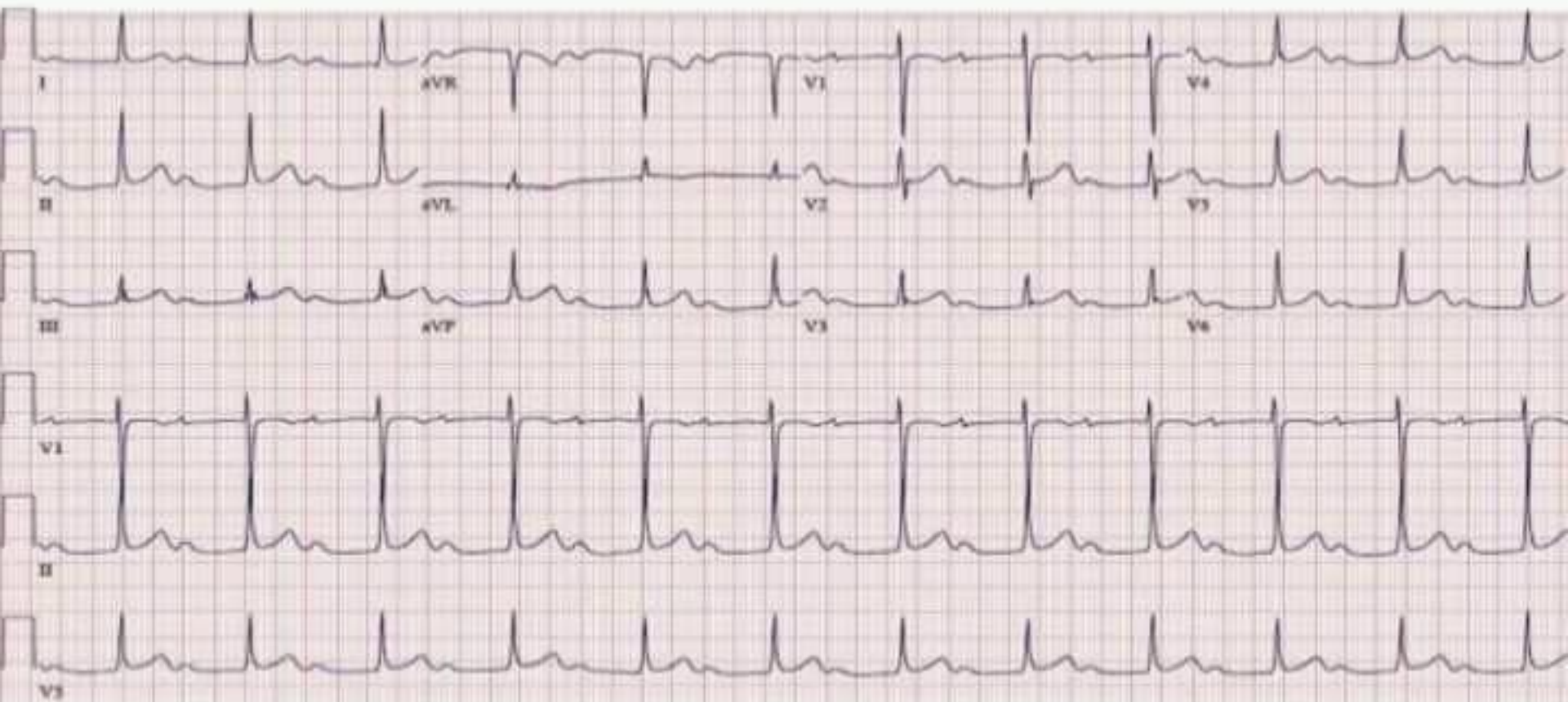


Most common causes of first-degree AV block: **a)** Intrinsic AVN disease **b)** Enhanced vagal tone **c)** Acute myocardial infarction (MI), particularly acute inferior wall MI **d)** Myocarditis **e)** Electrolyte disturbances (eg, hypokalemia, hypomagnesemia) **f)** Drugs (especially those drugs that increase the refractory time of the AVN, thereby slowing conduction)

Causes :

- 1 : drugs : eg, B Blockers , CCB , digoxin
- 2 : degenerative changes in AV node
- 3 : acute inferior MI

RX : Usually it need no treatment



BAV DE 2° degré

```
graph LR; A[BAV DE 2° degré] --- B[Mobitz type 1(luciani Wenckebach)]; A --- C[Mobitz type 2];
```

Mobitz type 1(luciani Wenckebach)

Mobitz type 2

BAV 2eme degré

Mobitz type I

L'allongement des périodes réfractaires et absolues. Le bloc est situé au-dessus du Faisceau de His.

un allongement progressif du PR, jusqu'à la **survenue** d'une onde P bloquée. Cette séquence se répète: ce sont les périodes de Luciani-Wenckebach.

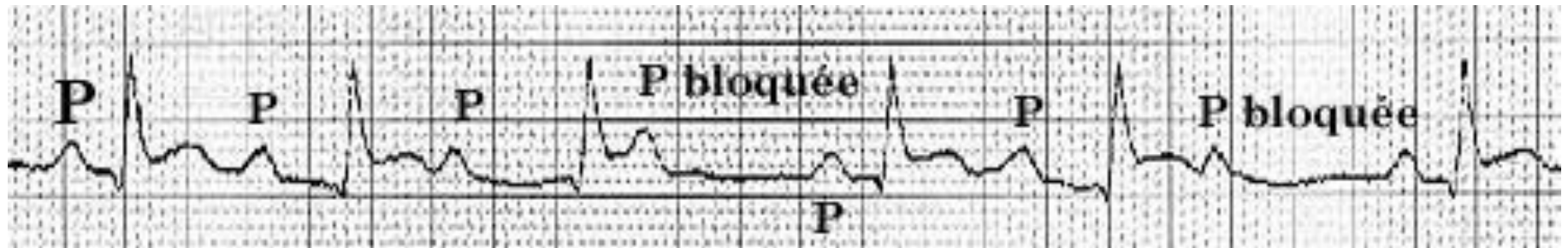
AV Block

- 2nd degree AV Block
 - Type 1 (Mobitz)
 - Progressive increased PR interval (gradually) until there is a P wave which is not followed by QRS complex



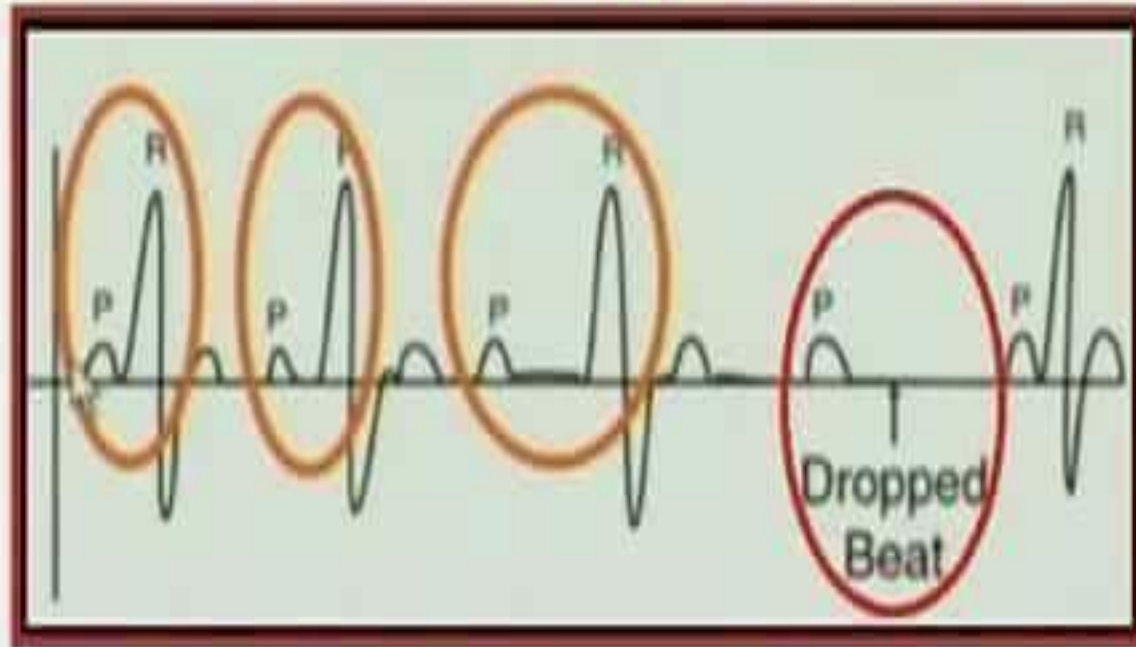
BAV 2éme degré Mobitz I périodes de Luciani-Wenckebach

- BAV 2 mobitz I:



Second-degree AV block

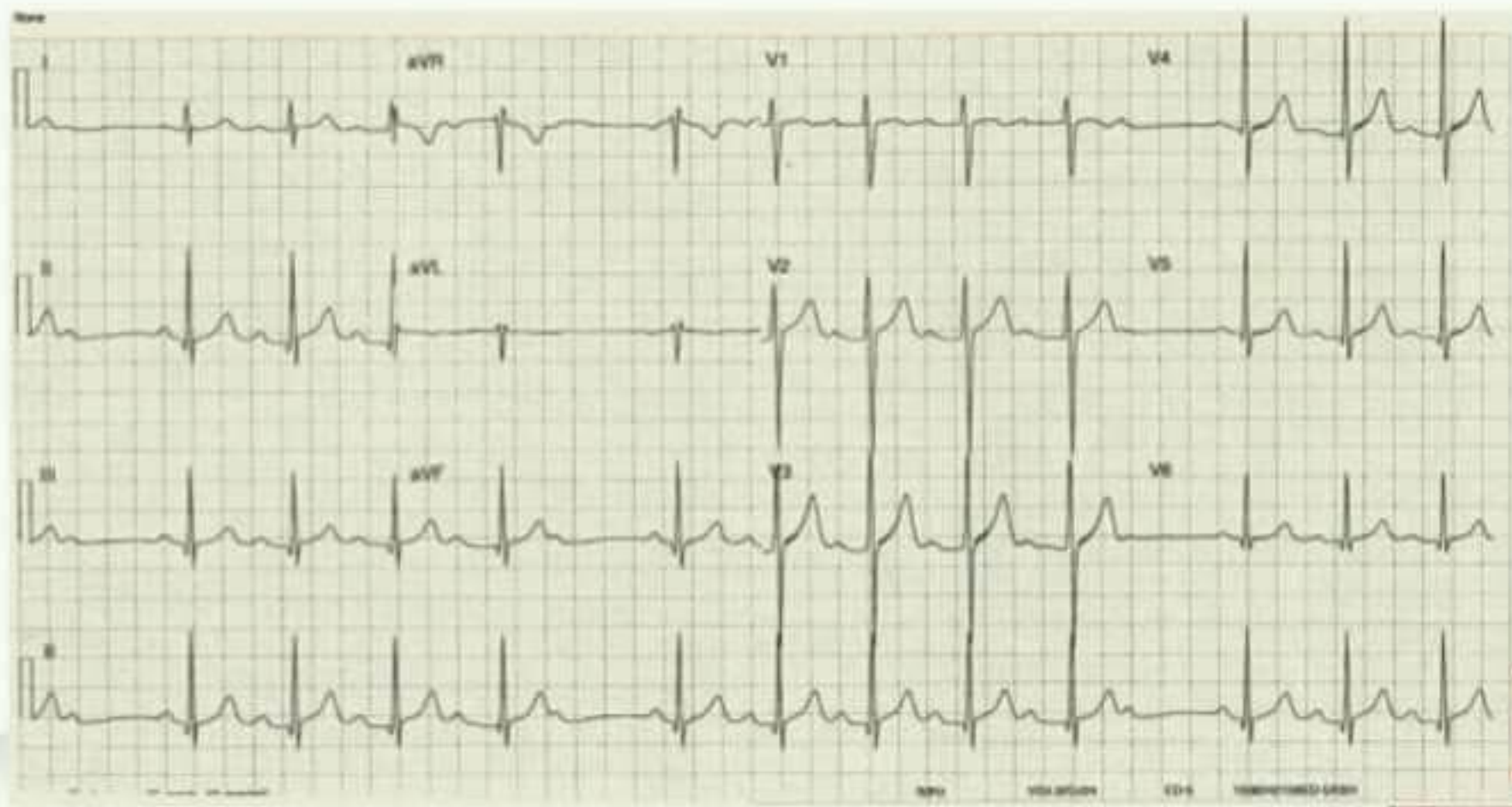
Wenckebach (Type I Mobitz)



Progressive prolongation of PR interval on consecutive beats followed by a blocked P wave (i.e., a '**dropped**' QRS complex). After the dropped QRS complex, the PR interval resets and the cycle repeats. One of the baseline assumptions when determining if an individual has Mobitz I heart block is that the atrial rhythm has to be regular. If the atrial rhythm is not regular, there could be alternative explanations as to why certain P waves do not conduct to the ventricles.

Another example of Wenckebach cycle :

Note the difference in PR interval between the first and last QRS complex of each group.



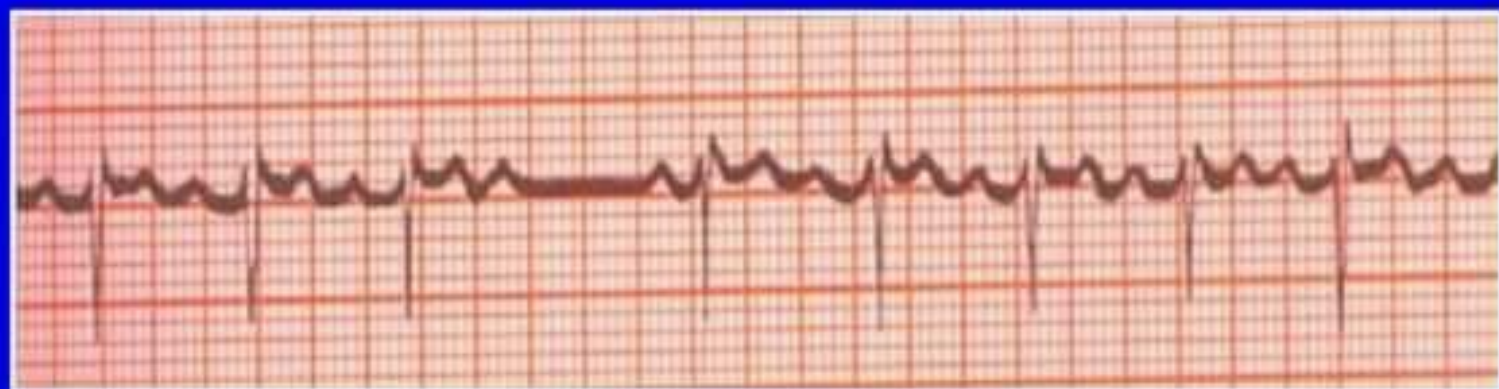
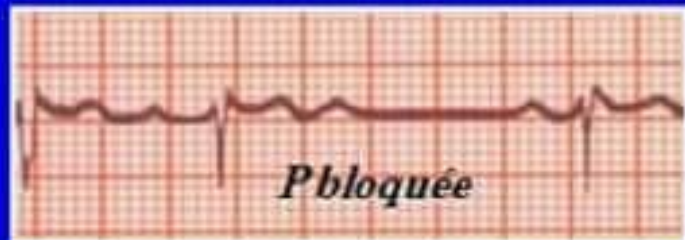
Type mobitz 2

- Mobitz type II

- Le siège du bloc est au niveau du faisceau de His ou des branches.
- Ces structures obéissent au principe du " tout ou rien ".
- une onde P bloquée. L'intervalle PR reste constant.

BAV 2° de degré Mobitz 2

PR constant
P bloquée

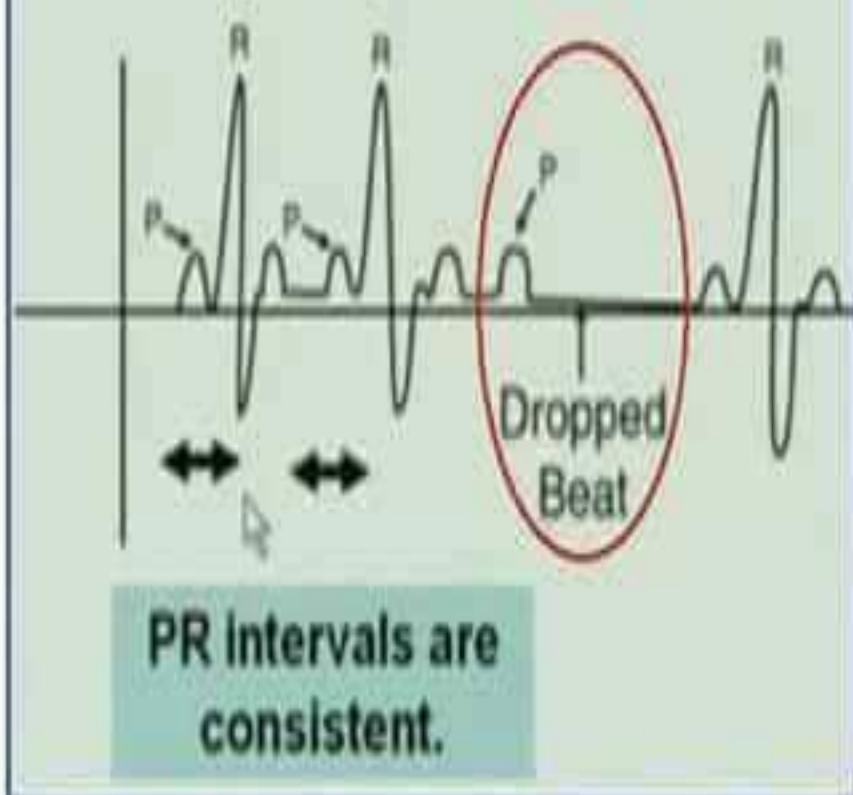


AV Block

- 2nd degree AV Block
 - Type 2 (Mobitz)
 - No progressive increased of PR interval
 - Suddenly there is a p wave which is not followed by QRS complex



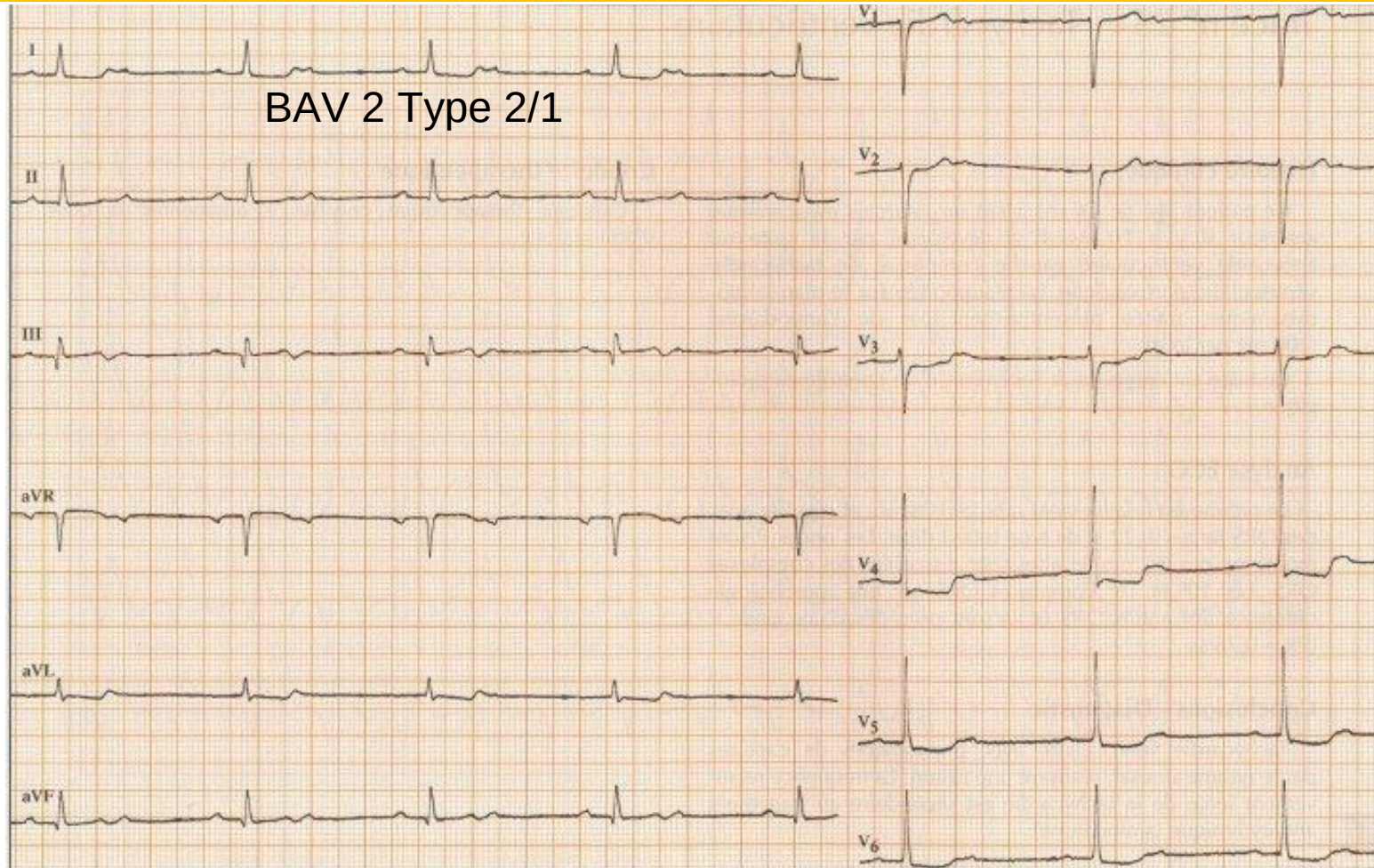
Mobitz Type II



2nd Degree AV block Type 2, also known as "Mobitz II"

- Intermittent non-conducted P waves without progressive prolongation of the PR interval. The PR interval in the conducted beats remains constant. The P waves 'march through' at a constant rate. The RR interval surrounding the dropped beat(s) is an exact multiple of the preceding RR interval (e.g. double the preceding RR interval for a single dropped beat, treble for two dropped beats, etc).

Les troubles de la conduction



B.A.V 2eme degré.

Mobitz I or Wenckebach



Mobitz II



2:1 block



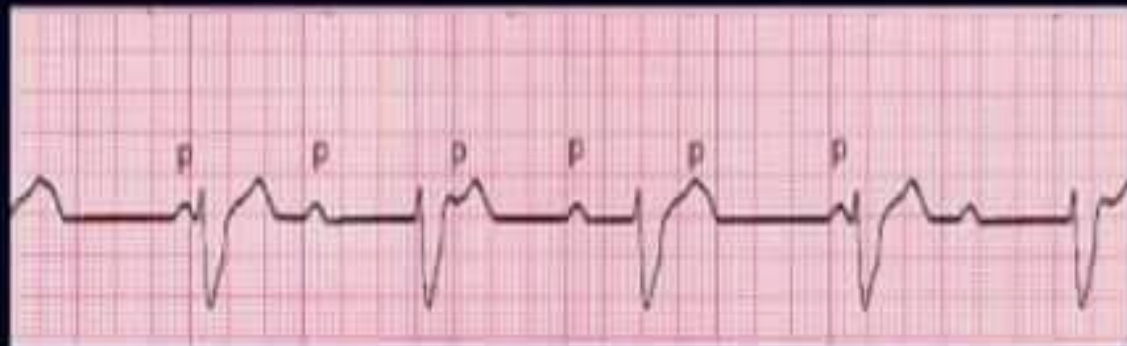
BAV 3 degré

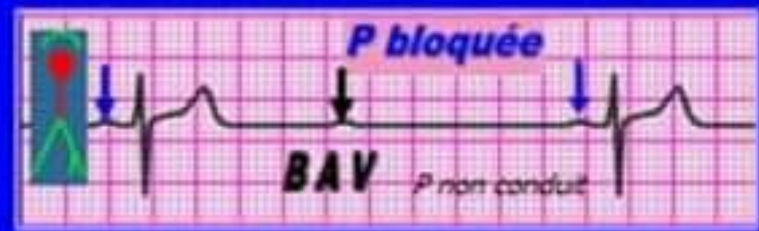
ou BAV Complet

- Absence complète de conduction entre les oreillettes et les ventricules.
- totale dissociation auriculo-ventriculaire.
- Si les oreillettes sont activées par un influx sinusal, les ventricules sont alors dépolarisés par un autre centre d'automatisme (soit la jonction, soit le réseau de Purkinje).

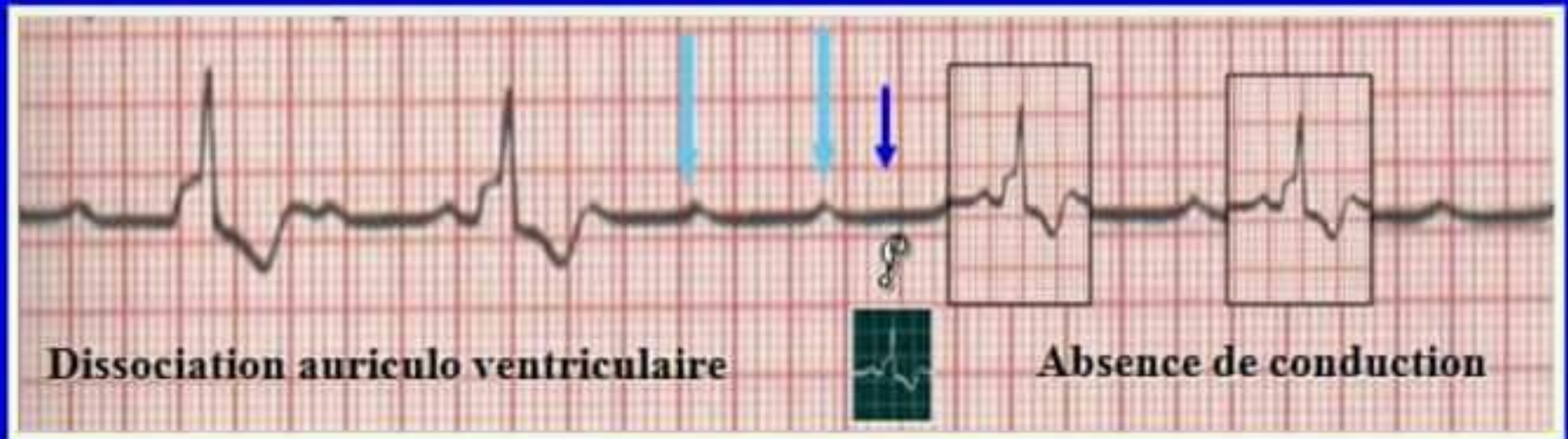
AV Block

- 3rd degree AV Block
 - No communication between P wave and QRS Complex
 - P wave rate is different than QRS rate





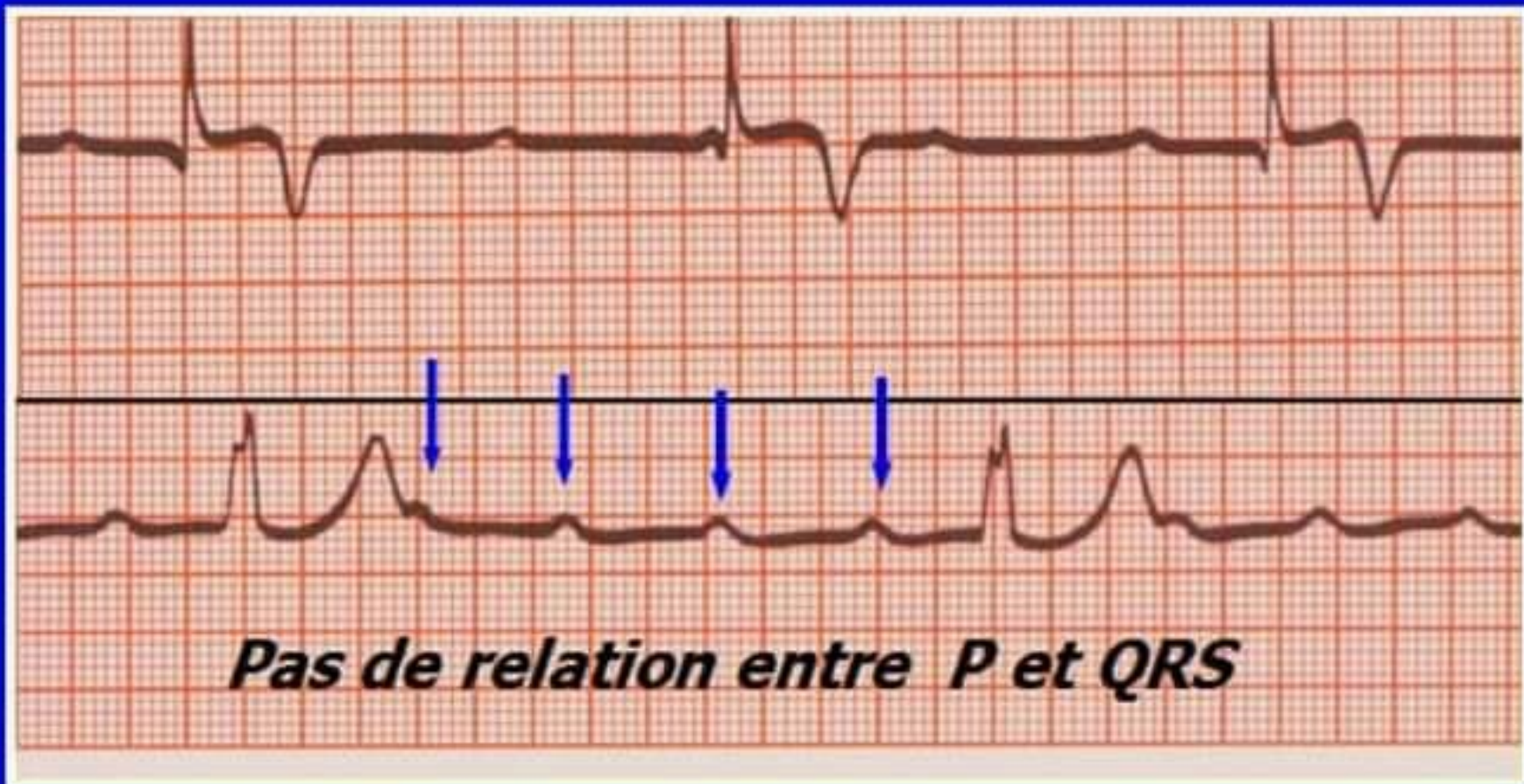
BAV Complet



BAV Complet



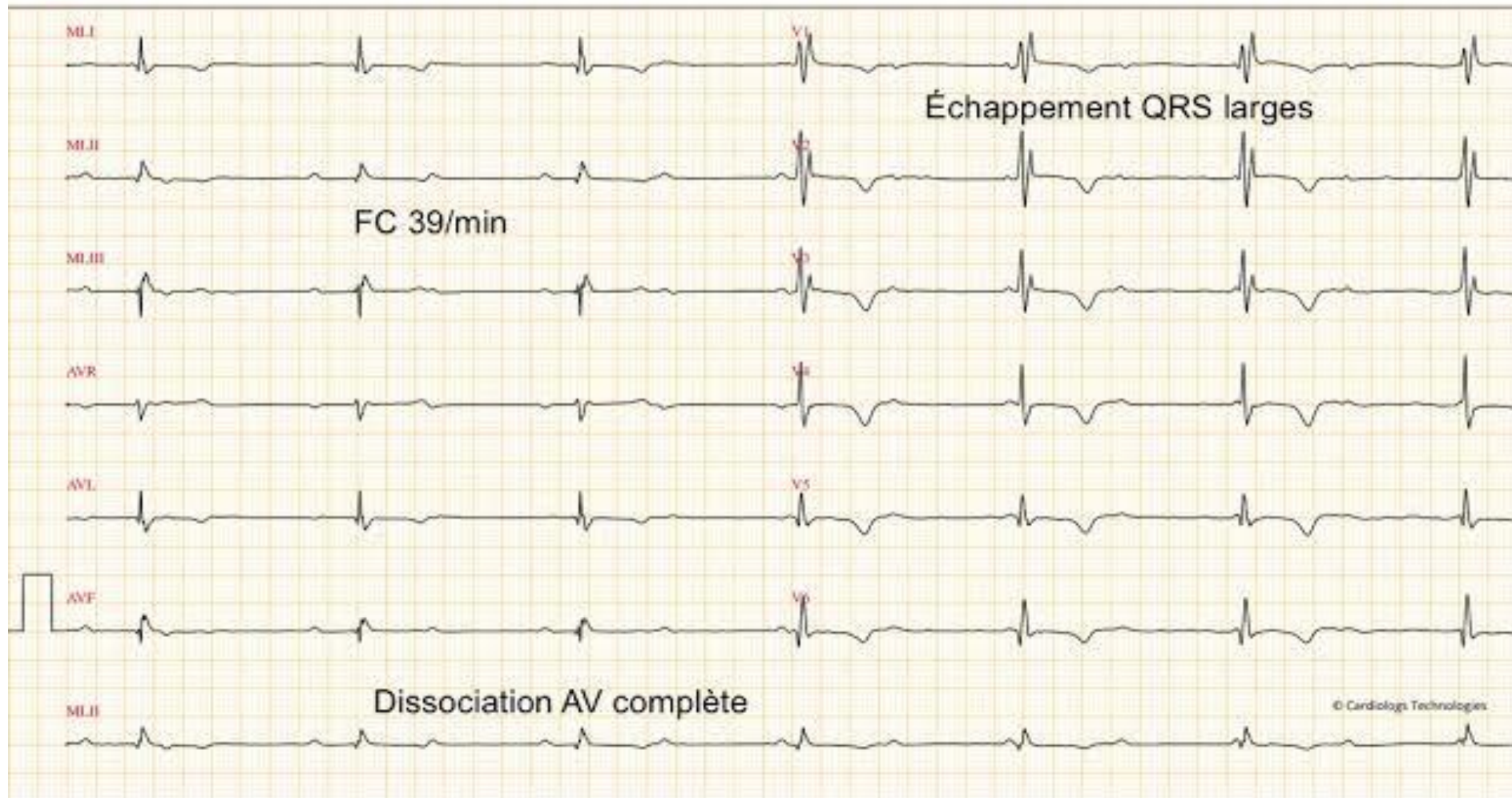
Fréq. ventriculaire très lente

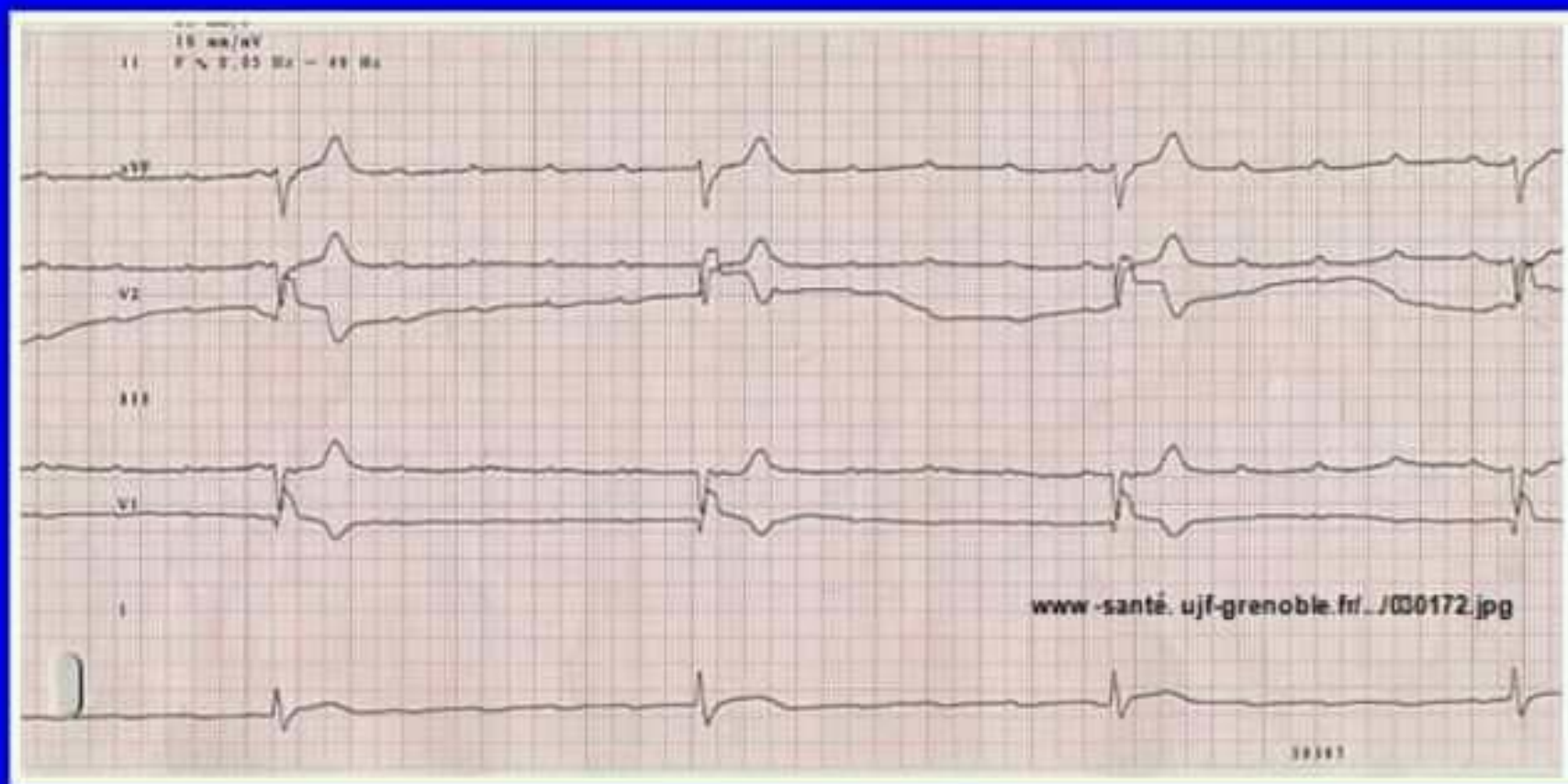


Dissociation auriculo ventriculaire

BAV 3^{ème} degré

BAV complet avec rythme échappement à QRS larges et lent
(en faveur d'un bloc infranodal)

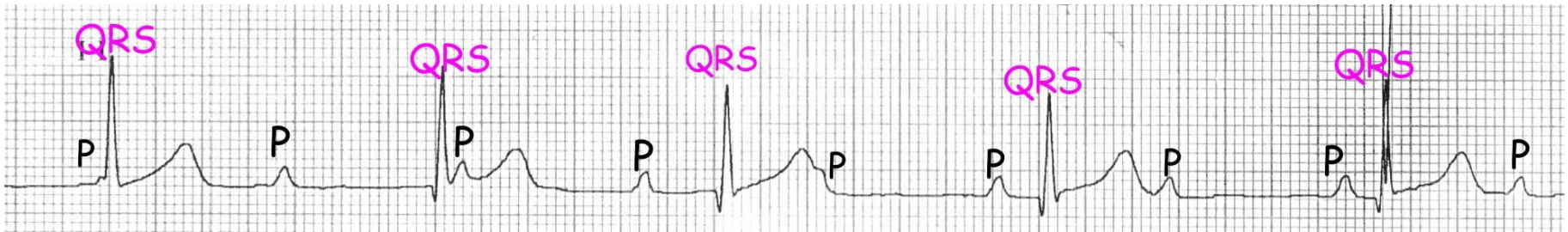




bloc auriculo-ventriculaire complet (3ème degré)

Les troubles de la conduction

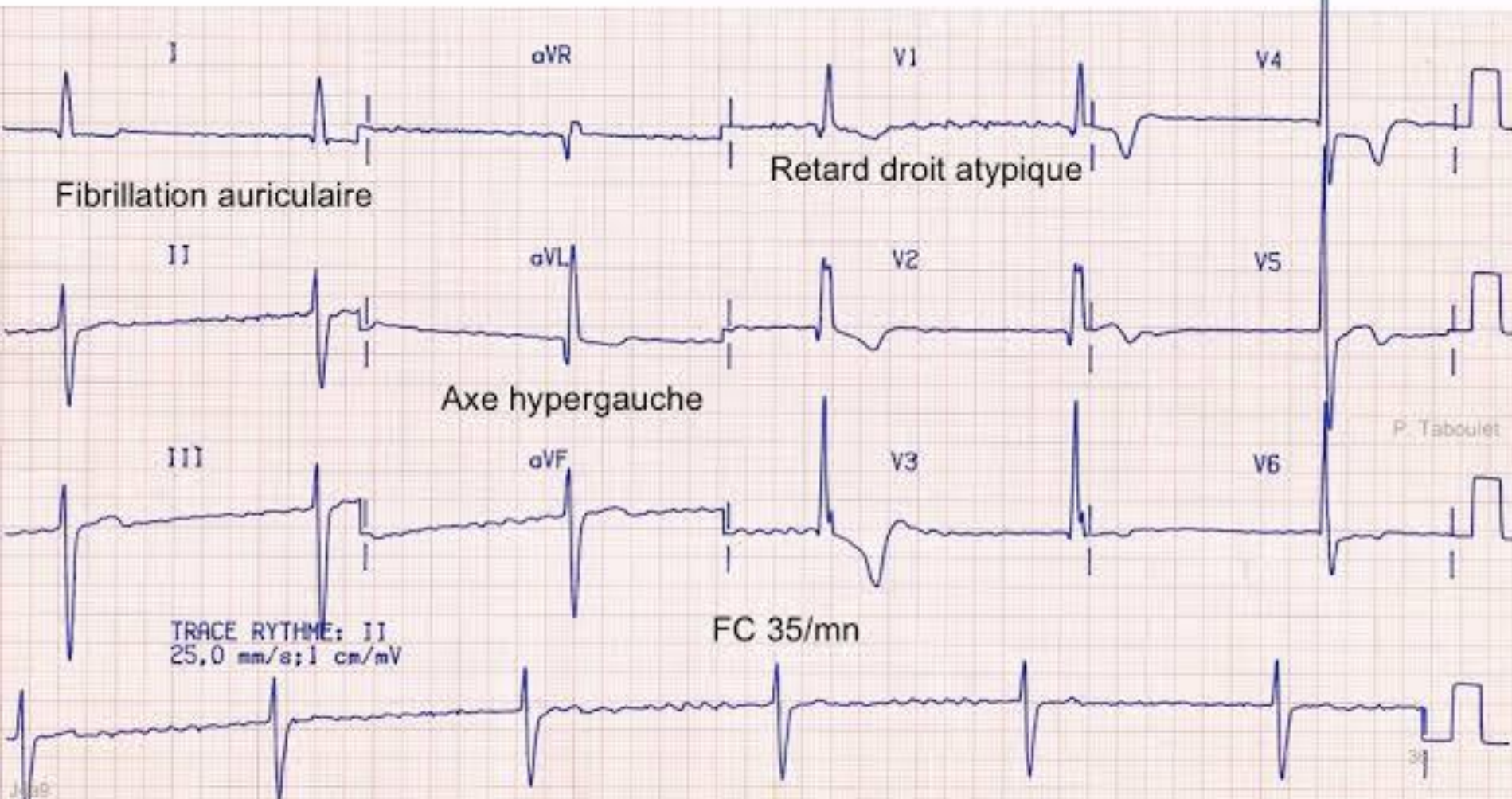
■ BAV 3 :

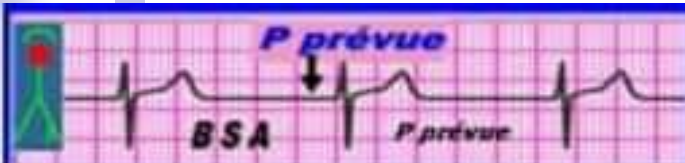


- Piège : FA + BAV3 = FA lente et régulière
- Piège 2 : BBG alternant avec BBD = BAV3

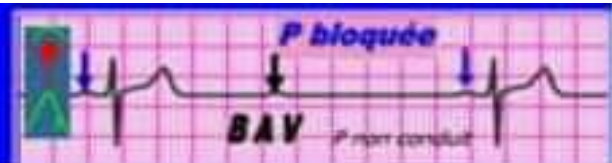
BAV 3^{ème} degré

Bradycardie à QRS réguliers au cours d'une FA
→ Rythme d'échappement ventriculaire sur BAV complet (infranodal)





A retenir



BAV 1er degré : $PR > 0,20-0,22$ seconde mais chaque P est suivi d'un QRS

BAV 2ème degré : certains P ne sont pas suivi d'un QRS. l'aspect est souvent répétitif 2/1 (2P pour 1 QRS), 3/2 (3P pour 2 QRS), etc...
On distingue :

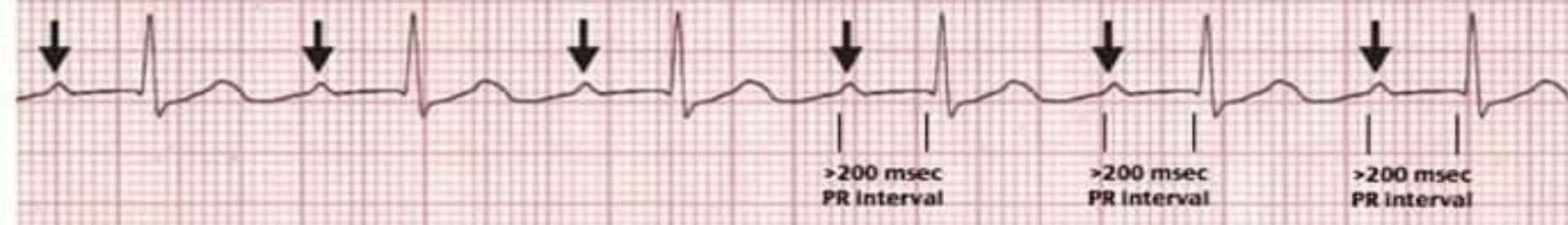
Type Möbitz 1 qui est caractérisé par un allongement progressif du PR (avec un QRS chaque fois) jusqu'à une onde P non suivie d'un QRS et ensuite retour au PR initial (Luciani Wenckebach).

Type Möbitz 2 caractérisé par un blocage périodique d'une onde P sans modification du PR précédents.

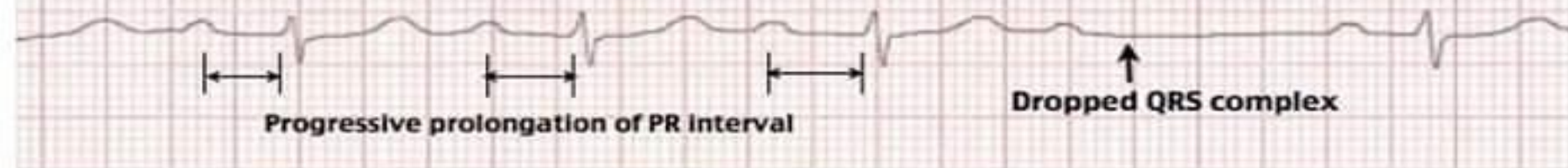
Type haut degré qui se caractérise par plus de 2 ondes P sans QRS

BAV complet : les ondes P battent régulièrement mais les QRS sont indépendants (arrêt complet de la conduction). Même complet, un BAV peut être bien toléré, tout dépend du rythme autonome des QRS.

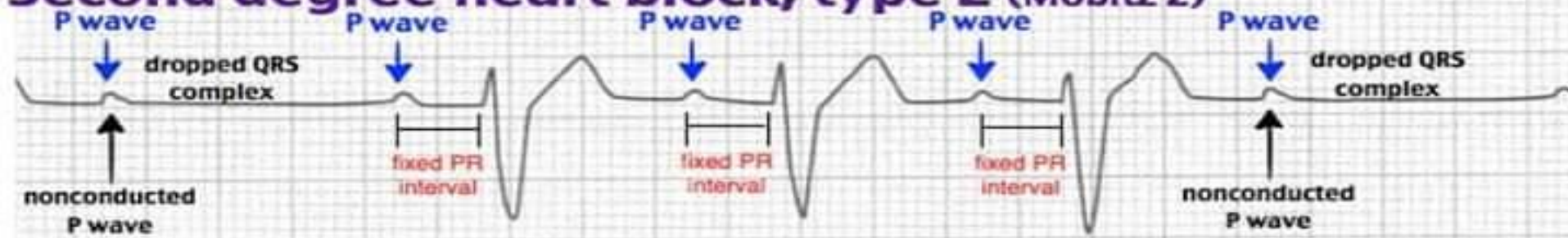
First Degree Heart Block



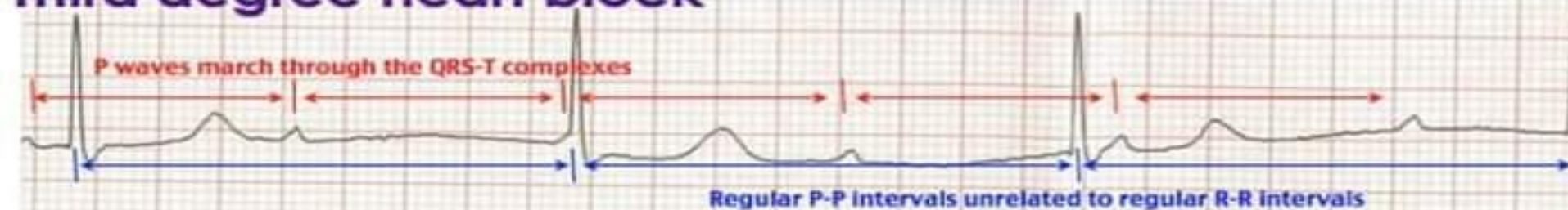
Second degree heart block, type I (Wenckebach/Mobitz I)



Second degree heart block, type 2 (Mobitz 2)

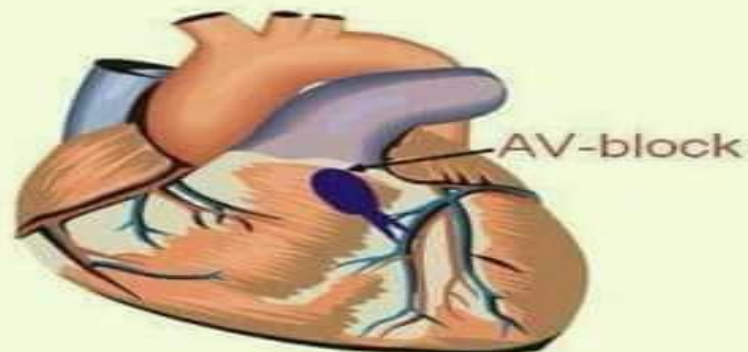
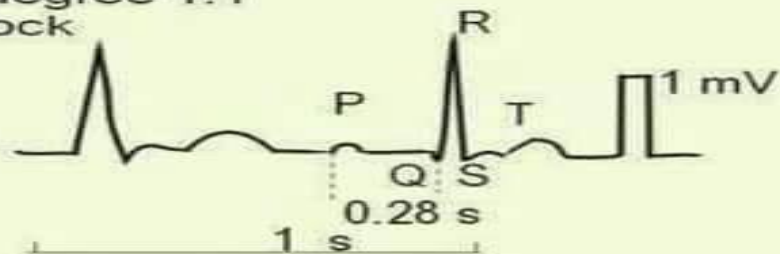


Third degree heart block

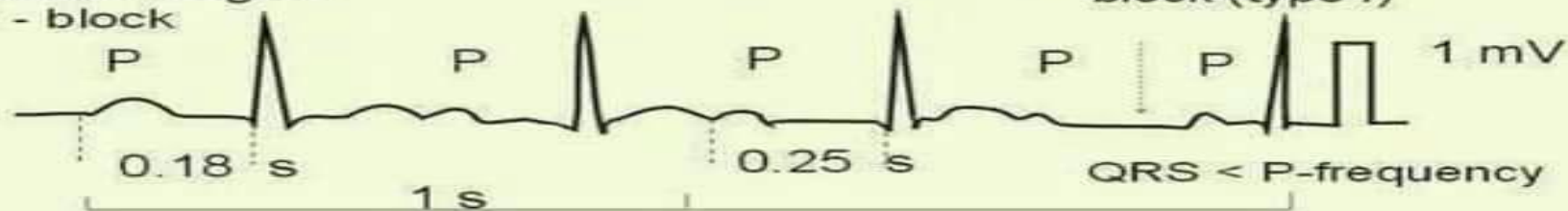


Four Types of AV-block (PQ interval > 0.2 s)

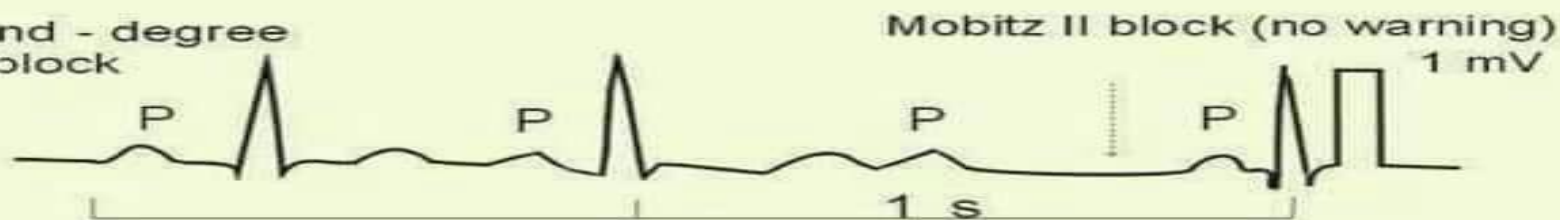
First - degree 1:1
AV - block



Second - degree
AV - block

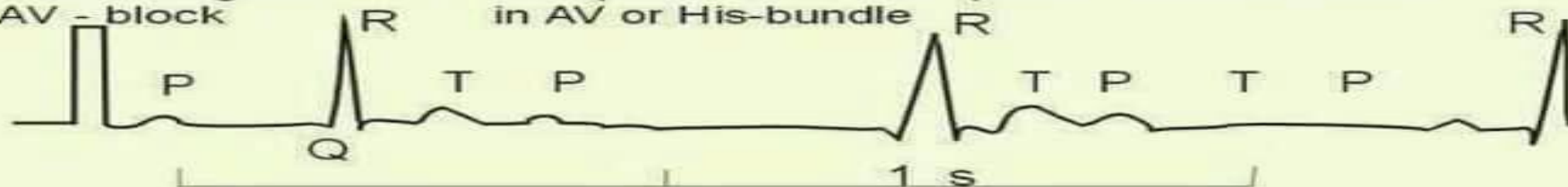


Second - degree
AV - block



Third - degree
AV - block

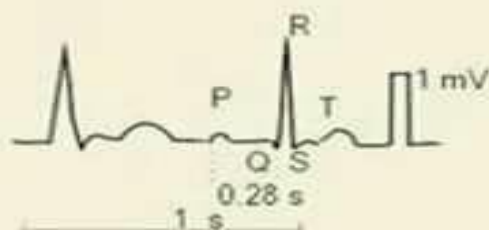
Complete AV - block (Adam Stokes disease
in AV or His-bundle)



The Heart Block Poem

by the Princeton Surgical Group & [nurseslabs](#)

If the **R** is far from **P**,
then you have a **FIRST DEGREE**.



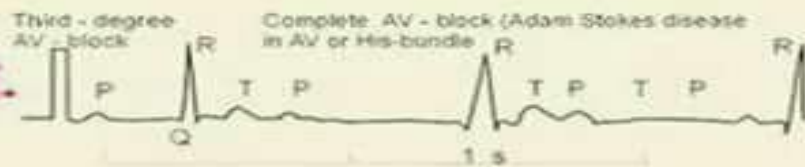
Longer, longer, longer, drop!
Then you have a **WENKEBACH**.



If some **P**s don't get through,
then you have **MOBITZ II**.

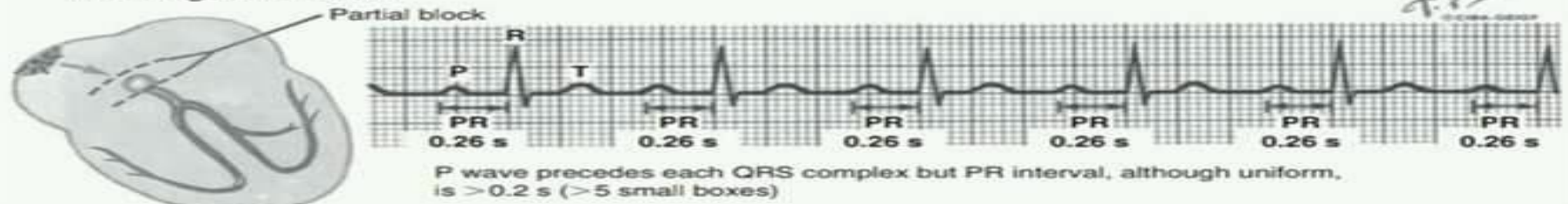


If **P**s and **Q**s don't agree,
then you have a **THIRD DEGREE**.

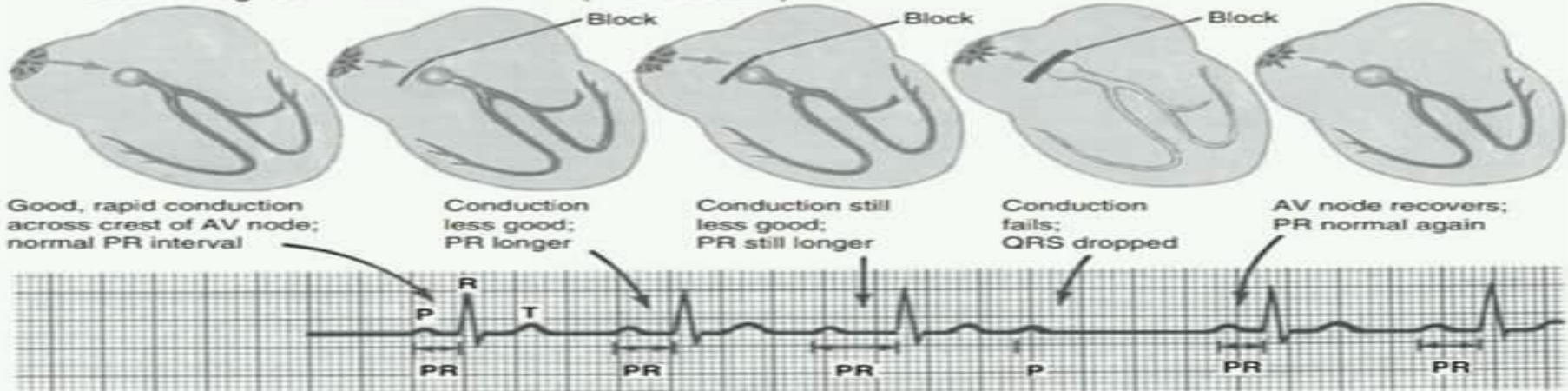


Atrioventricular conduction variations

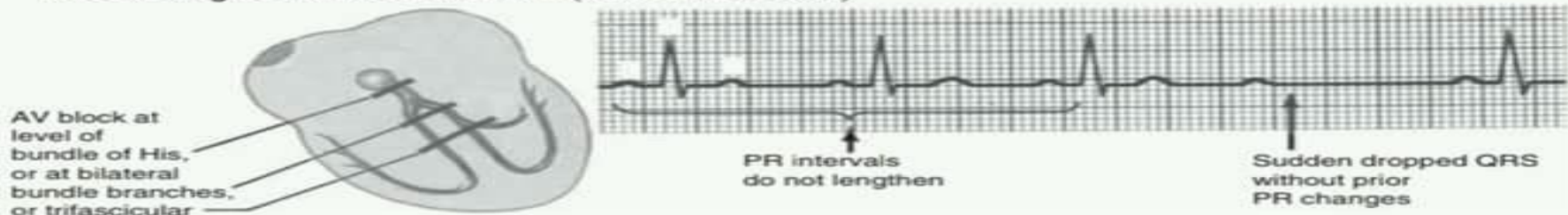
- a. Fixed normal PR interval
Sinus rhythm
- b. Fixed but short PR interval
 1. Junctional or coronary sinus rhythm
 2. Wolff-Parkinson-White syndrome
- c. P wave related to each QRS complex, but variable PR interval
 1. Wandering atrial pacemaker
 2. Multifocal atrial tachycardia
- d. Fixed but prolonged PR interval
First-degree AV block



- e. Progressive lengthening of PR interval with intermittent dropped beats
Second-degree AV block: Mobitz I (Wenckebach)



- f. Sudden dropped QRS without prior PR lengthening
Second-degree AV block: Mobitz II (non-Wenckebach)



Traitement des B.A.V

■ En pré-hospitalier :

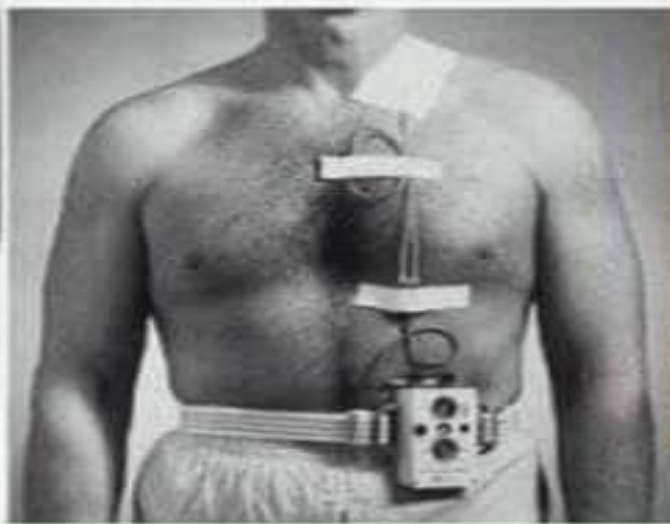
□ TDC de haut degré :

- Réanimation cardiorespiratoire, Scope, 1 à 2 voies veineuses
- Traitement médicamenteux symptomatique:
 - Atropine 1mg, très efficace si IDM inférieur ou vaso-vagal.
 - Isuprel 5 ampoules dans 250ml de SG5%.
- Traitement spécifique : traitement de l'IDM, Hyperkaliémie, Intox aux digitaliques...
- ATTENTION: L'intox aux digitaliques est une contre indication formelle à l'ISUPREL (TDR graves); ttt=DIGIDOT
- Si inefficacité médicamenteuse : patchs d'entraînement externes et stimulation cardiaque temporaire.
- Prise en charge en USIC +/- structure permettant la coronarographie et éventuelle revascularisation coronaires.
- Discuter une stimulation définitive : Pacemaker.

□ TDC de bas degrés : pas de traitement en urgence



**First pacemaker
1950's**



**Wearable pacemaker
mid 1950's**



**First implantable
pacemaker 1958**



**Modern devices 1990's
onwards**



**The future: Leadless,
wireless pacemakers**

CENTRAL ILLUSTRATION An Overview of the History of Cardiac Pacing

Paradigm Shifts in Cardiac Pacemakers

1950s

AC-powered pacemakers tethered to an extension cord (Furman)



1950s

Battery-powered transistorized "wearable" pacemakers (Lillehei/Bakken)



1958

First fully implantable pacemaker (Elmqvist/Senning)



2015

Implantable pacemaker—basic system had not evolved significantly



2016

Leadless pacemaker—the entire device is placed within cardiac chambers



Future

Batteryless devices, which harvest cardiac motion to power pacing circuits



Mulpuru, S.K. et al. *J Am Coll Cardiol.* 2017;69(2):189-210.

Historically, pacing developed using large, external, alternating current (AC)-powered devices, which subsequently evolved to "wearable" transistorized battery powered pacemakers—both comprise the era of external devices. A paradigm shift occurred with the introduction of the entirely implantable pacemaker, composed of an extravascular pulse generator connected to a transvenous lead in contact with the myocardium. This paradigm continues to this day. An emerging and rapidly developing new paradigm is that of leadless pacemakers, which are available for clinical use. Batteryless pacemakers that harvest cardiac mechanical motion to generate current, or that modify or add cells to introduce biological pacing activity, are under active investigation.

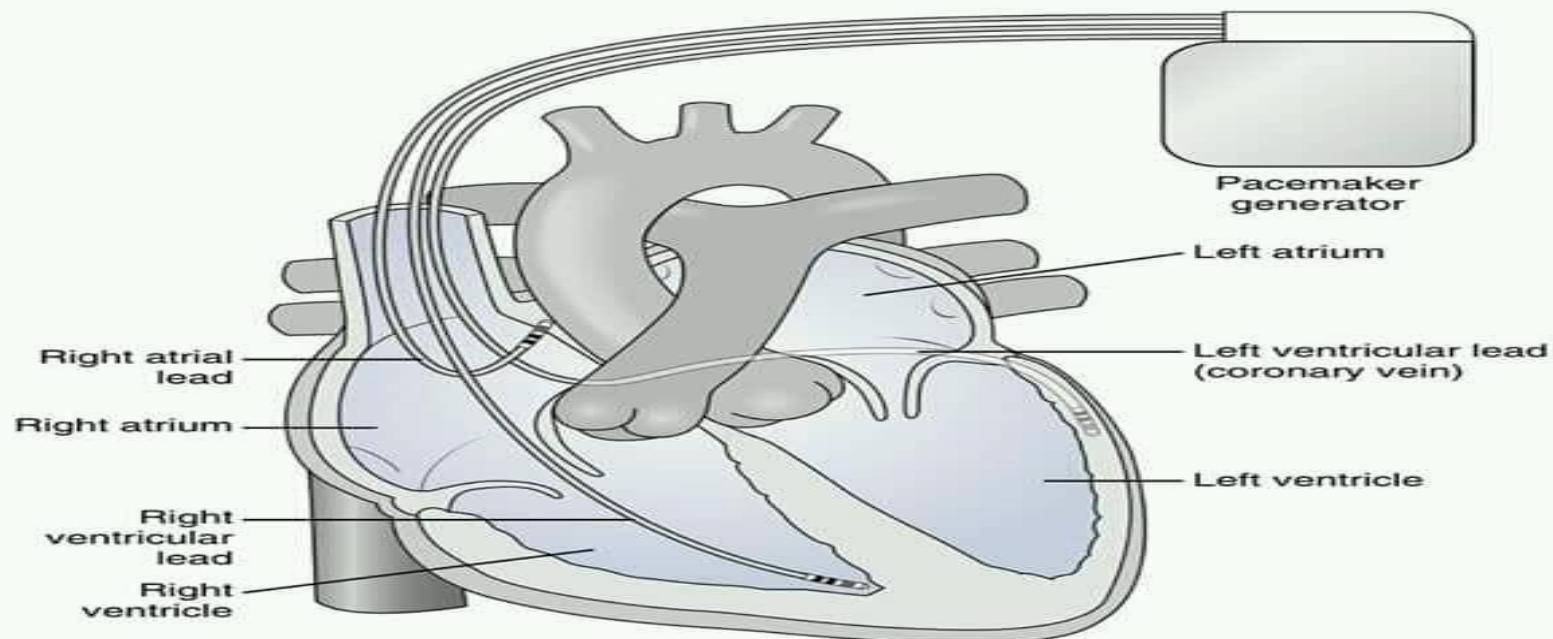


Fig. 38.3. Example of the components of a biventricular pacing system, with pacemaker generator a right atrial lead, a right ventricular lead, and a left ventricular lead. (Image adopted from Goldberger, A. L. [2006]. *Clinical electrocardiography: a simplified approach*, [7th ed.]. Philadelphia, PA: Mosby).

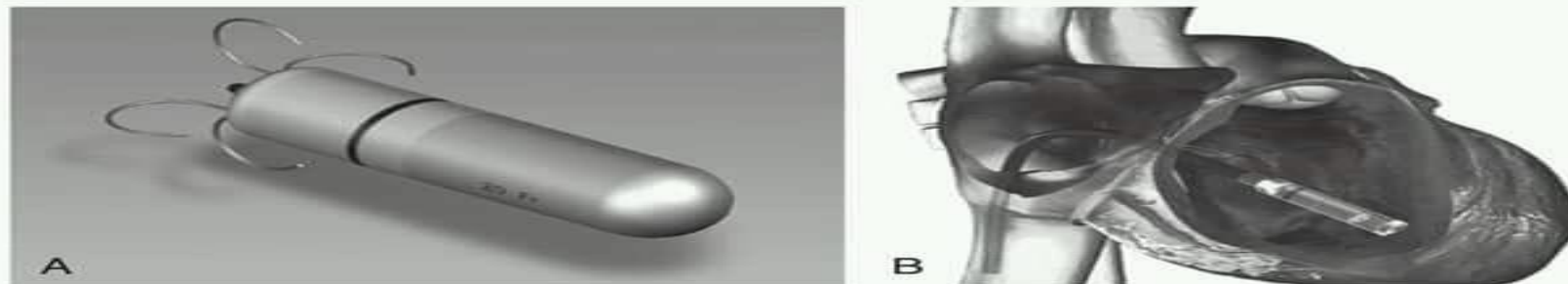


Fig. 38.4. Leadless pacemaker. **A**, Example of a leadless pacemaker. The pacemaker is only approximately 1 inch in length. **B**, illustration of implantation of the leadless pacemaker. The pacemaker is percutaneously inserted into the right ventricular apex via a femoral venous approach.



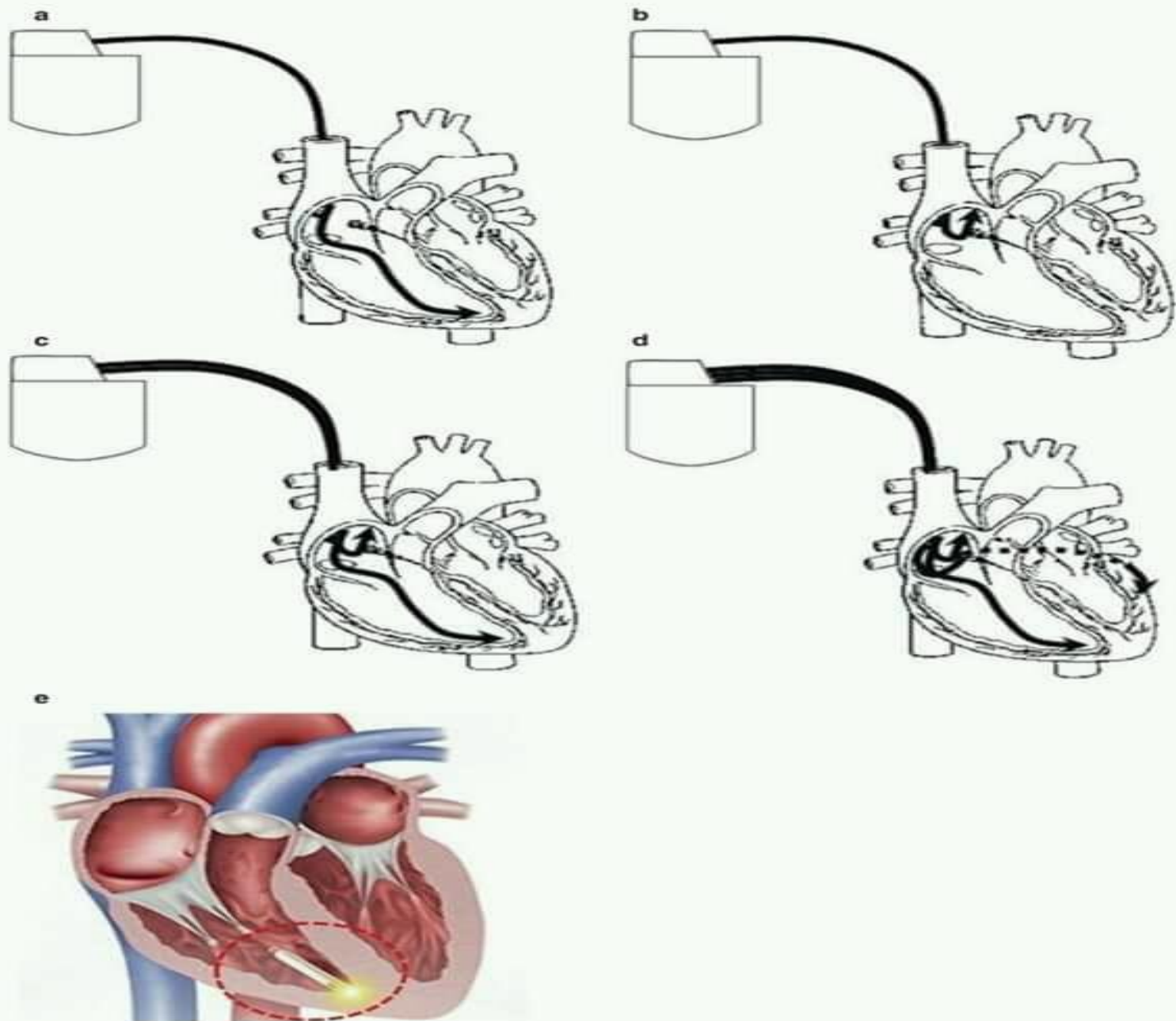
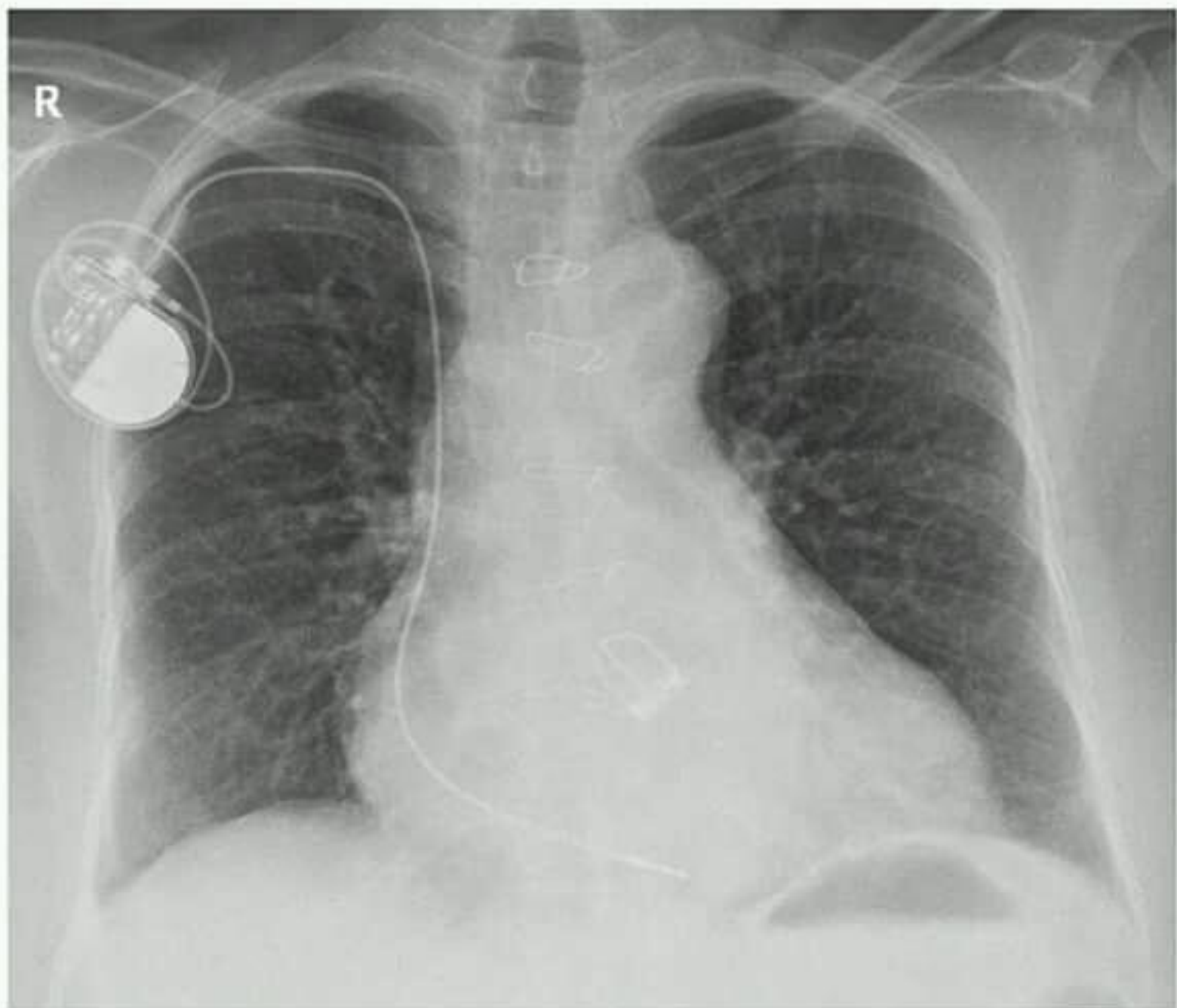
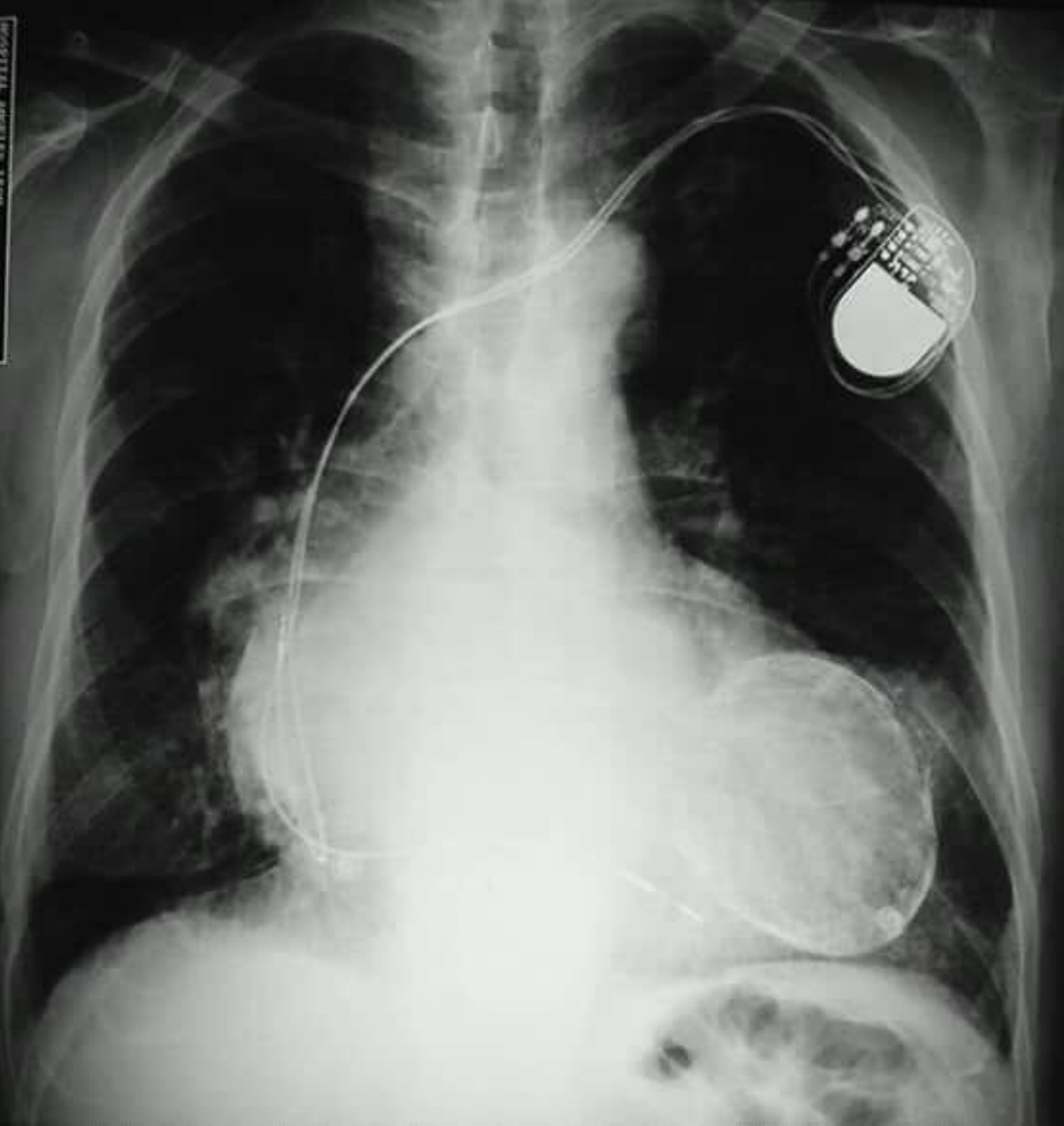
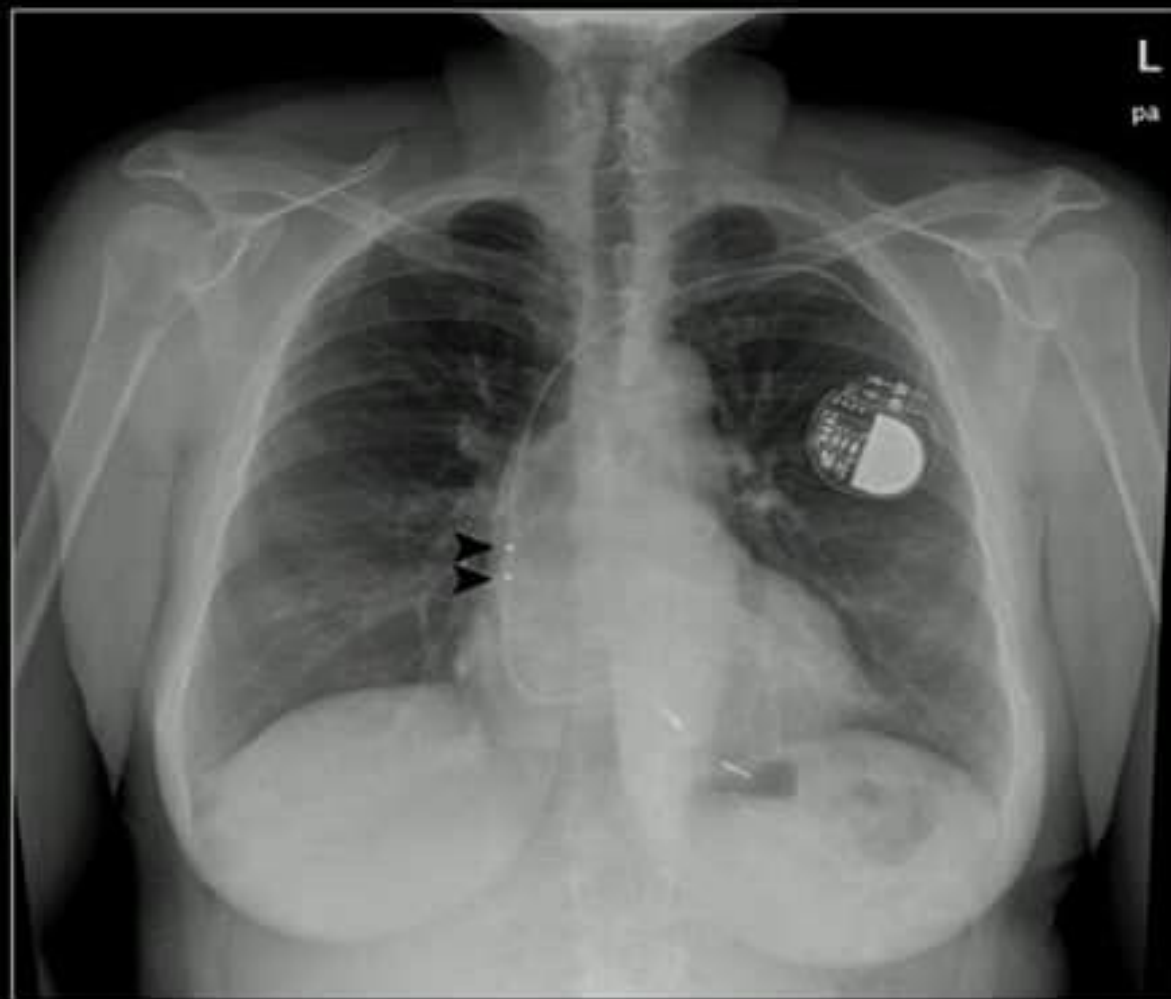


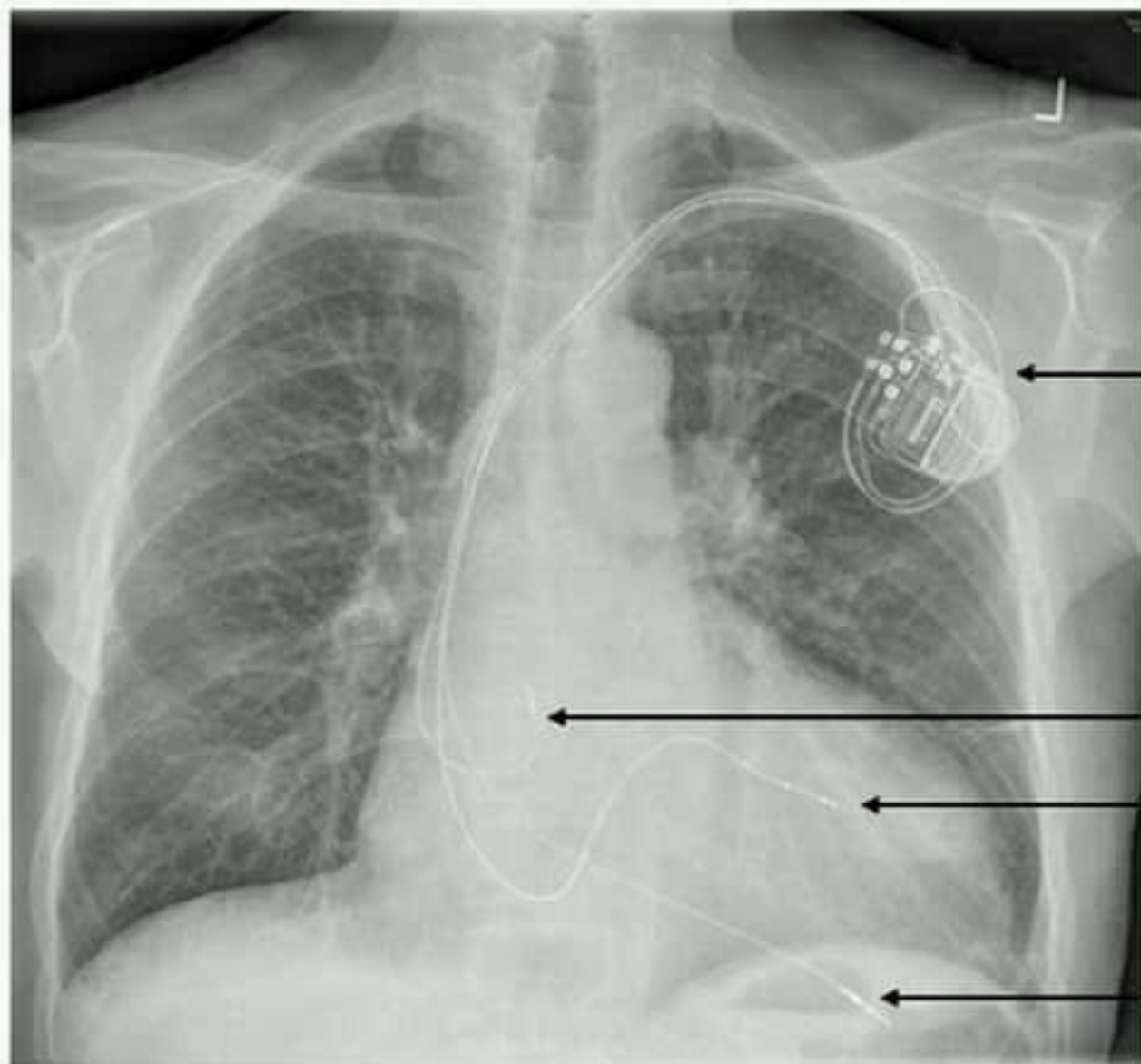
Fig. 10.1 Schematic of commonly used pacemaker systems and newly developed leadless pacemakers. (a) Single-chamber ventricular pacemaker; (b) Single-chamber atrial pacemaker; (c) Dual-chamber pacemaker; (d) Triple-chamber (biventricular) pacemaker (Permission

granted by Mayo Clinic Proceedings); (e) Nanostim leadless pacemaker (St. Jude Medical, Inc., Sylmar, CA). Small pen-size device which includes battery, electronic circuitry and pacing electrodes.









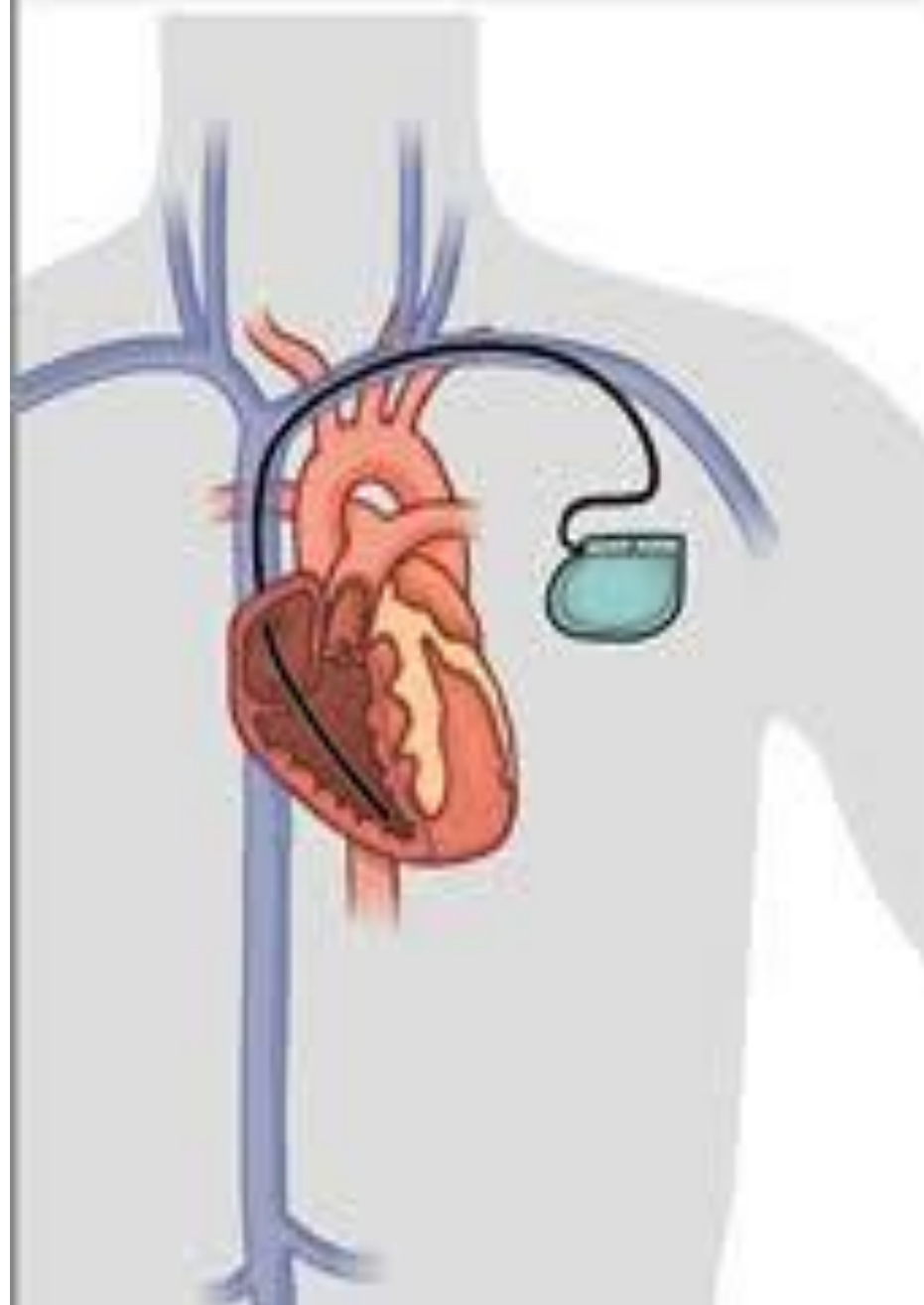
Pulse Generator

RA Lead

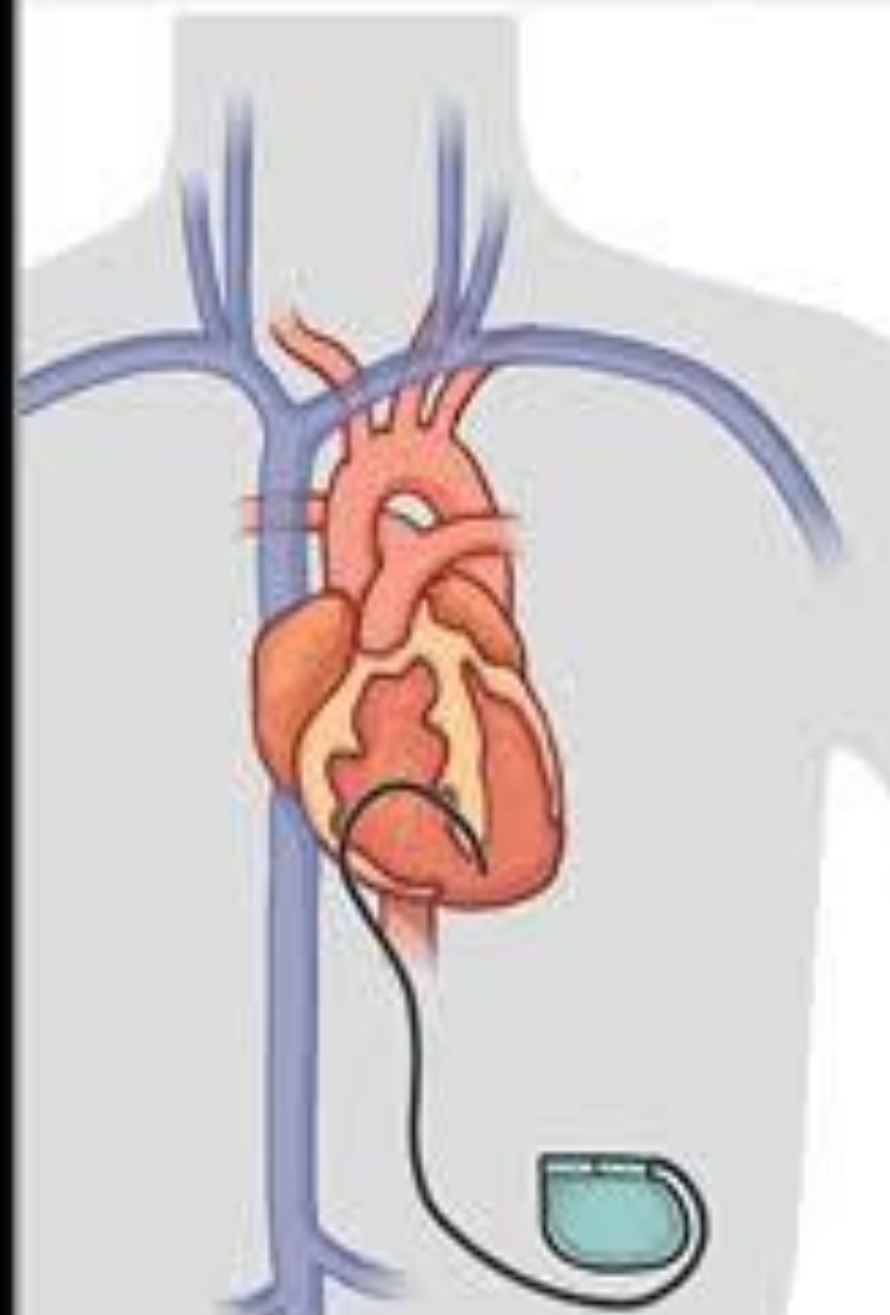
LV Lead through
the Coronary sinus

RV lead

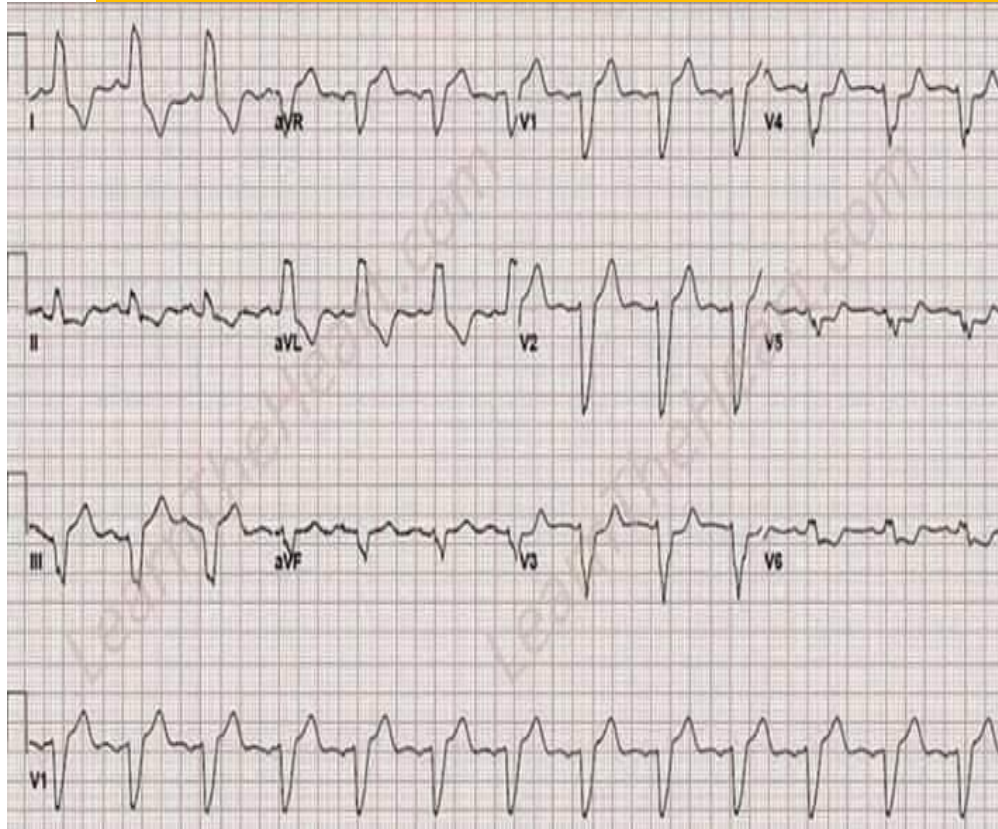
Stimulateur transveineux



Stimulateur épicardique



BBG complet



BBG :

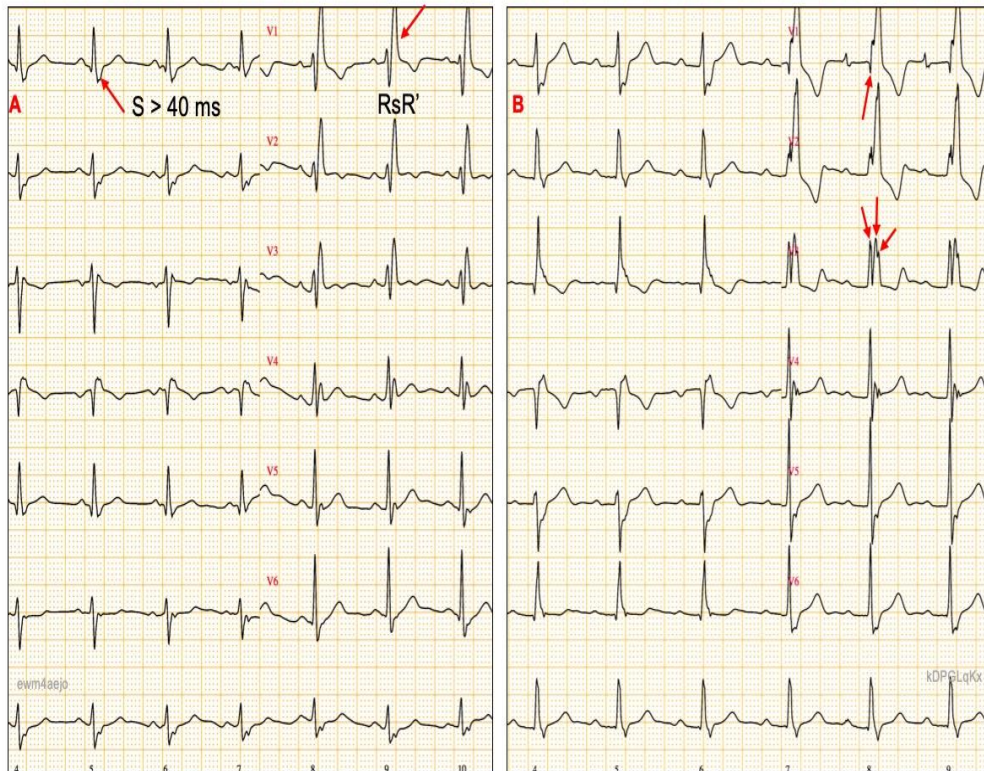
- rythme supra-ventriculaire $PR > 0.12s$
- Axe QRS a gauche .
- DIS gauche $> 0.50s$
- QRS large $> 0,12s$
- deformation morphologiques :
 - R exclusive en lat D1/AVL et V5V6 avec trouble de repolarisation secondaires(T-/ST- discordant).
 - QS en antérieur v1v2v3 et sus ST discordant .

BBD Complet

Bloc de branche droit

A - **BBD** : QRS 150 ms, axe 60°

B - **BBD HVD** : QRS 167 ms et fragmentés avec qRR' V1 et axe 100°



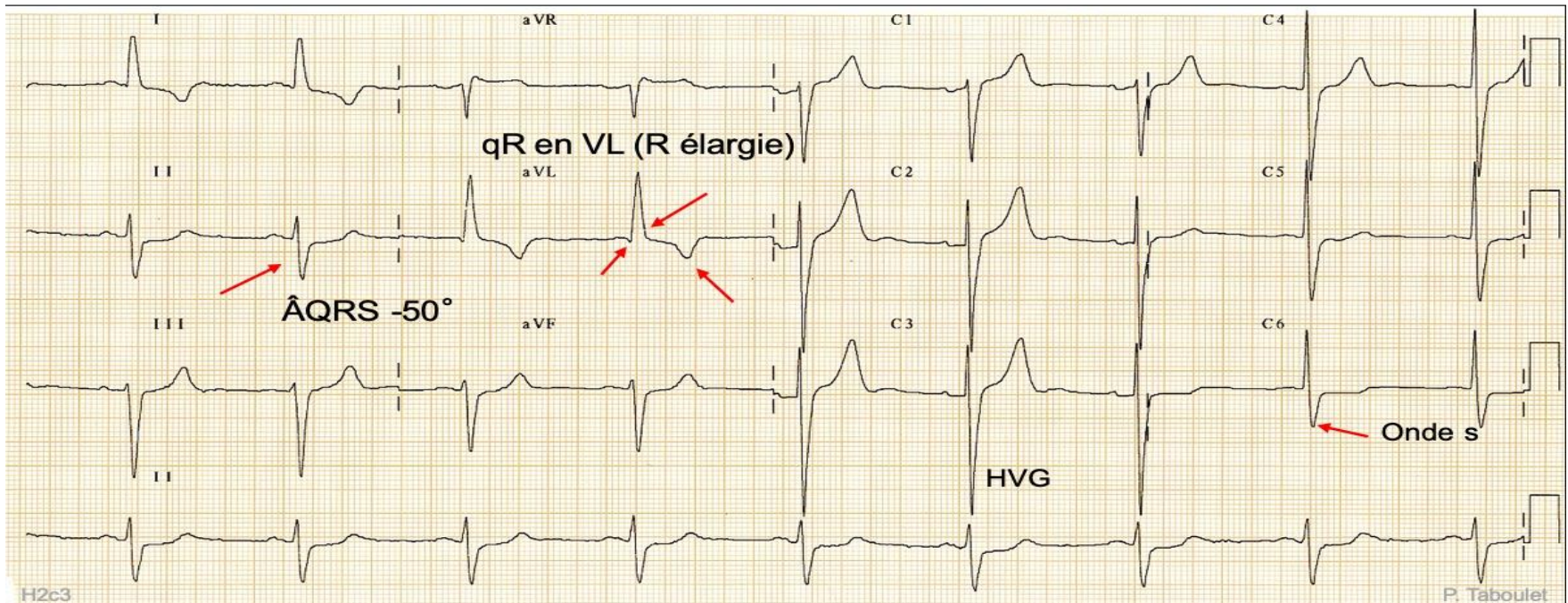
BBD :

- Rythme supraventriculaire
- PR > 0.12s.
- Axe QRS à droite.
- QRS large > 0.12s.
- DID > 50ms (déflexion interséconde d droite).
- Aspct morphologique : RsR' en V1+/-V2.
- S > 40 ms en V6 ou DI.
- trouble de repolarisation secondaire a type de ST-/T- en V1 V2.

HBAG (hémibloc antérieur gauche)

Bloc fasciculaire antérieur gauche

Axe hypergauche : $DII < 0$ (rS), $VL > 0$ (qR) avec R élargie (retard D.I. ≥ 45 ms)
(Surawicz et al. AHA... Circulation 2009)



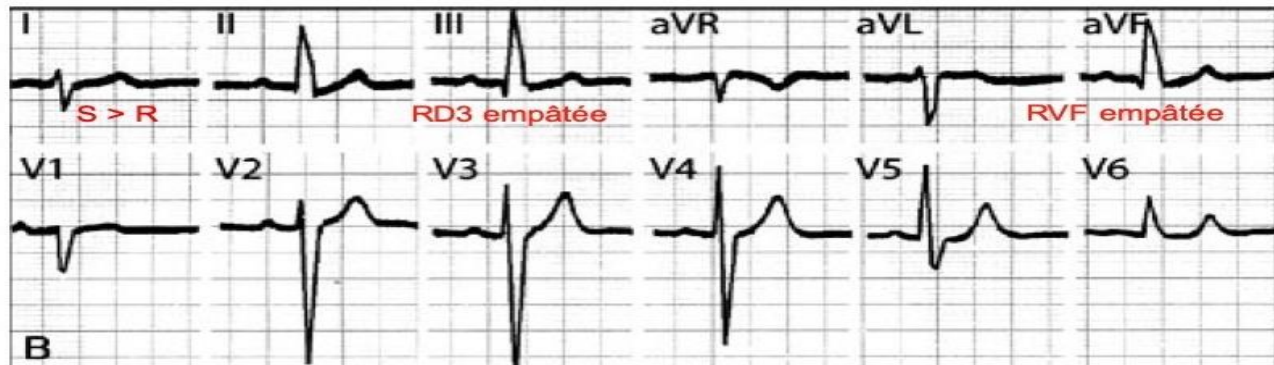
HBPG(hémibloc post gauche)

Elizari et al. Hemiblocks Revisited. Circulation 2007;115 1154-63

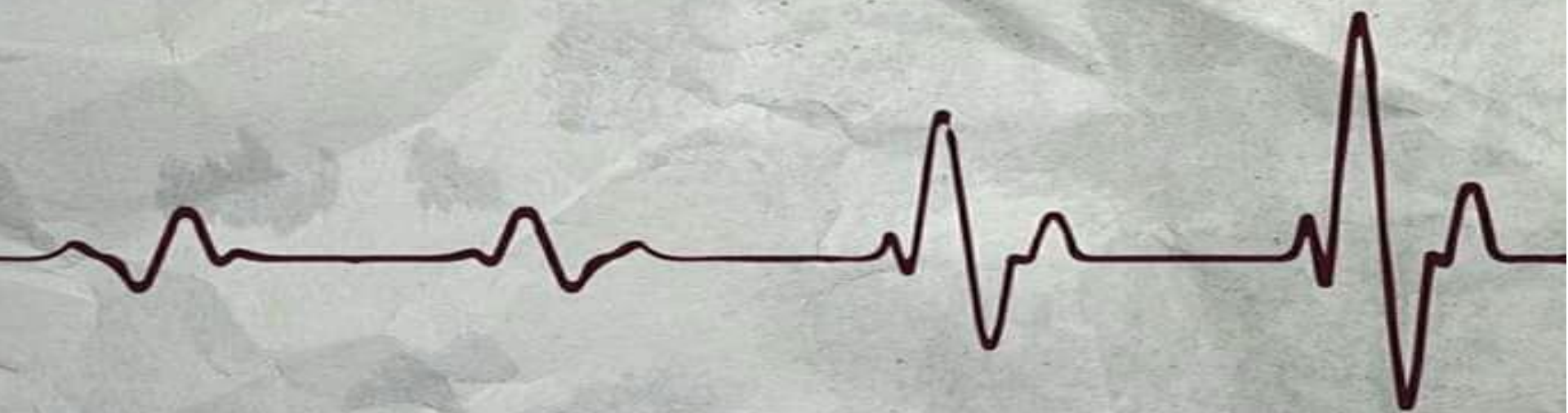
ECG N
ÂQRS 50°



HBPG
ÂQRS 110°



IF THERE ARE NO UPS AND DOWNS IN YOUR LIFE
IT MEANS YOU ARE DEAD





Bon courage.

Dr Chelouache.T
HMRUC

